

POWER IDEALS AND THE HILBERT BOUNDARY OF QUADRATIC PENCILS

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ABSTRACT. We identify the normalized Hilbert graph of lifted quadratic pencils on the full reduced corank-two incidence open with a Fitting-ideal blow-up. At a pencil with k incidence points its entire fibre is $(\mathbf{P}^1)^k$, and the universal curve is a reduced nodal tree. Independence of the incidence centres follows from the radicals of distinct pencil members, without an extra genericity hypothesis. For symmetric Jordan slices of every contact order $m \geq 2$, we compute the normalized Hilbert modification and its nonreduced curves: a main component meets an m -fold tail along a length- m scheme. The total multiplication graph has transverse singularity A_{m-1} . After ramified base change its normalization gives a stable-map tail of degree m , distinguishing the Hilbert and stable limits with the same cycle. The proofs use an exact power-ideal identity over arbitrary complex algebras. All-rank power graphs, singular-pencil data, and the complete earlier collision and reciprocal-fibre results are included in the appendices.

1. INTRODUCTION

A limiting pencil does not in general determine the embedded limit of its lift to complete quadrics. Even when the base modification is smooth, its universal Hilbert curve may be nonreduced and its total space singular. We determine these phenomena on the full reduced corank-two incidence open and on explicit nonreduced-contact slices of every order.

1.1. The full reduced-incidence modification. Let $G = \mathrm{Gr}(2, \mathrm{Sym}^2 V)$, $n = \dim V \geq 3$, and let U_{red} be the open of regular pencils that avoid rank at most $n - 3$ and whose intersections with rank at most $n - 2$ are reduced. The number of intersection points is unrestricted. With $q : D \rightarrow U_{\mathrm{red}}$ the finite incidence scheme, our principal identification is

$$\hat{G}|_{U_{\mathrm{red}}} \simeq \mathrm{Bl}_{\mathrm{Fitt}_0(q_* \mathcal{O}_D)} U_{\mathrm{red}}$$

(Theorem 6.3). The left side is the normalization of the reduced graph closure of lifted pencil curves in the specified Hilbert scheme. Its fibre at a k -incidence pencil is the whole scheme $(\mathbb{P}^1)^k$. Each factor parametrizes lines through a fixed point in an ambient exceptional $\mathbb{P}^2$; the parameter factor and the ambient plane are different spaces. The corresponding curve has one main component and k independently attached embedded tails. Its nodal local equations are $x_i y_i = t_i$, with independent exceptional parameters t_i .

The local arrangement is not assumed to be transverse. Lemma 6.2 proves the needed independence for every pencil in this open: radicals at distinct spectral points lie in a direct sum, and an arbitrary symmetric variation prescribes the forms on those radicals independently. The general transverse-incidence lemma is stated separately, so the classical blow-up mechanism is not confused with this pencil calculation. The product description gives every fibre and every local boundary intersection, rather than a family contained in one fibre.

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1.2. Nonreduced contact and the universal singularity. For every $m \geq 2$, a two-parameter family of $2m$ -dimensional symmetric linear pencils has exact residual Schur complement

$$\begin{pmatrix} x^m + u & v \\ v & x^m - u \end{pmatrix}.$$

The special pencil has contact length m and local Smith exponents m, m . The normalized graph of this *slice* is $\mathrm{Bl}_0 \mathbb{A}^2$. In the chart $u = t, v = tw$, its universal curve is the blow-up of (x^m, t) , with exact Rees equation $tU - x^m V = 0$ (Theorem 7.2). The special curve is $C_0 \cup T_m$, where

$$T_m = \mathbb{P}_{\mathbb{C}[x]/(x^m)}^1, \quad C_0 \cap T_m = \mathrm{Spec} \mathbb{C}[x]/(x^m).$$

It has no embedded associated points but is nonreduced. Its total space has transverse equation $x^m = tz$ and hence singularity A_{m-1} , with local cotangent deformation space of dimension $m - 1$ (Proposition 7.3). These assertions include arbitrarily high contact order in genuine linear pencils; they are not residual tangencies on an already reduced single-contact exceptional plane.

After $t = s^m$, normalization produces an ordinary point blow-up and a reduced stable-map tail mapping with degree m onto the residual line (Theorem 7.4). Thus the Hilbert limit has a thickened tail whereas the stable limit has a covered tail. Their cycles agree but their objects do not. This is an explicit comparison, not a claim that the Hilbert and stable-map compactifications are generally isomorphic. Nor do we identify a slice normalization with an unproved base change of the full ambient Hilbert normalization.

1.3. The multiplication input. The finite algebra

$$B_h(V) = \mathrm{Sym}(\mathrm{Sym}^2 V) / \langle (v^2)^{h+1} : v \in V \rangle$$

has coefficient ideals $J_{hq+s}(A) = I_q(A)^{h-s} I_{q+1}(A)^s$, $0 \leq s < h$, over every commutative complex algebra (Theorem 4.1). The first four sections fix its twists and coefficient map and prove the identity and graph description. On our rank open, $J_{h(n-2)+1} = I_{n-1}$, with $h = 1$ giving the distinguished power. All tail power systems are complete systems of degree

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+.$$

For a thick tail this is a statement over $\mathbb{C}[x]/(x^m)$, not merely on the reduced support. The exact coefficient space and the image Rees algebra retain information that radicals and cycles discard.

1.4. Classical geometry, scope, and dependencies. Li's wonderful-model Theorems 1.2–1.3 [Li(2009)] already identify arrangement modifications with product-ideal blow-ups. The strict-transform image-algebra construction is standard [Stacks Project, Tag 080C]. Our assertions identify the particular pencil incidence ideal, prove independence at all its reduced contacts, and identify the resulting space with the specified normalized Hilbert graph. The familiar equations of smooth blow-ups are not asserted to be new general geometry. Gathmann's tangency theorem [Gathmann(1998), Theorem 7.1] uses stable maps to blow-ups to express enumerative conditions. Our covered-tail calculation distinguishes that moduli object from a flat embedded Hilbert limit; it is not an enumerative formula.

The boundary results above start with V and its pencil. Their interpretation on the effective failure-algebra stack invokes the companion's sharp inverse and scalar-normalized envelope. That envelope is constructed on the recovered source, not contained in the original finite algebra. The companion gives its precise stack home, multi-Rees presentation, and relative lifting equations. The two core proofs are logically independent of these applications.

The appendices retain all-rank complete quadrics, the finite syzygy/contact recovery of arbitrary symmetric pencils, ordinary transverse collisions of arbitrary multiplicities, the earlier single-incidence theory, conductors, and reciprocal-fibre homology. The word ordinary in that collision theorem retains its squarefree residual and transverse hypotheses. The new

reduced-incidence open still excludes higher corank; the nonreduced result classifies the displayed slices, not every ambient higher-contact fibre. The accessible classical classification is [De Teran–Dmytryshyn–Dopico(2018), Theorem 2.1]. The comparison with Ballico’s 1993 failure-locus paper [1] remains limited by unavailable theorem/proof-level text; the separately available [Ballico(1996)] does not remove that limitation.

2. POWER COEFFICIENTS AND THEIR TWISTS

For a vector space V , put $M(V) = \text{Sym}^2 V$, with congruence action $A \mapsto gAg^\dagger$, and define

$$B_h(V) = \text{Sym}(M(V)) / \langle (v^2)^{h+1} : v \in V \rangle. \quad (2.1)$$

The degree of $M(V)$ is one. The generating space of the ideal is the Cartan inclusion $\text{Sym}^{2h+2} V \hookrightarrow \text{Sym}^{h+1}(\text{Sym}^2 V)$, normalized by $v^{2h+2} \mapsto (v^2)^{h+1}$. This definition requires neither a volume form nor a complementary space. It agrees with $B_{n,h}$ in Theorem 3.1 when $W = V^*$.

Proposition 2.1 (The coefficient map and its twists). *Let $n = \dim V$ and $N = nh$. The quotient multiplication μ_p gives an equivariant injection*

$$\mu_p^* : B_h(V)_p^* \hookrightarrow \text{Sym}^p(M(V)^*), \quad (\mu_p^* \varphi)(A) = \varphi(A^p). \quad (2.2)$$

Its image is the full coefficient space defining J_p , without a choice of apolar coordinates. The socle and perfect pairings are

$$B_h(V)_N = (\det V)^{2h}, \quad B_h(V)_p^* \simeq B_h(V)_{N-p} \otimes (\det V^*)^{2h}. \quad (2.3)$$

For a line L , $B_h(V \otimes L)_p = B_h(V)_p \otimes L^{2p}$, compatibly with all products. On $\mathbb{P}(M(V))$, the p th power is a section of $B_h(V)_p \otimes \mathcal{O}(p)$, and its base ideal is the image of $B_h(V)_p^ \otimes \mathcal{O}(-p) \rightarrow \mathcal{O}$.*

Proof. A homogeneous polynomial of degree p is identified with its symmetric p -linear polarization, normalized so that its value on A is its value on $(A, \dots, A)$. Dualizing the surjection μ_p gives exactly (2.2). Multinomial coefficients appearing in a monomial basis express this normalization and are nonzero; no dual representation is silently replaced by the original one. Theorem 3.1 identifies the quotient with determinant apolarity. The determinant of $A : V^* \rightarrow V$ takes values in $(\det V)^2$. Choosing a trivialization gives a scalar polynomial whose inverse-system line has character $(\det V^*)^{2h}$; its dual socle therefore has character $(\det V)^{2h}$. The Gorenstein pairing gives the second formula. Each generator in $M(V)$ has scalar weight two, proving the line-twist assertion directly from (2.1). The tautological line $\mathcal{O}(-1) \rightarrow M(V) \otimes \mathcal{O}$ has p th product $\mathcal{O}(-p) \rightarrow B_h(V)_p \otimes \mathcal{O}$, which proves the last assertion. $\square$

2.1. Graph schemes and classical inputs. The projective graph argument uses the following precise statement. It applies to the ambient space and also to the two-dimensional total base of the degeneration computed below.

Lemma 2.2 (Simultaneous graph with invertible twists). *Let X be an integral noetherian complex scheme and let $W_i \otimes L_i^{-1} \rightarrow \mathcal{O}_X$ be nonzero coherent systems of sections, with finite-dimensional W_i and invertible L_i . Write K_i for their image ideals, and let $U = X \setminus \bigcup_i V(K_i)$. The scheme-theoretic graph closure of $U \rightarrow \prod_i \mathbb{P}(W_i^*)$ is $\text{Bl}_{\prod_i K_i} X$ with its specified graph embedding. This statement identifies the scheme and the maps, not only their normalizations. Replacing the product ideal by an invertible twist or a positive power does not change its blow-up over X .*

Proof. The open U is dense because all K_i are nonzero and X is integral. After the Segre embedding the simultaneous map is given by all products of one section from each system. On a trivializing affine open, these products generate $K = \prod_i K_i$. The homogeneous coordinate algebra of the graph is the image of the polynomial algebra on these products in $\mathcal{O}_X[t]$, namely the Rees algebra $\bigoplus_{m \geq 0} K^m t^m$. It is a subalgebra of the function-field polynomial ring. Thus its Proj is

exactly the schematic closure of the graph, with no extra component. The Segre equations hold on the dense graph and hence on this integral closure scheme, so the graph lands in the product, not merely in its ambient projective space. Local trivializations glue. An invertible ideal changes the degree- m piece by its m th tensor power and does not change relative Proj; a positive power replaces the Rees algebra by a Veronese subalgebra. $\square$

For the power system, $L_i = \mathcal{O}(p)$ and $W_i = B_h(V)_p^*$; for the exterior system of order q , $L_i = \mathcal{O}(q)$ and its sections are the q -minors. The determinant system is invertible on the integral ambient space. It may be omitted from the blow-up product, although its zero divisor must still be retained in contact formulas. Likewise the first-coordinate identity map has unit base ideal. These observations specify the projective twists in the proof of Theorem 5.1.

The precise classical scheme-level complete-quadric construction is [33, Theorem 6.3]; the graph of exterior-power maps and its successive symmetric-rank blow-ups are stated in [34, Remark 2.5 and Construction 2.6]. The relative rank version is recalled in [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. Consequently smoothness, the flag description, and the orbit stratification in Corollary 5.2 are classical consequences of the graph identification, not new classifications.

There is also a precise invariant-ideal predecessor to the universal identity. Henriques–Varbaro [32, §2.4, Theorems 2.6–2.8] recall Abeasis’s classification for the symmetric coordinate ring, its products of determinantal ideals and their integral closures, and the passage between invariant generators and products indexed by diagrams. Our coefficient-space comparison uses this same multiplicity-free setting. Theorem 4.1 identifies a particular multiplication coefficient map with the balanced determinantal product as an ordinary ideal; it is not an independent classification of invariant ideals. The exact coefficient identification in Proposition 2.1 explains which map is being specialized on nonreduced bases. We assert no exhaustive historical nonanticipation of the displayed identity in all invariant-ideal terminology. Similarly the multiplier calculation in Corollary H.1 is the substitution of its exponents in the symmetric determinantal formula [32, Theorem 4.8], not an independent multiplier-ideal theorem.

3. QUADRATIC SECTIONS OF RECIPROCAL NORMALIZATION FIBRES

The reciprocal presentation has a distinguished linear coefficient section on which the entire algebra, rather than only its minimal relations, can be determined. Throughout this section a decomposition $E = \mathbb{C}a \oplus W$ is fixed. The section is equivariant for $\mathrm{GL}(W)$; it is not asserted to be independent of an arbitrary change of that decomposition.

Put $d = \dim W$, $g \geq k \geq 2$, and $h = \lfloor g/2 \rfloor$. In the algebra $F_{g,k}(W)$ of (Q.1), let $\mathfrak{l}$ be the ideal generated by the coefficients of all Q_i with $i \neq 2$. Write $Q = Q_2$ and

$$B_{d,h} = \mathrm{Sym}((\mathrm{Sym}^2 W)^*) / (\mathrm{coeff} Q^{h+1}). \quad (3.1)$$

Here the ordinary degree of a coefficient of Q is one; its reciprocal weight is two. These two gradings will always be specified separately. A partition $\lambda \subseteq (h^d)$ has at most d rows, each of length at most h .

Theorem 3.1 (The entire quadratic coefficient section). *There is an isomorphism of $\mathrm{GL}(W)$ -equivariant graded algebras*

$$F_{g,k}(W)/\mathfrak{l} \simeq B_{d,h}. \quad (3.2)$$

Under the coefficient/differential pairing, $B_{d,h}$ is the apolar algebra of the h th power of the determinant of a generic symmetric $d \times d$ matrix. Its ordinary-degree decomposition is

$$(B_{d,h})_p \simeq \bigoplus_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \mathbb{S}_{2\lambda}(W^*). \quad (3.3)$$

In particular it is Gorenstein of socle degree dh , with one-dimensional socle representation $(\det W^*)^{2h}$. Its Hilbert function and length are

$$H_{d,h}(p) = \sum_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \prod_{1 \leq i < j \leq d} \frac{2\lambda_i - 2\lambda_j + j - i}{j - i}, \quad (3.4)$$

$$L_{d,h} = \prod_{i=1}^d \frac{h + d - i + 1}{d - i + 1} \prod_{1 \leq i < j \leq d} \frac{2h + 2d - i - j + 2}{2d - i - j + 2}. \quad (3.5)$$

The Hilbert function is symmetric. A degree-one element is strong Lefschetz precisely when its associated symmetric matrix has full rank. More generally, an element associated with a rank- r matrix has nilpotence index $hr + 1$ (index one for the zero element). Consequently

$$\text{length } F_{g,k}(W) \geq L_{d,h}, \quad \mathfrak{m}^{dh} \neq 0 \text{ in } F_{g,k}(W). \quad (3.6)$$

The Gorenstein and Lefschetz assertions concern the section $B_{d,h}$, not the full fibre $F_{g,k}(W)$.

Proof. After setting the indicated coefficients to zero, the reciprocal series is $(1 + Qz^2)^{-1}$. The first even integer in $\{g + 1, \dots, g + k\}$ is $2(h + 1)$; it belongs to this interval because $k \geq 2$. Every odd coefficient vanishes, and the even ones are signed powers of Q . Thus the specialized ideal is exactly $(\text{coeff } Q^{h+1})$: all its higher powers have coefficients in this ideal. This proves (3.2) as an equality of ideals, not only an equality of radicals.

For clarity about the pairing, write $Q(x) = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$ and let X have independent symmetric entries X_{ij} , $i \leq j$. Send y_{ij} to $\partial/\partial X_{ij}$. Then $Q(x)$ acts by differentiation in the rank-one direction xx^t . Since $\det(X + zxx^t)^h$ has degree at most h in z , the coefficients of $Q(x)^{h+1}$ annihilate $(\det X)^h$. The reverse containment is the symmetric-matrix case of [30, Remark 11, pp. 8–9]: that apolar ideal is generated by the full general-linear orbit of the $(h + 1)$ st power of a rank-one differentiation direction. Powers of linear forms span their symmetric power, so this orbit span is precisely the coefficient space just displayed. No assertion that an arbitrary quotient of an apolar algebra is Gorenstein is used.

We record explicitly how the classical representation formulas translate here. In [30, Example 9 and Proposition 10], the surviving highest weights are indexed by $k_1, \dots, k_d \geq 0$ with $\sum k_i \leq h$, and their degrees are $\sum ik_i$. Set $\lambda_i = \sum_{j \geq i} k_j$. These are exactly the partitions in (h^d) , of size $\sum ik_i$, and their polynomial representations are $\mathbb{S}_{2\lambda}(W^*)$. This proves (3.3); the ordinary Weyl dimension formula gives (3.4). The same apolar module is the finite-dimensional highest-weight $\mathfrak{sp}_{2d}$ -module of weight $h\omega_d$, with its Levi character shifted by $\det^h$; see [30, Proposition 16 and Lemma 13]. For the positive roots $e_i - e_j$, $e_i + e_j$, and $2e_i$, its Weyl dimension factors are respectively 1, $(2h + 2d - i - j + 2)/(2d - i - j + 2)$, and $(h + d - i + 1)/(d - i + 1)$. Their product is (3.5). Apolarity to one homogeneous polynomial of degree dh gives the perfect pairings and the socle. Equivalently, the complementary partition $(h - \lambda_d, \dots, h - \lambda_1)$ gives the dual summand with the indicated determinant twist.

The strong Lefschetz assertion is the symmetric-matrix case of [30, Theorem 4 and Example 5]. There is a direct argument for the asserted nilpotence index. If $\ell_A = \sum_{i \leq j} A_{ij} y_{ij}$, then $\ell_A^p = 0$ in the apolar algebra exactly when $\partial_z^p \det(X + zA)^h = 0$ as a polynomial in X . Congruence reduces a rank- r symmetric matrix to $\text{diag}(1, \dots, 1, 0, \dots, 0)$. The last determinant power has degree exactly hr in z , as is seen by taking X diagonal with independent entries. This proves the index. Finally a surjective local algebra map surjects on every power of its maximal ideal. The section has nonzero degree dh , proving both assertions in (3.6). $\square$

Corollary 3.2 (Size and Loewy depth of the actual common boundary fibre). *In the notation of Theorem W.1, put $d = n^2 - 1$ and $h = n(n - 1)/2$. The entire normalization fibre at K_∞*

has length at least $L_{d,h}$ and its maximal ideal has nonzero dh th power. For the ternary pencil boundary,

$$\text{length } F_{6,2}(\mathbb{C}^8) \geq 922268360, \quad \mathfrak{m}^{24} \neq 0. \quad (3.7)$$

Its quadratic coefficient section has exactly that length, ordinary socle degree 24, and reciprocal-weight socle 48.

Proof. The common boundary fibre is $F_{g,k_n}(\mathbb{C}^{n^2-1})$, with $g = n(n-1) \geq k_n \geq 2$. Apply the theorem. For $n = 3$, substitute $(d, h) = (8, 3)$ into (3.5). This computes a whole closed section, not the length of the original fibre. The previously obtained 44 coefficient variables and 9867 minimal equations remain counts of different invariants. $\square$

4. UNIVERSAL POWER IDEALS

The power-zero construction admits an equality of ideals before any restriction to a pencil or to a valuation ring. This is the point at which the reciprocal section acquires a global boundary geometry. Let V have dimension $n \geq 2$, put $M = \text{Sym}^2 V$, and use the algebra $B_{n,h}$ of Theorem 3.1 with $W = V^*$. Thus $(B_{n,h})_1 = M$. If A is a symmetric matrix with entries in a commutative $\mathbb{C}$ -algebra T , denote its element of $M \otimes T$ by ℓ_A , and set

$$J_p(A) = \text{im}((B_{n,h})_p^* \otimes T \longrightarrow T, \quad \varphi \longmapsto \varphi(\ell_A^p)). \quad (4.1)$$

Write $I_q(A)$ for the ideal of all q -minors, with $I_0(A) = T$. In this section powers of ideals are ordinary powers unless parentheses explicitly indicate symbolic powers.

Theorem 4.1 (Universal determinantal formula). *For every commutative $\mathbb{C}$ -algebra T , every symmetric $n \times n$ matrix A over T , and $0 \leq p \leq nh$, write $p = hq + s$ with $0 \leq s < h$. Then*

$$J_p(A) = I_q(A)^{h-s} I_{q+1}(A)^s. \quad (4.2)$$

When $s = 0$ the second factor is omitted. For $p > nh$, $J_p(A) = 0$. These are equalities of ideals, not of radicals or integral closures, and commute with every homomorphism of $\mathbb{C}$ -algebras. In particular, for every symmetric map $A : T^n \rightarrow T^n$,

$$J_{h(n-j)}(A) = \text{Fitt}_j(\text{coker } A)^h, \quad 0 \leq j \leq n. \quad (4.3)$$

The formulas also hold locally, and hence as ideal sheaves, for symmetric maps of vector bundles with a line-bundle twist on an arbitrary $\mathbb{C}$ -scheme.

We separate the classical determinantal input. For an arbitrary matrix X over a characteristic-zero polynomial ring, the balancing inclusion

$$I_u(X)I_v(X) \subseteq I_{u+1}(X)I_{v-1}(X), \quad 0 \leq u \leq v-2, \quad (4.4)$$

is the determinantal symmetrization lemma. See [31, Lemma 2.5 and Theorem 2.4], which attributes the characteristic-zero product theorem to De Concini–Eisenbud–Procesi. Because this is a polynomial ideal inclusion, it remains true after specialization to a symmetric matrix; no flatness of that specialization is needed. This classical lemma is not a claim of the present paper.

Lemma 4.2 (The bounded-row generating space). *Let $S = \text{Sym}(\text{Sym}^2 V^*)$ and let A be its universal symmetric matrix. For $p = hq + s$, the degree- p part generating $I_q(A)^{h-s} I_{q+1}(A)^s$ is*

$$\bigoplus_{\substack{|\lambda|=p, \ell(\lambda) \leq n \\ \lambda_1 \leq h}} \mathbb{S}_{2\lambda}(V^*) \subseteq S_p. \quad (4.5)$$

Proof. The symmetric Cauchy decomposition of S_p is multiplicity free, with summands $\mathbb{S}_{2\lambda}(V^*)$, $|\lambda| = p$; see [32, §2.4]. We may use the dual general linear group to regard these as polynomial representations. Every minor has torus weight at most two in any one coordinate: its row set and column set each use that coordinate at most once. A product of h minors therefore has no weight with a coordinate exceeding $2h$. Its invariant span cannot contain a highest weight 2λ with $\lambda_1 > h$. This proves one inclusion.

For the converse, let $c_1 \geq \dots \geq c_h \geq 0$ be the column lengths of a partition λ occurring on the right, padded with zeros. The product of the leading principal minors of orders c_i is a nonzero highest-weight vector of weight 2λ in S_p . It belongs initially to $\prod_{i=1}^h I_{c_i}(A)$. Whenever two column lengths differ by at least two, apply (4.4) to increase the smaller length by one and decrease the larger by one. The sum of the squares of the lengths strictly decreases, so the procedure terminates. At termination the lengths are $q+1$ repeated s times and q repeated $h-s$ times, since their sum is p . Thus that highest-weight vector belongs to $I_q^{h-s} I_{q+1}^s$. The latter ideal is invariant; its degree- p part contains the full irreducible summand. Multiplicity freeness proves the equality. The case $q = 0$ includes zero-length minors, which are the constant 1. $\square$

Proof of Theorem 4.1. First work over S with its universal matrix. The coefficient map of ℓ_A^p is the dual of the multiplication quotient $\text{Sym}^p M \rightarrow (B_{n,h})_p$, up to nonzero multinomial constants. Consequently its degree- p coefficient span is exactly $(B_{n,h})_p^* \subseteq S_p$. The character formula (3.3) identifies that span with (4.5). Both ideals in (4.2) are generated in degree p , so the lemma proves their equality in S . This step uses the full coefficient space, not just evaluation on diagonal matrices or closed points.

For a matrix over T , substitute its entries in this polynomial identity. Taking the image of the fixed finite-dimensional coefficient map commutes with scalar extension, as does taking minors and products of ideals. This proves the assertion even when T is nonreduced, non-Noetherian, or not a domain. The Fitting formula follows from its minor presentation. Changing a vector-bundle frame applies an invertible congruence to A and an invertible linear transformation to each coefficient space; changing a local frame of the twisting line multiplies generators by a unit. Therefore the local ideal identities glue. Finally, $(B_{n,h})_p = 0$ for $p > nh$. $\square$

Example 4.3 (A nonreduced test base). Take $T = \mathbb{C}[\epsilon]/(\epsilon^3)$, $n = h = 2$, and $A = \text{diag}(\epsilon, 0)$. Then $J_1(A) = (\epsilon)$ and $J_2(A) = (\epsilon^2) \neq 0$, whereas $J_3(A) = J_4(A) = 0$. Every closed-point matrix is zero. Thus equality on geometric fibres would not establish the theorem. Nor does (4.3) license taking unique ideal roots: for example, in $\mathbb{C}[\epsilon]/(\epsilon^2)$ the distinct ideals (ϵ) and 0 have the same square. At $h = 1$ the power ideals are the Fitting ideals themselves over every base.

5. THE RECIPROCAL POWER GRAPH AND COMPLETE QUADRICS

Let $P = \mathbb{P}(M)$, let P^{nd} be the nonsingular-quadric open, and define $\mathcal{G}_h$ as the scheme-theoretic closure of the graph

$$P^{\text{nd}} \longrightarrow P \times \prod_{p=1}^{nh-1} \mathbb{P}((B_{n,h})_p), \quad [A] \longmapsto ([A], [\ell_A], \dots, [\ell_A^{nh-1}]). \quad (5.1)$$

The first factor records the base point. We use the reduced, integral graph closure, equivalently its scheme-theoretic image; no additional normalization is inserted in this definition. Let $\text{CQ}(V)$ be the classical space of complete quadrics: the graph closure of $[A] \mapsto ([\wedge^2 A], \dots, [\wedge^{n-1} A])$ over P . For $n = 2$ the empty product gives $\text{CQ}(V) = P$.

Theorem 5.1 (A compactification determined by multiplication). *For every $h \geq 1$ there is a canonical $\text{GL}(V)$ -equivariant isomorphism over P*

$$\mathcal{G}_h \simeq \text{CQ}(V). \quad (5.2)$$

In particular the graph is smooth and projective, is independent of h as a scheme over P , and has a simple-normal-crossing boundary. Write $\pi : \text{CQ}(V) \rightarrow P$, and let E_r be its boundary divisor of rank break r , $1 \leq r < n$; for $r < n - 1$ it is exceptional over rank at most r , and E_{n-1} is the strict transform of the determinant divisor. Then every power ideal has the exact total transform

$$J_p \mathcal{O}_{\text{CQ}(V)} = \mathcal{O}_{\text{CQ}(V)} \left(- \sum_{r=1}^{n-1} (p - hr)_+ E_r \right), \quad 0 \leq p \leq nh, \quad (5.3)$$

where $(a)_+ = \max(a, 0)$. Removing this common divisor from the universal p th power gives a nowhere-zero projective section on the whole compactification.

The smoothness, boundary and flag interpretation of complete quadrics are classical, due to the complete-form construction; see [33, 34]. The assertion here identifies that particular compactification, with all of its scheme structure, from the multiplication in one reciprocal normalization-fibre section. It does not assert that complete quadrics is a newly constructed variety.

Proof. We first recall the graph calculation with its scheme structure. If rational maps from an integral scheme are given by generators of nonzero ideals $K_1, \dots, K_a$, their simultaneous graph closure is $\text{Bl}_{K_1 \dots K_a}$ of the base. Indeed, after the Segre embedding the coordinates are all products of one generator from each ideal. The closure of this graph is the Proj of the Rees algebra of that product: on a generator chart its coordinate ring is the subring of the function field obtained by adjoining the generator ratios. These are exactly the affine charts of both constructions. Tensoring an ideal by an invertible ideal, or replacing it by a positive power, does not change the blow-up, the latter by the Veronese description of Proj.

Apply this calculation on P . The sheaf $I_1 \mathcal{O}_P$ is the unit ideal, and $I_n \mathcal{O}_P$ is invertible. For each $2 \leq q \leq n - 1$, the exponent of I_q in $\prod_{p=1}^{nh-1} J_p$ is

$$\sum_{s=0}^{h-1} (h - s) + \sum_{s=1}^{h-1} s = h^2.$$

Theorem 4.1 therefore gives, up to an invertible ideal,

$$\prod_{p=1}^{nh-1} J_p = \left(\prod_{q=2}^{n-1} I_q \right)^{h^2}.$$

The same graph calculation for the exterior-power maps identifies $\text{CQ}(V)$ with the blow-up of $\prod_{q=2}^{n-1} I_q$. This proves (5.2) as an isomorphism of schemes, not only after normalization. Both isomorphisms are the identity on the common dense open, hence are canonical and equivariant.

For completeness, the local calculation behind the boundary formula is as follows. The complete-quadric construction blows up the strict rank loci successively. On its Gaussian charts, after a congruence and projective normalization, its matrix has the form

$$U \text{diag}(1, \tau_1, \tau_1 \tau_2, \dots, \tau_1 \cdots \tau_{n-1}) U^t, \quad (5.4)$$

with U lower unitriangular and $E_r = (\tau_r = 0)$. One obtains these charts inductively by making the first pivot one, taking a Schur complement, blowing up the zero residual matrix, and repeating. At each step the next nonzero residual quadratic form has a nonisotropic vector, so congruence translates of these charts cover the construction. This also explains the smooth charts and the normal crossings stated in the classical construction.

Cauchy–Binet, applied to (5.4), shows that every q -minor is divisible by $\tau_1^{q-1}\tau_2^{q-2}\cdots\tau_{q-1}$. The leading principal q -minor is exactly that product. Thus, on these charts and therefore globally,

$$I_q \mathcal{O}_{\text{CQ}(V)} = \mathcal{O} \left(- \sum_{r < q} (q-r) E_r \right). \quad (5.5)$$

Using (4.2), its exponent along E_r is $(h-s)(q-r)_+ + s(q+1-r)_+ = (p-hr)_+$. This proves (5.3). The coefficient sections generate the transformed ideal; division by its local generator makes them generate the unit ideal, proving the last assertion. $\square$

Corollary 5.2 (Native boundary strata). *The power graph has one congruence orbit for every subset $I \subseteq \{1, \dots, n-1\}$. It is the stratum $E_I^\circ = (\bigcap_{r \in I} E_r) \setminus \bigcup_{r \notin I} E_r$, of codimension $|I|$, and its closure is $\bigcap_{r \in I} E_r$. Its points are flags with nondegenerate projective quadratic forms on successive quotients; the quotient dimensions are the successive gaps in $\{0\} \cup I \cup \{n\}$. The simultaneous normalized power values recover this complete-quadric datum. They are not the one common limit for the full cotangent-coordinate group.*

Proof. The flag description and boundary stratification are the classical complete-quadric description. The general linear group acts transitively on flags of fixed dimensions, and over $\mathbb{C}$ every nondegenerate quadratic form on a fixed-dimensional quotient is congruent to the identity. Independent diagonal scalings remove the projective scalars. The chart (5.4) realizes each subset of vanishing τ_r and permits further coordinates to vanish, giving the stated codimensions and closures. The graph is a closed subscheme of the product in (5.1); its projections therefore retain the point, including its flag and quotient forms. The group here is the native congruence group of V . $\square$

Remark 5.3 (Relative construction and base change). For a vector bundle $\mathcal{V}$ on any $\mathbb{C}$ -scheme S , the associated bundle with fibre $\text{CQ}(\mathbb{C}^n)$ is a relative smooth projective scheme over S . Equivariance glues the power maps and the boundary formula. Formation of this associated bundle and the polynomial ideal identity commute with arbitrary base change. This does not say that taking the graph closure of an arbitrary restricted family commutes with nonflat base change. The relative ambient resolution and the closure of an individual family's graph are different constructions.

6. MULTIPLE REDUCED INCIDENCE AND ITS COMPLETE HILBERT BOUNDARY

We remove the restriction to a single incidence point. The independence of the incidence conditions is proved from the radicals of distinct members of a regular pencil; it is not added as a transversality hypothesis. The resulting modification has a global Fitting-ideal description even though the incidence points need not be globally labelled.

Put $M = \text{Sym}^2 V$, $P = \mathbb{P}(M)$ and $G = \text{Gr}(2, M)$, with $n = \dim V \geq 3$. Let $Z_j \subset P$ have its symmetric determinantal scheme structure. When $n = 3$ put $Z_0 = \emptyset$. Let G_{reg} be the open of regular pencils and let $X_G = \mathbb{P}(\mathcal{R})$ be the universal line. The image of $X_G \times_P Z_{n-3}$ is closed, since the projection from this incidence is proper. Remove it and call the resulting open $G_* \subset G_{\text{reg}}$. Every line over G_* meets Z_{n-2} in a finite scheme: a positive-dimensional intersection would contain the line and contradict regularity. Consequently

$$q_*^{\text{inc}} : D_* := X_{G_*} \times_P Z_{n-2} \longrightarrow G_*$$

is proper and quasi-finite, hence finite. Define

$$U_{\text{red}} = G_* \setminus q_*^{\text{inc}}(\text{Supp } \Omega_{D_*/G_*}), \quad D = D_* \times_{G_*} U_{\text{red}}, \quad q : D \rightarrow U_{\text{red}}. \quad (6.1)$$

This is open because q_*^{inc} is finite. Its geometric incidences are precisely the finite reduced intersections: for a finite algebra over $\mathbb{C}$, vanishing of relative differentials is equivalent to being a

product of copies of $\mathbb{C}$. Unlike the single-incidence open, it allows every possible number of such points. Set

$$\mathcal{Q} = q_* \mathcal{O}_D, \quad \mathcal{I} = \text{Fitt}_0(\mathcal{Q}), \quad B_{\text{red}} = \text{Bl}_{\mathcal{I}} U_{\text{red}}. \quad (6.2)$$

The zeroth Fitting ideal of the zero module is the unit ideal.

6.1. The general incidence calculation.

Lemma 6.1 (Transverse incidence of a family of curves). *Let S, Y be smooth complex varieties, let $Z \subset Y$ be smooth of codimension $c \geq 2$, and let $X \rightarrow S$ be a smooth family of embedded curves in Y . Suppose $D = X \times_Y Z \rightarrow S$ is finite and unramified. Fix $s \in S$ with incidence points $p_1, \dots, p_k$. Write $a_i \subset N_{Z/Y, p_i}$ for the normal image of $T_{p_i} X_s$. Assume $a_i \neq 0$ and that the joint normal evaluation map*

$$T_s S \longrightarrow \bigoplus_{i=1}^k N_{Z/Y, p_i} / a_i$$

is surjective. Etale locally at s , the incidence splits into closed immersions $D_i \simeq \Sigma_i \hookrightarrow S$, with independent smooth centres of codimension $c - 1$. Near p_i there are coordinates in which

$$I_D = (x_i, u_{i1}, \dots, u_{i,c-1}), \quad I_{\Sigma_i} = (u_{i1}, \dots, u_{i,c-1}), \quad (6.3)$$

where all base parameters u_{ij} are independent. If $\mathcal{Q}_S = q_ \mathcal{O}_D$, then*

$$\text{Fitt}_0(\mathcal{Q}_S) = \prod_i I_{\Sigma_i} = \bigcap_i I_{\Sigma_i}.$$

The blow-up of this ideal is the fibre product of the individual blow-ups. It is smooth, with etale normal-crossing exceptional boundary and fibre $\prod_i \mathbb{P}^{c-2}$ at s .

Proof. Over the strict henselization at s , the finite algebra splits into factors indexed by its geometric fibre. These factors and their idempotents descend to an etale neighbourhood. For each factor the unit map from the base ring has cokernel whose fibre at s is zero: finite base change identifies that fibre with the corresponding one-dimensional residue algebra. Nakayama's lemma, followed by shrinking, makes the unit map surjective. Each factor is therefore a closed immersion. This argument uses the finite algebra, not just the set of its geometric points.

The map of the curve tangent into the normal space is nonzero, so one local equation of Z has nonzero relative derivative. Its zero set is etale over the base. After an etale base change choose this as a section and use that equation as the relative coordinate x_i . Modulo x_i , the other equations become base functions u_{ij} . Their differentials give exactly the joint normal evaluation in the statement. Surjectivity makes them independent regular parameters, proving the normal form and smoothness of all intersections of the centres. In particular the length-one factors in this argument are reduced on complex geometric fibres; reducedness has not been inferred from an arbitrary higher fibre length.

Now $\mathcal{Q}_S = \bigoplus_i \mathcal{O}_S / I_{\Sigma_i}$. Fitting ideals of a direct sum multiply. In the displayed independent coordinates, intersection equals product: in a polynomial coordinate ring a monomial in every ideal uses a variable from each disjoint group, and the equality persists under etale flat extension. Equivalently one may verify it in the faithfully flat completion. The simultaneous blow-up is covered by charts in which, for each i , one u_{ij} is retained and the others are its multiples by new ratios. These charts are smooth and identify the fibre product with the blow-up of the product ideal. The latter equality also follows from the simultaneous Rees graph lemma. There are k independent exceptional parameters, and their zero sets have normal crossings. The fibre consists of independent choices of a line in each normal space of dimension $c - 1$, proving the last assertion. $\square$

The arrangement blow-up mechanism is classical. Li's wonderful-model Theorems 1.2 and 1.3 [Li(2009)] give the normal-crossing and product-ideal description for a building set. Here we have proved the independent-centre case explicitly. The pencil-specific assertion needed to apply it at every point of U_{red} is the following.

Lemma 6.2 (Independence at distinct spectral points). *For every $R \in U_{\text{red}}$, with incidence points $p_1, \dots, p_k$, the joint normal evaluation of Lemma 6.1 is surjective. In particular the local incidence centres have codimension two and meet transversely, without any additional genericity assumption on the other elementary divisors of R .*

Proof. Choose a nonsingular member Q_0 of the pencil and write its other members as $Q_1 - \lambda Q_0$. Put $T = Q_0^{-1}Q_1$ on V^* . The radicals $K_i = \ker(T - \lambda_i)$ at distinct spectral points lie in distinct generalized eigenspaces. Their sum is direct. Restriction therefore gives a surjection

$$\text{Sym}^2 V \longrightarrow \bigoplus_i \text{Sym}^2(K_i^*) :$$

extend a basis of the direct sum of the K_i to a basis of V^* and prescribe the indicated diagonal blocks of a symmetric form. Varying Q_1 with Q_0 fixed realizes every one of these restrictions. At a corank-two point the normal space to the rank locus is the three-dimensional space of forms on K_i , with its harmless local projective twist. The tangent of the pencil maps to the nonzero line a_i , because the incidence is reduced. Quotienting by a_i gives the joint evaluation in the statement. A change of basis of the pencil contributes only multiples of its tangent image in each summand, so the map factors through $T_R G = \text{Hom}(R, M/R)$. It remains surjective after this quotient. This proves the claim also when the tangent form on K_i has rank one rather than rank two. $\square$

6.2. Identification of the full Hilbert graph. Let $G^\circ \subset U_{\text{red}}$ be the open with no corank-two incidence. A pencil over it has a unique lifted embedded curve in $Q = \text{CQ}(V)$. Use the polarization $H = \bigotimes_{q=1}^{n-1} H_q$, where H_q is the resolved exterior hyperplane bundle, and put $P_H(l) = \binom{n}{2}l + 1$. Let Γ_{red} be the reduced closure of this graph in $U_{\text{red}} \times \text{Hilb}^{P_H}(Q)$, and let $\widehat{\Gamma}_{\text{red}}$ be its normalization. Thus $\widehat{\Gamma}_{\text{red}} = \widehat{G} \times_G U_{\text{red}}$ with the conventions of Proposition D.1. This equality uses localization of integral closure [Stacks Project, Section 29.55, Lemmas 29.55.3–4]; it is an open restriction, not an arbitrary base-change assertion.

Theorem 6.3 (The complete multiple-incidence modification). *There is a canonical congruence-equivariant isomorphism*

$$\widehat{\Gamma}_{\text{red}} \simeq B_{\text{red}} = \text{Bl}_{\text{Fitt}_0(q^* \mathcal{O}_D)} U_{\text{red}}. \quad (6.4)$$

The space is smooth, and its exceptional boundary has normal crossings etale locally. Over a pencil with k distinct reduced corank-two incidences the whole scheme fibre is $(\mathbb{P}^1)^k$. A point of that fibre specifies independently, at each p_i , a line through $[a_i]$ in

$$\pi^{-1}(p_i) = \mathbb{P}(N_{Z_{n-2}/P, p_i}) \simeq \mathbb{P}^2.$$

Its Hilbert curve is the strict transform C_R of the pencil line, with these k embedded tails attached at their respective points. It is reduced and nodal, with no embedded components. For each tail F_i and each $1 \leq q < n$,

$$\deg_{F_i} H_q = \mathbf{1}_{q=n-1}, \quad \deg_{C_R} H_q = q - k \mathbf{1}_{q=n-1}. \quad (6.5)$$

The incidence scheme satisfies $k \leq \lfloor n/2 \rfloor$.

On a labelled etale neighbourhood the exceptional divisors E_i meet like coordinate hyperplanes; every subset occurs locally, and

$$K_{B_{\text{red}}} = \beta^* K_{U_{\text{red}}} + \sum_i E_i.$$

No global labelling, or global simplicity of each irreducible boundary component, is asserted. The universal curve is a flat nodal family; near its i th node its equation is $x_i y_i = t_i$, with $t_i = 0$ the corresponding exceptional divisor. The base blow-up has its usual effective-Cartier image-ideal universal property. These statements classify all fibres over U_{red} , not only the fibres of selected smoothing arcs.

Proof. Apply the preceding two lemmas with $c = 3$. Locally write $I_i = (u_i, v_i)$. In the simultaneous blow-up chart put $u_i = t_i$, $v_i = t_i w_i$. Near the i th incidence the image of the ambient rank-locus ideal on the pulled-back universal line is (x_i, t_i) . Its blow-up has charts

$$\mathcal{O}_B[x_i, y_i]/(x_i y_i - t_i), \quad \mathcal{O}_B[z_i], \quad x_i = t_i z_i.$$

The incidence points have disjoint neighbourhoods on the curve, so these modifications do not interfere. Outside them the ideal is the unit ideal. The local hypersurfaces are flat over B : they are relative Cartier divisors in a smooth relative plane and their nonzero defining equations remain nonzerodivisors on every fibre. The fibrewise Cartier flatness criterion applies. These are the standard nodal local models, so all the stated fibre properties follow. Their total spaces are smooth in the displayed independent coordinates.

Here is the scheme-level embedding needed for the Hilbert argument. If $\mathcal{K}$ is the rank-locus ideal on the open ambient space and $f : X_B \rightarrow P$ is evaluation, there is a surjection of graded algebras

$$f^* \left(\bigoplus_{d \geq 0} \mathcal{K}^d \right) \rightarrow \bigoplus_{d \geq 0} (\mathcal{K} \mathcal{O}_{X_B})^d. \quad (6.6)$$

Its kernel removes precisely the relations annihilated on passing to the image ideal. The target is the image Rees algebra inside $\mathcal{O}_{X_B}[T]$; it is not an assumed tensor-product identification. Taking Proj gives a closed immersion into the base-changed ambient blow-up, which is Q over rank at least $n - 2$. The universal line is already embedded in $P \times B$. We obtain a flat closed family of curves in $Q \times B$. This agrees with the lifted pencil on the dense nonincident open.

In normal coordinates its exceptional map at p_i is $[x_i : t_i] \mapsto [x_i a_i + t_i b_i]$. The vectors a_i, b_i are independent, and their span determines an embedded line, whose homogeneous ideal is its linear annihilator. Distinct directions modulo a_i therefore give distinct closed subschemes of the exceptional plane, not merely different parametrizations of one curve. Tails at different p_i cannot be confused. The induced proper map $B \rightarrow \Gamma_{\text{red}}$ has finite geometric fibres and is birational. It is consequently finite, and the smooth source is the normalization of the reduced integral graph. This proves (6.4), including its scheme fibres. Both maps restrict to the identity on the dense graph, giving canonicity and equivariance and gluing the etale calculations.

The smaller minor systems have no base point, while the last minor system has one simple base point at each p_i . Subtracting these base points and adding one unit of degree on each tail gives (6.5). The determinant has vanishing order at least two at each distinct corank-two point, so $2k \leq n$. The nodal tree has Euler characteristic one and total H -degree $\binom{n}{2}$. This proves the Hilbert polynomial used above. Finally the product blow-up charts give independent parameters t_i ; the Jacobian for each codimension-two blow-up contributes one E_i to the canonical divisor. These computations also prove the local intersection statement. $\square$

Corollary 6.4 (Multiplication systems at every node). *The ideal $\mathcal{I}$ in (6.2) is also obtained by taking the zeroth Fitting ideal of the pushforward of the zero scheme of $J_{h(n-2)+1}$ on the universal line, for every $h \geq 1$. At $h = 1$ this is the distinguished construction without an exponent choice. On every tail the resolved j th power is the complete system of degree*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+.$$

The exceptional parameter space at a k -incidence pencil has Chow ring $\mathbb{Z}[\xi_1, \dots, \xi_k]/(\xi_1^2, \dots, \xi_k^2)$. In particular at a double incidence the full fibre is a smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$, not merely two independent one-parameter examples.

Proof. On this rank open the universal formula gives $J_{h(n-2)+1} = I_{n-1}$ because $I_{n-2} = \mathcal{O}$. Its zero scheme is exactly D . Finite pushforward and Fitting ideals give the specified centre. The power assertion is Proposition E.3 at each disjoint tail. For clarity, a coefficient ideal generated in degree b has degree- b part equal to the linear span of its actual generators. In the residual binary ring this part is $f^{(b-h)+} \text{Sym}^{b-2(b-h)+}(\text{Sym}^2 W^*)$; division by the common factor therefore leaves all forms, not merely a subspace with the same base locus. The Chow ring is the projective bundle formula applied to the identified scheme fibre $(\mathbb{P}^1)^k$. $\square$

7. HIGHER CONTACT, THICK TAILS, AND SINGULAR MULTIPLICATION GRAPHS

Nonreduced incidence has a different Hilbert boundary from reduced multiple incidence. We give a two-parameter family of genuine symmetric pencils for every contact order $m \geq 2$, determine its entire normalized Hilbert graph, and compute the nonreduced fibres and the singularities of the universal curve. The result is a slice theorem, not an assertion that the whole ambient Hilbert modification commutes with restriction to that slice.

7.1. Symmetric linear pencils of every contact order. Let E_m be the reversal matrix, N_m the nilpotent matrix with ones on its superdiagonal, and put

$$L_m(x) = E_m(xI_m - N_m), \quad e = e_1.$$

This is symmetric. In dimension $n = 2m$, consider the affine family of pencils, with x the pencil parameter and $(u, v) \in S = \mathbb{A}^2$,

$$A_{u,v}(x) = \begin{pmatrix} L_m(x) + uee^t & vee^t \\ vee^t & L_m(x) - uee^t \end{pmatrix}. \quad (7.1)$$

The projective pencil is obtained by homogenizing every entry to degree one. Its leading matrix is $E_m \oplus E_m$, so infinity is nonsingular. For $n > 2m$ one may adjoin fixed scalar blocks $x - \lambda_j$ with distinct nonzero λ_j and restrict the parameter base to a neighbourhood where these additional simple roots stay disjoint from the cluster. All assertions below then hold near the same cluster; we state the global slice formula first for $n = 2m$.

Lemma 7.1 (Exact contact slice). *For the family (7.1), the complement of the two e_1 directions is a nonsingular block over $\mathbb{C}[x]$, of constant nonzero determinant. Its Schur complement is exactly*

$$\begin{pmatrix} x^m + u & v \\ v & x^m - u \end{pmatrix}. \quad (7.2)$$

Consequently, on the affine pencil chart,

$$I_{n-1}(A) = (x^m, u, v), \quad \det A = c_m(x^{2m} - u^2 - v^2), \quad c_m \in \mathbb{C}^*.$$

All pencils are regular and have rank at least $n - 2$ everywhere. Only the origin of S has a corank-two incidence; its intersection scheme has length m . Its two positive local Smith exponents are m, m . The finite incidence algebra pushed to S is $(\mathcal{O}_S/(u, v))^{\oplus m}$, so its zeroth Fitting ideal is $(u, v)^m$.

Proof. Delete row and column one from $L_m(x)$. The remaining matrix is anti-triangular with -1 on its antidiagonal, hence has constant nonzero determinant. Since $\det L_m(x) = \det(E_m)x^m$, its scalar Schur complement is x^m ; the ratio of the two constant determinants is one. This can also be read directly by eliminating along the superdiagonal. The two kernel directions receive exactly the displayed perturbation, proving (7.2). The unit-block minor identity then gives

$I_{n-1} = I_1(\text{Schur complement}) = (x^m, u, v)$ in characteristic zero. All smaller required minors are units. The determinant and rank claims follow from the same block decomposition, including infinity. At $u = v = 0$ the residual matrix is $x^m I_2$, which proves both the intersection length and the Smith exponents. The incidence algebra is $\mathbb{C}[x, u, v]/(x^m, u, v)$; its $\mathbb{C}[u, v]$ -module basis is $1, x, \dots, x^{m-1}$ with (u, v) annihilating every basis vector. The direct-sum formula for Fitting ideals proves the last assertion. $\square$

Let Γ_m be the reduced closure of the lifted-pencil graph over $S \setminus \{0\}$ in $S \times \text{Hilb}^{P_H}(\text{CQ}(V))$, with $P_H(l) = \binom{n}{2}l + 1$, and let $\hat{\Gamma}_m$ be its normalization. This definition is independent of whether normalization of the global pencil graph commutes with this particular slice; no such commutation is assumed.

Theorem 7.2 (The full contact-slice Hilbert modification). *For every $m \geq 2$ the slice graph is*

$$\hat{\Gamma}_m \simeq \text{Bl}_0 \mathbb{A}^2 = \text{Bl}_{\text{Fitt}_0(q_* \mathcal{O}_D)} S. \quad (7.3)$$

Its whole exceptional fibre is $\mathbb{P}^1$; the Hilbert curves over that exceptional fibre are nonreduced. In a base chart $u = t$, $v = tw$, the total multiplication graph is the blow-up of (x^m, t) on the pulled-back pencil surface. With $R = \mathbb{C}[x, t, w]$, its Rees presentation and its two projective charts are

$$\begin{aligned} \mathcal{R}(x^m, t) &= R[U, V]/(tU - x^m V), \\ D_+(U) : \quad t &= x^m y, \quad D_+(V) : \quad x^m = tz. \end{aligned} \quad (7.4)$$

Here U represents $x^m T$ and V represents tT in $R[T]$. The charts glue by $y = z^{-1}$. The universal curve is flat over the base blow-up, is normal, and has a transverse A_{m-1} surface singularity along the section $x = t = z = 0$ of its exceptional divisor.

For every direction $[\alpha : \beta] \in \mathbb{P}^1$, its special Hilbert curve is

$$C_0 \cup T_m, \quad T_m \simeq \mathbb{P}_{\mathbb{C}[x]/(x^m)}^1, \quad C_0 \cap T_m \simeq \text{Spec } \mathbb{C}[x]/(x^m). \quad (7.5)$$

The intersection is at one section of T_m ; the support of T_m is the residual line through $[I_2]$ with second member $\begin{pmatrix} \alpha & \beta \\ \beta & -\alpha \end{pmatrix}$ in the exceptional plane. The fibre has no embedded associated points. Its cycle is $[C_0] + m[F]$, where $F = (T_m)_{\text{red}}$. Every direction occurs, and these embedded thickened curves are pairwise distinct. The multidegrees are

$$\deg_{C_0} H_q = q - m \mathbf{1}_{q=n-1}, \quad \deg_F H_q = \mathbf{1}_{q=n-1}.$$

For $\delta_j = (j - h(n-2))_+ - 2(j - h(n-1))_+$, the j th resolved multiplication system on T_m is, over $\mathbb{C}[x]/(x^m)$, the complete system $\mathcal{O}_{T_m}(\delta_j)$. In particular its reduced degree is δ_j and its degree as a cycle is $m\delta_j$.

Proof. All rank loci below corank two are avoided, so the relevant ambient complete-quadric map is the blow-up of the rank- $(n-2)$ locus. Pull the exact ideal in Lemma 7.1 to $\text{Bl}_0 S$. In the stated chart it is (x^m, t) . These two elements are a regular sequence. The only homogeneous Rees relation is $tU - x^m V$: successive comparison of powers of T , using that x^m and t are relatively prime, expresses every relation as a multiple of this one. Thus the relation is saturated in the image subalgebra of $R[T]$, and the stated Proj charts follow. The other base chart is identical with the roles of u, v exchanged.

Both chart equations remain nonzerodivisors on every fibre of their smooth relative affine planes, so the fibrewise Cartier flatness criterion proves flatness. The first chart eliminates t and is smooth. The second is a hypersurface with singular locus $x = t = z = 0$, of codimension two in its total space. It is a domain and Cohen–Macaulay, so regularity in codimension one proves normality. Its transverse completed equation is $x^m = tz$, the A_{m-1} surface singularity. This describes a singular universal curve over a smooth parameter blow-up; the two smoothness assertions are not conflated.

At $t = 0$ the first chart has ring $\mathbb{C}[x, y]/(x^m y)$ and the second has ring $\mathbb{C}[x, z]/(x^m)$. In the first ring the ideals (y) and (x^m) intersect in $(x^m y)$. These pieces glue to the strict transform C_0 and to the thick tail whose two rings are $(\mathbb{C}[x]/(x^m))[y]$ and $(\mathbb{C}[x]/(x^m))[z]$, with $y = z^{-1}$ and x unchanged. This proves (7.5), including its length- m intersection. The local hypersurfaces of the fibre are Cohen–Macaulay of dimension one and have only their component primes as associated primes. There are no embedded points.

The Rees quotient (6.6) embeds this flat family into the fixed complete-quadric space times the base blow-up. Its generic curve is the prescribed lift. The support of each tail is the indicated residual line. Distinct $[\alpha : \beta]$ give distinct residual lines, hence distinct embedded closed subschemes. The resulting proper map to Γ_m is quasi-finite and birational, so the smooth base blow-up is its normalization. This proves the whole fibre statement in (7.3). The Fitting formula follows from $(u, v)^m$ and invariance of Proj under a positive Veronese power.

The last minor system loses a base divisor of length m on C_0 , while the smaller systems lose none. The graph line bundle restricts to $\mathcal{O}(1)$ on the residual projective line over $\mathbb{C}[x]/(x^m)$. It follows that $\chi(\mathcal{O}_{T_m} \otimes H^l) = m(l + 1)$ and the intersection has Euler characteristic m . Adding to the main component gives $\binom{n}{2}l + 1$, as required. More generally the unit-block congruence is invertible over $\mathbb{C}[x]$ and thus over $\mathbb{C}[x]/(x^m)$. The binary multiplication coefficient calculation of Corollary 6.4, extended to this Artin ring, spans every degree- δ_j binary form. Multiplication by the unit-block socle is a split injection on these coefficient modules: it has a unit coordinate in a trivializing frame. This proves the relative complete-system assertion, not just its reduced-fibre version. The degree- j coordinates originally have the twist $\mathcal{O}(j)$ on the pencil line; it is trivial on this local thickening. Removing their Cartier common factors produces the stated graph line bundle. $\square$

7.2. The singularity and the stable-map comparison.

Proposition 7.3 (Local deformation module of the multiplication graph). *At a singular point of a one-parameter direction in Theorem 7.2, let $A_m = \mathbb{C}[[x, t, z]]/(x^m - tz)$. The abstract local cotangent deformation modules are*

$$T^1(A_m) = \mathbb{C}[[x]]/(x^{m-1}), \quad T^i(A_m) = 0 \quad (i \geq 2). \quad (7.6)$$

The first equality has dimension $m - 1$. A formally versal family of hypersurface deformations is obtained by adding $\sum_{a=0}^{m-2} c_a x^a$ to $x^m - tz$. These are deformation modules of the total graph singularity, not of its fixed-base relative curve or of the raw failure algebra.

In a transverse surface the reduced tail is $\mathbb{Q}$ -Cartier with $F^2 = -1/m$. Its minimal resolution introduces the A_{m-1} chain; the total fibre has multiplicities m on the strict transform of F and $m - 1, m - 2, \dots, 1$ along the consecutive chain, and multiplicity one on C_0 .

Proof. The cotangent complex of a hypersurface is represented by the conormal free module mapping to the three ambient differentials. Dualizing gives cokernel by the partial derivatives $mx^{m-1}, -z, -t$, hence (7.6); there are no higher terms. Lifting an equation across a small extension has no obstruction, and the displayed monomials form a basis of the cokernel, giving the formal versality assertion. This is the classical hypersurface calculation applied to the exact Rees equation.

In the first Rees chart the exceptional Cartier ideal is generated by x^m , so its Cartier divisor is mF . Its associated ideal bundle restricts to $\mathcal{O}_F(1)$; thus $mF^2 = -1$. The standard cyclic quotient presentation can be checked directly:

$$x = ab, \quad t = a^m, \quad z = b^m, \quad A_m = \mathbb{C}[[a, b]]^{\mu_m}, \quad (a, b) \mapsto (\zeta a, \zeta^{-1} b).$$

Subdivide its two-dimensional cone by the primitive rays $((m - i)/m, i/m)$, $1 \leq i < m$, in the quotient lattice. Consecutive cones are smooth. The internal curves form the chain with

self-intersection -2 , and the orders of $t = a^m$ are $m - i$. The end attached to the tail has multiplicity $m - 1$; the next ones decrease by one. The main component meets the other, already smooth, end of the tail, where $t = x^m y$ gives multiplicities $m, 1$. These charts prove the stated fibre description. $\square$

Theorem 7.4 (A thick Hilbert tail and a covered stable-map tail). *Fix a nonzero direction $(u, v) = \tau(\alpha, \beta)$ in the slice, with contact order m . After the base change $\tau = s^m$, the normalization of its total Hilbert graph is the blow-up of the smooth pencil surface at $(x, s) = (0, 0)$. Its special fibre is a reduced nodal union $C_0 \cup F'$. The induced map to complete quadrics is an embedding on C_0 and a degree- m map on F' onto F ,*

$$[X : S] \longmapsto [X^m : S^m] \quad (7.7)$$

in the residual line coordinates. It is a stable map with tail deck group μ_m . It has the same cycle $[C_0] + m[F]$ as the Hilbert limit, but the latter has the nonreduced scheme structure (7.5), not the reduced image of this map.

Proof. The base-changed graph is the blow-up of (x^m, s^m) . This particular base change is flat, and the Rees presentation also verifies it directly. On the chart $x^m = s^m z$ adjoining $w = x/s$ gives the integral element $w^m = z$ and the normal chart $\mathbb{C}[s, w]$, with $x = sw$. On the other chart $s^m = x^m y$ adjoining $r = s/x$ gives $r^m = y$ and the normal chart $\mathbb{C}[x, r]$, with $s = xr$. These charts are finite and birational over the original charts and glue by $w = r^{-1}$. They are exactly the point blow-up charts. They also show a reduced nodal fibre after normalization.

The old graph coordinates $[x^m : s^m]$ restrict to (7.7) on the exceptional line of the point blow-up. Away from the contact point normalization does not alter the original lifted line. Both components are nonconstant; the main component is embedded and the nonconstant tail has precisely the finite deck group μ_m , which fixes the attachment point. Thus the map is stable. The image of F' is reduced while the flat Hilbert fibre has the explicit nilpotent x of order m , so the two objects differ scheme-theoretically even though pushforward cycles and degrees agree. This comparison is calculated here; it is not deduced from a general identification of Hilbert and stable-map compactifications. $\square$

For the reduced-incidence local model ($m = 1$), the two local descriptions coincide with the nodal model. For every $m \geq 2$, the length of the contact, the tail multiplicity, the transverse graph singularity A_{m-1} , and the stable tail covering degree record the same integer in four different objects. The assertions concern all orders in the specified symmetric Jordan slices. They do not classify arbitrary higher-corank contacts or the entire ambient Hilbert fibre over a nonreduced-incidence pencil.

APPENDIX A. POWER GRAPHS OF EVERY RANK

For $1 \leq r \leq n$, let $P^{(r)} \subset \mathbb{P}(\text{Sym}^2 V)$ be the rank-exactly- r locus. Define $\mathcal{G}_{r,h}$ as the scheme-theoretic closure of

$$P^{(r)} \longrightarrow \mathbb{P}(\text{Sym}^2 V) \times \prod_{p=1}^{hr} \mathbb{P}(B_h(V)_p), \quad [A] \longmapsto ([A], [A], \dots, [A^{hr}]). \quad (\text{A.1})$$

Keeping the final power is essential when $r < n$. Write $\mathcal{U}$ for the tautological rank- r bundle on $\text{Gr}(r, V)$ and $\text{CQ}(\mathcal{U})$ for its relative complete-quadric space. For rank one this means $\mathbb{P}(\text{Sym}^2 \mathcal{U}) = \text{Gr}(1, V)$.

Theorem A.1 (The rank-stratified power graph). *There is a canonical isomorphism*

$$\mathcal{G}_{r,h} \simeq \text{CQ}(\mathcal{U}) \longrightarrow \text{Gr}(r, V). \quad (\text{A.2})$$

The final power in (A.1) recovers the image r -plane by the degree- $2h$ Pluecker embedding. In particular the graph is smooth and projective, independently of h . For its relative boundary divisors E_i , $1 \leq i < r$, its power base ideals have total transforms

$$J_p \mathcal{O}_{\mathcal{G}_{r,h}} = \mathcal{O}_{\mathcal{G}_{r,h}} \left(- \sum_{i=1}^{r-1} (p - hi)_+ E_i \right), \quad 0 \leq p \leq hr. \quad (\text{A.3})$$

The projective power maps in this formula have their common divisors removed. All powers $p > hr$ vanish on the rank- r locus.

Proof. For $U \subset V$, the map $B_h(U) \rightarrow B_h(V)$ from the presentation (2.1) is injective: any linear projection $V \rightarrow U$ splitting the inclusion induces a retraction on the quotient algebras. The injection itself is independent of that projection. Locally this gives subbundles $B_h(\mathcal{U})_p \subset B_h(V)_p$. If A has image U and rank r , it is a nonsingular element of $\text{Sym}^2 U$, and A^{hr} generates the socle $B_h(U)_{hr} = (\det U)^{2h}$. As U varies these lines give the homogeneous embedding

$$\text{Gr}(r, V) \hookrightarrow \mathbb{P}(\mathbb{S}_{(2h)r} V) \hookrightarrow \mathbb{P}(B_h(V)_{hr}).$$

Indeed the line at the coordinate r -plane is a highest-weight line of weight $(2h)^r$; its orbit map is the Pluecker embedding followed by its degree- $2h$ linear system. In particular this is a closed immersion, not just a recovery of the image plane on closed points.

Over the recovered Grassmannian, the remaining graph is precisely the relative simultaneous power graph for $B_h(\mathcal{U})$. By Theorem 5.1 on each frame chart, it is $\text{CQ}(\mathcal{U})$. The relative graph embedding, the subbundle injections above, and the final-power Grassmannian embedding together give a closed immersion into the product in (A.1). Its dense open is the graph of $P^{(r)}$. Both its source and that graph closure are integral, so equality of their scheme-theoretic images proves (A.2) without a normalization step. Applying (5.3) in rank r proves the transform formula. Local split inclusion of the coefficient bundles ensures that ambient and relative power ideals agree. The vanishing assertion follows from the socle degree of $B_h(U)$. $\square$

The varieties of complete singular quadrics are classical: see [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. The point of the theorem is the exact recovery of their image-plane projection and entire scheme structure from the power maps in one fixed algebra $B_h(V)$, simultaneously for every r . The classical orbit stratification is background, not an additional novelty claim.

APPENDIX B. CONTACTS AND MINIMAL INDICES OF ARBITRARY PENCILS

Let $R = \langle A_0, A_1 \rangle \subset \text{Sym}^2 V$ be a genuine pencil, $\Lambda = \mathbb{P}(R)$, and r its normal rank. Its tautological symmetric map is $A : V^* \otimes \mathcal{O}_\Lambda \rightarrow V \otimes \mathcal{O}_\Lambda(1)$. Write

$$K = \ker A, \quad U = K^\perp \subset V \otimes \mathcal{O}_\Lambda.$$

The kernel is saturated and hence a subbundle on the smooth curve; U is the saturation of the generic image. Thus A is a generically nonsingular section of $\text{Sym}^2 U \otimes \mathcal{O}_\Lambda(1)$. The map on the rank- r open lifts uniquely to $\tilde{\Lambda} \rightarrow \mathcal{G}_{r,h}$ by properness. Its Grassmannian projection is the map defined by U , and its source is still Λ . Let D_i be its contact divisor with E_i .

Lemma B.1 (Symmetric diagonalization with a kernel). *Over $\mathbb{C}[[t]]$, a symmetric matrix of rank r over the fraction field is congruent to $\text{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0)$, with $e_1 \leq \dots \leq e_r$. For the local matrix of a genuine pencil, $e_1 = 0$. The exponents are its nonzero Smith exponents.*

Proof. Choose a matrix entry of smallest valuation. If a diagonal entry has that valuation, it is a pivot. Otherwise an off-diagonal entry does, and the value of the quadratic form on $e_i + ce_j$ has the same valuation for some $c \in \mathbb{C}$: its leading coefficient is a nonzero polynomial in c , since 2 is invertible. Extend this primitive vector to a basis. The pivot divides every entry in its

row, so symmetric row-and-column elimination clears the row and column over the ring. All remaining entries have valuation at least that of the pivot. Repeat until the remaining block is zero. Units have square roots in $\mathbb{C}[[t]]$ and can be absorbed into the basis. Invertible congruence preserves determinantal ideals, which identify the resulting exponents with Smith exponents. A genuine pencil has a nonzero matrix at every projective parameter, giving a unit entry and $e_1 = 0$. This proof supplies the local congruence step rather than relying on left-right Smith form alone; compare the classical pencil congruence treatment in [36]. $\square$

For $j \geq 0$ define the finite multiplication-syzygy map

$$C_j(R) : V^* \otimes \text{Sym}^j R^* \longrightarrow V \otimes \text{Sym}^{j+1} R^*, \quad q \longmapsto Aq, \quad \kappa_j = \dim \ker C_j(R). \quad (\text{B.1})$$

Set $\kappa_{-1} = \kappa_{-2} = 0$. Under a basis of R this is the block Toeplitz map with rows $A_0 q_0$, $A_0 q_i + A_1 q_{i-1}$, and $A_1 q_j$; its definition is independent of that basis.

Theorem B.2 (Complete power-and-syzygy data). *For every complex symmetric pencil, including every singular pencil, the following formulas recover its complete congruence data. If $\varepsilon_1, \dots, \varepsilon_c$ are its minimal indices, where $c = n - r$, then*

$$K \simeq \bigoplus_{a=1}^c \mathcal{O}_\Lambda(-\varepsilon_a), \quad \kappa_j = \sum_{a=1}^c (j - \varepsilon_a + 1)_+, \quad (\text{B.2})$$

$$\#\{a : \varepsilon_a = j\} = \kappa_j - 2\kappa_{j-1} + \kappa_{j-2}. \quad (\text{B.3})$$

It is enough to use $0 \leq j \leq \lfloor (n-1)/2 \rfloor$. At $x \in \Lambda$, let $0 = e_1(x) \leq \dots \leq e_r(x)$ be the nonzero local Smith exponents. With $v_0 = 0$ and $v_p(x) = \text{length}(\mathcal{O}_{\Lambda, x}/J_p)$ for $p \leq hr$, one has

$$v_{hq+s}(x) = h \sum_{i=1}^q e_i(x) + s e_{q+1}(x), \quad (\text{B.4})$$

$$e_i(x) = \frac{v_{hi}(x) - v_{h(i-1)}(x)}{h}, \quad \text{mult}_x D_i = e_{i+1}(x) - e_i(x). \quad (\text{B.5})$$

In the first formula $s = 0$ at $q = r$ and the last term is omitted. The positive exponents at each supported point, together with the minimal indices, determine the pencil up to congruence and projective reparametrization of Λ . The spectral datum includes the positions of the points, not only their local partitions.

There is also a global conservation law:

$$\sum_{i=1}^{r-1} (r-i) \deg D_i + 2 \deg(\Lambda \longrightarrow \text{Gr}(r, V)) = r, \quad \deg(\Lambda \longrightarrow \text{Gr}(r, V)) = \sum_a \varepsilon_a. \quad (\text{B.6})$$

The Grassmannian degree uses its Pluecker polarization. The first summand is the dimension of the regular summand in the Kronecker congruence form. Every object and number in the theorem is invariant under the ungraded failure-algebra isomorphisms of Theorem I.B.1.

Proof. We use the classical complex symmetric Kronecker congruence theorem [36]. A singular block of minimal index ε has size $2\varepsilon + 1$ and form

$$S_\varepsilon(s, t) = \begin{pmatrix} 0 & L_\varepsilon(s, t)^t \\ L_\varepsilon(s, t) & 0 \end{pmatrix}, \quad L_\varepsilon = \begin{pmatrix} s & t & 0 \\ & s & t \\ & & \ddots & t \end{pmatrix},$$

where L_ε has ε rows and $\varepsilon + 1$ columns; S_0 is a zero 1×1 block. Its kernel is generated by $((-t)^\varepsilon, s(-t)^{\varepsilon-1}, \dots, s^\varepsilon)$ in the first block. These components have no common projective zero, so it gives $\mathcal{O}(-\varepsilon)$ as a kernel subbundle. A regular block has zero kernel as a sheaf. Direct sums prove the first formula in (B.2). Taking sections after twist by $\mathcal{O}(j)$ identifies them with (B.1),

proving the second formula. The second finite difference of $(j - \varepsilon + 1)_+$ is one exactly at $j = \varepsilon$ and zero otherwise. Each singular block uses $2\varepsilon + 1 \leq n$ dimensions, proving the stated finite bound. The canonical-form classification is used here as a classical input, not reproved or claimed as new.

Locally, symmetric diagonalization factors the matrix through the saturated r -plane U with an $r \times r$ symmetric block. Theorem 4.1, or its rank- r version via the split inclusion $B_h(U) \hookrightarrow B_h(V)$, yields (B.4). Finite differences recover the e_i . In the relative complete-quadric chart the successive ratios of diagonal entries have valuations $e_{i+1} - e_i$, proving the contact formula. The torsion of the pencil cokernel has precisely the positive Smith exponents. The regular blocks of the classical congruence form are specified by these elementary divisors, and its singular blocks by the minimal indices; over $\mathbb{C}$ there are no additional signature parameters. Retaining the supported points on Λ makes the statement compatible with arbitrary projective changes of the pencil basis.

The exact sequence $0 \rightarrow U \rightarrow V \otimes \mathcal{O} \rightarrow K^* \rightarrow 0$ gives $\deg U = -\sum \varepsilon_a$, so the Grassmannian degree is $\sum \varepsilon_a$. The determinant of the induced symmetric map $U^* \rightarrow U(1)$ is a nonzero section of $(\det U)^2 \otimes \mathcal{O}(r)$; its zero divisor has degree $r - 2\sum \varepsilon_a$. At each point its order is $\sum e_i = \sum_{i < r} (r - i) \text{mult}_x D_i$. This proves (B.6). The block-size identity $n = n_{\text{reg}} + \sum (2\varepsilon_a + 1)$ identifies the first summand with n_{reg} . Finally the intrinsic first relation extracts Λ_A on its source and preserves all the maps involved; changes of source lift or pencil basis give isomorphic kernel maps and equal ideals, hence the asserted invariance. $\square$

Example B.3 (Power contacts alone miss a singular invariant). The size-six pencils $S_0 \oplus S_2$ and $S_1 \oplus S_1$ both have constant rank four and no spectral torsion. All their power ideals are the unit ideal for $p \leq 4h$ and zero for $p > 4h$, and every contact divisor is empty. Their kernel splittings are respectively $\mathcal{O} \oplus \mathcal{O}(-2)$ and $\mathcal{O}(-1)^2$. Thus their first three syzygy nullities are $(1, 2, 4)$ and $(0, 2, 4)$, distinguishing the two congruence types. Both have Grassmannian degree two, as required by (B.6); the full splitting, rather than just its degree, is necessary.

Proposition B.4 (Finite closed tests in families). *For $j \geq 0$ and $0 \leq a \leq n(j+1)$, on the pencil Grassmannian each condition $\kappa_j \geq a$ is the closed determinantal condition on the bundle map C_j given by minors of size $n(j+1) - a + 1$. These conditions commute with arbitrary base change as schemes. Their fibrewise ranks need not be constant, so no unqualified assertion that the kernel bundles commute with every base change is made. Along a specialization over an integral curve, κ_j can only increase.*

Proof. Use the tautological rank-two subbundle in (B.1). Its source has rank $n(j+1)$ and its target rank $n(j+2)$. Nullity at least a is equivalent to rank at most $n(j+1) - a$, giving the stated minors. The ideals are polynomial in the bundle map and therefore commute with arbitrary substitution. Upper semicontinuity of nullity, or specialization of these minors, gives the last assertion. $\square$

APPENDIX C. BOUNDARY CONTACTS AND SPECTRAL SCHEMES

Let $R \subset M$ be a regular pencil and $\Lambda = \mathbb{P}(R)$. Its map to P lifts uniquely to a morphism $\tilde{\iota}_R : \Lambda \rightarrow \text{CQ}(V)$: it lifts on the nonsingular open, and properness extends it uniquely across the missing points of the smooth curve. Put $D_r(R) = \tilde{\iota}_R^* E_r$. These are effective divisors because the generic member is nonsingular.

Theorem C.1 (Contact-divisor reconstruction of spectral data). *For every $h \geq 1$, every regular pencil, and $0 \leq p \leq nh$,*

$$J_p|_{\Lambda} = \mathcal{O}_{\Lambda} \left(- \sum_{r=1}^{n-1} (p - hr)_+ D_r(R) \right). \quad (\text{C.1})$$

At a point $x \in \Lambda$, let $0 = e_1 \leq \dots \leq e_n$ be the symmetric Smith exponents of the local pencil matrix. Then

$$\text{mult}_x D_r(R) = e_{r+1} - e_r \quad (1 \leq r < n). \quad (\text{C.2})$$

Consequently the supported contact divisors determine every elementary divisor, hence the full Segre symbol. If $v_p(x) = \text{length}(\mathcal{O}_{\Lambda, x}/J_p)$ and $v_0(x) = 0$, then

$$\text{mult}_x D_r(R) = \frac{v_{h(r+1)}(x) - 2v_{hr}(x) + v_{h(r-1)}(x)}{h}. \quad (\text{C.3})$$

Proof. Pull back (5.3); equality of extended ideals under composition gives (C.1). The pencil matrix is nonzero at every point, since R is a genuine two-dimensional subspace; hence its entries generate the unit ideal locally and $e_1 = 0$. Over $\mathbb{C}[[t]]$ symmetric elimination gives a congruence to $\text{diag}(t^{e_1}, \dots, t^{e_n})$, absorbing unit factors by square roots. Its unique lift has, in (5.4), $\tau_r = t^{e_{r+1} - e_r}$ up to a unit. This proves (C.2); faithful flatness of completion gives the local divisor statement. Taking valuations in (C.1) and second differences at multiples of h proves (C.3). Recover e_i by summing the first $i - 1$ contact multiplicities. The positive exponents, with their points on Λ , are precisely the elementary-divisor partitions of the spectral cokernel. $\square$

Example C.2 (A boundary point and its contact order are distinct data). For a symmetric germ with exponents $(0, 1, 3)$ and $h = 2$, the valuations of $J_1, \dots, J_6$ are $(0, 0, 1, 2, 5, 8)$. Its two boundary contacts are 1 and 2. The germs $\text{diag}(1, t, t^3)$ and $\text{diag}(1, t^2, t^5)$ reach the same full-flag complete-quadric point but have contacts $(1, 2)$ and $(2, 3)$. The compactification retains the flag; the arc with its divisorial contacts retains the elementary exponents. These assertions must not be conflated. For singular pencils the universal ideal theorem still holds and detects all Fitting schemes and the normal rank; Fitting ideals alone do not determine Kronecker minimal indices.

APPENDIX D. A PROJECTIVE INCIDENCE COMPACTIFICATION FOR PENCILS

The complete-quadric graph concerns members of pencils. There is also a projective parameter space carrying both the original finite failure algebra and flat limits of their resolved member curves. This construction keeps the pencil and does not identify distinct pencils at a common full-coordinate-change limit.

Proposition D.1 (Resolved pencil curves and the failure family). *Let $n \geq 3$, $G = \text{Gr}(2, M)$, and let G° be the open set of regular pencils whose lines avoid the rank-at-most- $(n - 2)$ locus in P . There is a normal projective $\text{GL}(V)$ -equivariant scheme $\hat{G}$ with a projective birational morphism $\rho : \hat{G} \rightarrow G$, an isomorphism over G° , and a flat family of subschemes $\mathcal{C} \subseteq \hat{G} \times \text{CQ}(V)$ with the following properties. Over G° it is exactly the lifted pencil line. Every fibre over R lies scheme-theoretically in $\pi^{-1}(\mathbb{P}(R))$. All the resolved power maps restrict to $\mathcal{C}$, for every h . The sharp finite failure algebra on G pulls back to a finite locally free algebra on $\hat{G}$, with its original multiplication.*

Proof. The rank-at-most- $(n - 2)$ locus has codimension three in P , so a general line avoids it; a general pencil is regular. Thus G° is nonempty and dense. Over that open, π is an isomorphism along every pencil line and these lifts form a flat family of embedded projective lines.

For $1 \leq q < n$, let H_q be the pullback to $\text{CQ}(V)$ of the hyperplane bundle for its q th exterior-power projection, taking $q = 1$ to mean P . The bundle $L = \bigotimes_{q=1}^{n-1} H_q$ is very ample, by the graph embedding and Segre embedding. On a lifted line over G° , $\deg H_q = q$, since its q -minors have no common zero. The Hilbert polynomial with respect to L is therefore $\binom{n}{2}m + 1$. The lifted family defines a morphism

$$G^\circ \longrightarrow \text{Hilb}^{\binom{n}{2}m+1}(\text{CQ}(V)).$$

Take the reduced closure of its graph in G times this projective Hilbert scheme, and then its normalization; this is $\widehat{G}$. Normalization is finite here, so the result is projective and normal. The graph is already closed over G° and that open is smooth, so ρ is an isomorphism there. Equivariance of the family and the uniqueness of normalization give the stated group action.

Pull back the universal Hilbert family to obtain $\mathcal{C}$; it is flat. To verify the asserted scheme-theoretic incidence, pull back the linear equations for the universal pencil line to $\mathcal{C}$. They vanish over the dense open G° . On an affine open of the integral base $\widehat{G}$, flatness makes the coordinate modules of $\mathcal{C}$ torsion free over the base. A section which becomes zero at the generic point is therefore zero. Applied locally to these linear equations, this proves the incidence on the whole family. The power maps are everywhere defined on $\text{CQ}(V)$ by Theorem 5.1, so their restrictions exist. Finally pull back the finite locally free algebra of Corollary I.7.5 along ρ ; arbitrary base change preserves its rank and multiplication. $\square$

This is an incidence compactification over the Grassmannian, with its native congruence action, not a properness assertion for the quotient stack by that action. A limiting Hilbert fibre may have extra or nonreduced components; no classification of all such fibres is used. The construction is independent of h : its target, polarization and Hilbert polynomial were specified using the h -independent complete quadric graph. The unmarked inverse in the companion reconstruction paper recovers the underlying pencil, so the incidence construction is compatible with that inverse rather than a replacement of it.

APPENDIX E. THE SIMPLE CORANK-TWO INCIDENCE MODIFICATION

The ordinary-collision calculation describes specified families of limits. We now identify the entire normalized Hilbert incidence space on a natural open of the pencil Grassmannian. The result holds in every dimension. It does not require a semisimple collision or a squarefree residual pencil.

Let $n \geq 3$, $M = \text{Sym}^2 V$, $P = \mathbb{P}(M)$, and $G = \text{Gr}(2, M)$. Write $Z_j \subset P$ for the symmetric rank-at-most- j scheme, and put $Z_0 = \emptyset$ when $n = 3$. Let $\mathcal{U} \subset G$ consist of regular pencils R such that their lines $L_R = \mathbb{P}(R)$ avoid Z_{n-3} and their scheme-theoretic intersections with Z_{n-2} are either empty or a single reduced point. Write $\mathcal{P}$ for the universal line over $\mathcal{U}$. We use projective spaces of lines, so that the exceptional fibre of a smooth blow-up is the projectivization of the normal space.

Lemma E.1 (The incidence centre and its normal space). *The locus $\mathcal{U}$ is open and contains G° of Proposition D.1. Its incidence scheme*

$$\mathcal{D} = \mathcal{P} \times_P Z_{n-2} \longrightarrow \mathcal{U}$$

is a closed immersion with smooth image Σ_2 of codimension two. At $R \in \Sigma_2$, let $p = L_R \cap Z_{n-2}$ and let $a_R \subset N_{Z_{n-2}/P, p}$ be the nonzero image of $T_p L_R$. There is a canonical isomorphism

$$N_{\Sigma_2/\mathcal{U}, R} \simeq N_{Z_{n-2}/P, p}/a_R. \quad (\text{E.1})$$

Near (p, R) , after an étale change of base and coordinates, the ideal of $\mathcal{D}$ in $\mathcal{P}$ is

$$I_{n-1}(A) = (x, u, v), \quad I_{\Sigma_2} = (u, v), \quad (\text{E.2})$$

where x is a relative line coordinate and u, v are part of a regular system of parameters on the base. The generators may be changed by an invertible matrix; the asserted equality is an equality of ideals.

Proof. On $P \setminus Z_{n-3}$ the scheme Z_{n-2} is smooth of codimension three. A nonsingular $(n-2)$ -block reduces its equations to the three entries of a symmetric 2×2 Schur complement. In particular these are regular local equations for its given determinantal scheme structure.

The universal line evaluation $\mathcal{P}_G \rightarrow P$ is smooth: over a point of P its fibre is the projective space of lines through that point. Hence its inverse image of the smooth rank locus is smooth and has dimension $\dim G - 2$. Lines contained in the rank locus, or meeting the lower rank locus, form closed exceptional sets after the usual proper incidence projection. On the complementary open the incidence projection is finite. The condition that its fibre have length at most one is open, by upper semicontinuity for a finite module. This proves the asserted openness. On this open the unit map $\mathcal{O}_{\mathcal{U}} \rightarrow q_* \mathcal{O}_{\mathcal{D}}$ is surjective: on each geometric fibre its target is either zero or a one-dimensional algebra with unit, and Nakayama's lemma applies to its coherent cokernel. Thus q is a closed immersion. Its source is smooth by the smooth evaluation just noted; its codimension is two.

First-order motion of a line evaluates surjectively onto $T_p P / T_p L_R$: in the description $T_R G = \text{Hom}(R, M/R)$ this is restriction to the one-dimensional subspace representing p . To remain incident to the rank locus, this evaluation must vanish in $T_p P / (T_p Z_{n-2} + T_p L_R)$. A reduced intersection means that the normal image of $T_p L_R$ is nonzero. The kernel is precisely $T_R \Sigma_2$, since the incidence point may move along both the line and the rank locus. The quotient is therefore (E.1).

For the local assertion choose a rank-locus equation with nonzero relative derivative. Its zero hypersurface is etale over the base near (p, R) ; after an etale base change it gives a section. Use that equation as the coordinate x . Subtract multiples of x from the other two equations. Their restrictions to the section are base functions u, v , and the ideal becomes (x, u, v) . The normal-space calculation proves that du, dv are independent. This proves (E.2). The argument is the smooth-centre coordinate calculation, not an assertion that arbitrary graph closures commute with base change. $\square$

The open is nonempty along its centre. For example, a diagonal pencil with two equal linear entries and all other entries having distinct zeros away from their common zero belongs to Σ_2 . Although $a_R \neq 0$, its residual binary quadratic form may have rank one. This case is allowed throughout the following theorem.

Theorem E.2 (The full simple-incidence Hilbert modification). *For the normalized Hilbert incidence modification $\rho : \widehat{G} \rightarrow G$ of Proposition D.1, there is a canonical isomorphism over $\mathcal{U}$*

$$\widehat{G} \times_G \mathcal{U} \simeq B := \text{Bl}_{\Sigma_2} \mathcal{U}. \quad (\text{E.3})$$

It is equivariant under source congruence. In particular this part of the incidence space is smooth. Its exceptional divisor $F = \mathbb{P}(N_{\Sigma_2/\mathcal{U}})$ is a $\mathbb{P}^1$ -bundle, and

$$K_B = \beta^* K_{\mathcal{U}} + F, \quad \mathcal{O}_B(F)|_{F_R} = \mathcal{O}_{\mathbb{P}^1}(-1), \quad (\text{E.4})$$

where $\beta : B \rightarrow \mathcal{U}$ is the blow-up.

For every $R \in \Sigma_2$ the entire fibre of ρ is $F_R \simeq \mathbb{P}^1$. Its point $[\bar{b}] \in \mathbb{P}(N_{Z_{n-2}/P, p}/a_R)$ represents the reduced nodal curve

$$C_R \cup \ell_{[\bar{b}]}, \quad \ell_{[\bar{b}]} = \mathbb{P}(a_R + \mathbb{C}b) \subset \pi^{-1}(p) = \mathbb{P}(N_{Z_{n-2}/P, p}). \quad (\text{E.5})$$

Here C_R is the strict transform of L_R and b is any lift of $\bar{b}$. The tail attaches to C_R at $[a_R]$. Every line through $[a_R]$ in this exceptional $\mathbb{P}^2$ occurs, and no further components, embedded points, or nonreduced fibres occur over this open. The two components have exterior multidegrees

$$\deg_{C_R} H_q = q - \mathbf{1}_{q=n-1}, \quad \deg_{\ell} H_q = \mathbf{1}_{q=n-1} \quad (1 \leq q < n). \quad (\text{E.6})$$

Their common Hilbert polynomial is $\binom{n}{2}m + 1$ for $\bigotimes_{q=1}^{n-1} H_q$.

The universal curve over B is the simultaneous multiplication graph on $\mathcal{P} \times_{\mathcal{U}} B$. Near a node it is etale locally $xy = t$, where $t = 0$ defines F . Thus it is a flat family of nodal curves, including along directions for which the residual determinant has a repeated zero. It commutes with base

change as this specified flat family. The blow-up is characterized by the universal property that $I_{\Sigma_2} \mathcal{O}_B$ is an invertible ideal; the corresponding factorization property holds for every $T \rightarrow \mathcal{U}$ on which that image ideal is an effective Cartier ideal.

Proof. On $P \setminus Z_{n-3}$, the complete-quadric morphism is exactly the blow-up of the smooth centre Z_{n-2} : all earlier rank blow-ups are isomorphisms there and the determinant divisor requires no blow-up. We use the scheme-level classical construction [34, Remark 2.5 and Construction 2.6], originating in [33, Theorem 6.3]. Nothing in this step declares the space of complete quadrics itself new.

Pull the universal line back to B . In a base blow-up chart $u = t$, $v = tw$, the ideal (E.2) becomes (x, t) . Its blow-up has the two charts

$$\mathcal{O}_B[x, y]/(xy - t), \quad \mathcal{O}_B[z], \quad x = tz.$$

They glue by the usual ratio change. Their total spaces are smooth, and they are flat over B . For example $\mathcal{O}_B[x, y]/(xy - t)$ is free as an $\mathcal{O}_B$ -module with basis $1, x, x^2, \dots, y, y^2, \dots$; reduction by the monic relation $xy - t$ gives this basis. Its fibres at $t = 0$ have two reduced branches and one ordinary node. The other base chart has the same description. Outside the incidence section the pulled ideal is the unit ideal. Blowing up the globally defined pulled ideal therefore gives a projective flat family $\mathcal{C}_B$ with these local charts. All smaller minor ideals are units on this open. The universal power identity makes the product of the nontrivial power ideals an h^2 th power of I_{n-1} , up to an invertible determinant ideal. Thus this same blow-up is the simultaneous multiplication graph for every h , not only the exterior graph.

There is a closed immersion $\mathcal{C}_B \hookrightarrow \text{CQ}(V) \times B$. Indeed the base-changed Rees algebra of the rank-locus ideal surjects onto the Rees algebra of its image on the universal line; taking Proj gives a closed immersion into the base-changed complete-quadric graph. The universal line is already closed in $P \times B$. Composing these two closed immersions proves the assertion. Equivalently, use the simultaneous graph lemma on the integral total space. This argument does not identify a general base change of a blow-up with the blow-up of the image ideal without the indicated Rees quotient.

In normal coordinates at p , the exceptional map is $[x : t] \mapsto [xa_R + tb]$, with b representing the normal direction of the base motion. Formula (E.1) implies that a_R, b are independent. This is a line embedded in $\mathbb{P}(N_{Z_{n-2}/P, p})$, not a multiple cover. The strict-transform branch meets it at $[a_R]$. This proves (E.5), including the converse parametrization of all lines through that point. The local charts show the absence of embedded components and multiplicities.

The minor systems of orders $q < n - 1$ have no base point on L_R . The $(n - 1)$ -minors have one simple common zero, by (E.2). Their degrees on C_R are therefore q and $n - 2$, respectively. On the exceptional $\mathbb{P}^2$ only the last exterior system has positive degree, namely one. This proves (E.6). The two projective lines have Euler characteristic $2 - 1 = 1$, since their intersection is one reduced point. Their total degree is $\binom{n}{2}$.

The flat embedded family defines a morphism from B to the reduced Hilbert graph closure over $\mathcal{U}$. It is proper because B is proper over $\mathcal{U}$ and the Hilbert graph is separated over that base. Its image is exactly the graph closure: it is closed and contains the dense graph over $\mathcal{U} \setminus \Sigma_2 = G^\circ$, while B is integral. Over an incident pencil the displayed exceptional lines are pairwise distinct as embedded curves; over a nonincident pencil the fibre is a point. The morphism thus has finite geometric fibres and is finite by [Stacks Project, Tag 02LS]. It is birational, and B is normal because its centre is smooth. A finite birational map from a normal integral scheme identifies it with the normalization: on affine charts its ring is the integral closure in the common function field. Normalization commutes with restriction to the open $\mathcal{U}$. This proves (E.3), without assuming that an arbitrary nondominant normal test scheme lifts to a normalization.

All maps are specified by the universal incidence and agree on the dense nonincident locus. Their uniqueness proves canonicity and congruence equivariance. For completeness, in coordinates $v = uw$, $du \wedge dv = u du \wedge dw$, giving the discrepancy one in (E.4). The exceptional normal bundle is tautological of degree -1 . The universal property is the usual Rees-algebra factorization for an effective Cartier image ideal [Stacks Project, Tag 0806]; here the preceding argument identifies the actual Hilbert incidence modification with that universal object. Flat pullback preserves the explicitly constructed nodal family. No assertion about unrestricted graph-closure base change is used. $\square$

E.1. Every power on every exceptional direction.

Proposition E.3 (Complete exceptional power systems). *Fix $R \in \Sigma_2$ and $p = L_R \cap Z_{n-2}$. For $h \geq 1$ and $1 \leq j \leq hn$, define*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+. \quad (\text{E.7})$$

On the full exceptional plane $\pi^{-1}(p) \simeq \mathbb{P}^2$, the resolved j th multiplication map is the complete degree- δ_j Veronese map into its linear span, with degree zero interpreted as constant. Its restriction to every tail in (E.5) is therefore the complete degree- δ_j system on $\mathbb{P}^1$. In particular $j = h(n - 2) + 1$ detects the tail as an embedded line. These assertions include tangent lines to the residual determinant conic and attachments on that conic.

Proof. A unit $(n - 2)$ -block leaves a residual two-dimensional space W . Lemma G.2 identifies the first nonzero normal coefficient at degree $j > h(n - 2)$ with the power of order $b = j - h(n - 2)$ in $B_h(W)$, multiplied by a nonzero unit-block socle. For $j \leq h(n - 2)$ the order-zero coefficient is constant on the exceptional fibre. In the three-variable coordinate ring of $\text{Sym}^2 W$, put $\mathfrak{a} = I_1$ and $f = \det$. The universal ideal formula gives

$$J_b = \mathfrak{a}^b \quad (b \leq h), \quad J_b = \mathfrak{a}^{2h-b} f^{b-h} \quad (h \leq b \leq 2h).$$

The generators all have degree b . Their span is the entire degree b part of the displayed ideal. After removal of the common factor $f^{(b-h)_+}$ it is the full space of ternary forms of degree $b - 2(b - h)_+ = \delta_j$. Hence it defines the complete Veronese system on the exceptional plane. Restriction of all ternary forms to any line gives all binary forms of that degree, independently of its intersection multiplicities with the conic $f = 0$. This proves all assertions, including the endpoint $j = hn$ where the map is constant. $\square$

APPENDIX F. A COMPUTED BOUNDARY OF THE PENCIL INCIDENCE SPACE

We give the incidence compactification a precise universal property and compute a family of its boundary fibres in every dimension. The new boundary is selected by one power base ideal, not by pulling an unrelated algebra back to a parameter space.

Proposition F.1 (The normal main-component flattening property). *Let $G^\circ \subset G = \text{Gr}(2, \text{Sym}^2 V)$ and $\widehat{G} \rightarrow G$ be as in Proposition D.1. Among normal integral schemes T with a dominant morphism $f : T \rightarrow G$ for which the closure of the lifted pencil family over $f^{-1}(G^\circ)$ is an embedded flat family in $\text{CQ}(V) \times T$ with Hilbert polynomial $\binom{n}{2}m + 1$, there is a unique morphism $T \rightarrow \widehat{G}$ over G , and that family is the pullback of its universal family. Consequently $\widehat{G}$ is the minimal normal dominant modification with this specified embedded-flat-closure property.*

Proof. The flat embedded family gives a morphism to the indicated Hilbert scheme. Together with f it agrees, over a dense open of T , with the graph defining the reduced graph closure. Since T is integral, its morphism factors through this closed graph scheme: each equation vanishing on the dense open vanishes on T . The induced morphism is dominant. A dominant morphism from a normal integral scheme to an integral scheme factors uniquely through the normalization:

locally, an element integral over the target ring becomes integral over the source ring and lies in its fraction field, hence belongs to the normal source ring. Thus the morphism factors uniquely through $\widehat{G}$. The Hilbert-scheme representing property identifies the families. Conversely its pulled-back universal family is flat; flatness over an integral base makes it the schematic closure of its restriction to the dense open. Uniqueness follows also from separatedness. $\square$

The hypothesis of dominance is part of this property; this proposition does not assert that normalization commutes with arbitrary base change. The property is a standard consequence of the Hilbert functor and normalization. Its usefulness here is that the following calculation identifies a nontrivial fibre of the resulting modification.

Assume $n \geq 3$. Choose distinct nonzero numbers $a_3, \dots, a_n$ and put $D = \text{Spec } \mathbb{C}[[\tau]]$. On $\mathbb{P}_D^1$, consider the symmetric pencil

$$A_\tau(s, t) = \text{diag}(s, s + \tau t, t - a_3 s, \dots, t - a_n s). \quad (\text{F.1})$$

For $n = 3$ there is just one number a_3 . Write $X = \mathbb{P}_D^1$ and let o be the point $s = \tau = 0$ in the special fibre. Its generic pencil has n distinct rank- $(n - 1)$ spectral points, so defines a point of G° .

Theorem F.2 (A transverse corank-two tail and its power maps). *The closure of the generic lifted curve of (F.1) in $\text{CQ}(V) \times D$ is*

$$Z = \text{Bl}_o X. \quad (\text{F.2})$$

It is flat over D , and its special fibre is the reduced nodal union $C \cup F$, where $C \simeq \mathbb{P}^1$ is the lifted special pencil and $F \simeq \mathbb{P}^1$ is an exceptional tail. There are no embedded components. If H_q is the exterior-power hyperplane bundle of order q , then

$$(\deg_C H_q, \deg_F H_q) = \begin{cases} (q, 0), & 1 \leq q \leq n - 2, \\ (n - 2, 1), & q = n - 1. \end{cases} \quad (\text{F.3})$$

Thus the special fibre has the same Hilbert polynomial $\binom{n}{2}m + 1$ as the generic lifted line.

The tail lies in E_{n-2} , its two intersections with E_{n-1} are distinct and reduced, and its attachment to C is neither of those two points. For any $h \geq 1$, let L_p be the hyperplane bundle of the resolved p th power map. Then

$$\deg_F L_p = \begin{cases} 0, & p \leq h(n - 2), \\ \min\{b, 2h - b\}, & p = h(n - 2) + b, \quad 0 \leq b \leq 2h. \end{cases} \quad (\text{F.4})$$

When this degree is positive, the restricted map on F is the complete Veronese map of that degree into its linear span. Most directly, in the local coordinate $z = s/t$ at o ,

$$J_{h(n-2)+1}(A_\tau) = (z, \tau). \quad (\text{F.5})$$

Hence a single multiplication power ideal selects the blow-up centre and its exceptional component.

Proof. The rank is at least $n - 2$ on X . In a neighbourhood of o the last $n - 2$ diagonal entries are units; all other spectral points have rank $n - 1$ and are pairwise distinct. Consequently the extended minor ideals I_q are the unit ideal for $q \leq n - 2$, while

$$I_{n-1} = (z, z + \tau) = (z, \tau), \quad I_n = (uz(z + \tau))$$

near o , with u a unit. Away from o , I_{n-1} is a unit ideal; I_n is everywhere invertible. The simultaneous exterior-power graph on the integral surface X is therefore the blow-up of (z, τ) . This is an application of the graph-by-generators lemma, not a claim that arbitrary base change of a blow-up is a blow-up. Since the universal family of pencil lines embeds X into $\mathbb{P}(\text{Sym}^2 V) \times D$, this graph embeds into $\text{CQ}(V) \times D$ and is the required scheme-theoretic closure.

On the chart with $\tau = zv$, the special fibre has equation $zv = 0$; on the other chart, $z = \tau w$, it has equation $\tau = 0$. These charts prove flatness over the DVR, multiplicity one of both components, their transverse intersection, and absence of embedded components. The map on C is the unique lift of the special pencil. The q -minors of that pencil have degree q . They have no common factor for $q \leq n - 2$, and precisely one factor s for $q = n - 1$. This proves their degrees on C . On F , all lower maps are constant, and the $(n - 1)$ st exterior map is linear. This gives (F.3). For $L = \bigotimes H_q$, the degrees on C and F add to $\binom{n}{2}$. The normalization sequence of $C \cup F$ subtracts one point, so its Euler characteristic is one and its Hilbert polynomial is as stated. Thus it is the Hilbert limit used by $\hat{G}$; equivalently the generic morphism from D extends to $\hat{G}$ by properness, with this universal fibre.

Use homogeneous coordinates $[\alpha : \beta] = [z : \tau]$ on F . After removing the exceptional factor, the residual quadratic form on the two-dimensional kernel is $\text{diag}(\alpha, \alpha + \beta)$. This is a line in $\text{CQ}(\mathbb{C}^2) = \mathbb{P}(\text{Sym}^2 \mathbb{C}^2)$, and gives the claimed image of F in the fibre of E_{n-2} . Its determinant $\alpha(\alpha + \beta)$ has two simple zeros $[0 : 1]$ and $[-1 : 1]$, which are its intersections with E_{n-1} . The main component attaches at $[1 : 0]$, where this residual form is nondegenerate.

For clarity, the whole list of extended power ideals near o is

$$J_p = \begin{cases} \mathcal{O}, & p \leq h(n-2), \\ (z, \tau)^b, & p = h(n-2) + b, \quad 0 \leq b \leq h, \\ (z, \tau)^{2h-b} [z(z + \tau)]^{b-h}, & p = h(n-2) + b, \quad h \leq b \leq 2h, \end{cases} \quad (\text{F.6})$$

up to an invertible factor. These are actual ideals, by Theorem 4.1. In particular $b = 1$ gives (F.5) for every h .

It remains to check the linear system on the tail, not only its degree. On diagonal matrices the coefficient polynomials of a p th power span exactly the monomials of total degree p in the diagonal entries with each exponent at most h . For containment, use the torus-weight bound in Lemma 4.2; for the reverse inclusion, take the coefficients of the independent diagonal entries of X in $\det(X + uA)^h$. Thus, for $p = h(n-2) + b$, the leading normal coefficients use exponent h in each of the $n - 2$ invertible entries and span

$$\alpha^j(\alpha + \beta)^{b-j}, \quad \max\{0, b - h\} \leq j \leq \min\{b, h\}.$$

If $b \leq h$ these span every binary form of degree b . If $b > h$, remove the common factor $[\alpha(\alpha + \beta)]^{b-h}$; the remaining monomials span every binary form of degree $2h - b$. Lower powers have nonzero constant normal term and are constant on F . This proves the complete-linear-system assertion and (F.4). $\square$

Remark F.3 (What the failure family contributes). Along (F.1), the original finite failure algebras form the finite locally free family already constructed in Corollary I.7.5. Its pullback to F is constant; we do not count that pullback as a new interaction. Instead, its first relation canonically recovers the source and the pencil line, and the source-normalized envelope has the total-base ideal (F.5). Resolving that multiplication ideal creates F , and the other products prescribe all the linear systems (F.4) on it. A degeneration arc, not only its closed failure algebra, supplies this total-base ideal. Thus the calculation distinguishes information in the family from information in the special point and supplies an explicit modular Hilbert limit.

APPENDIX G. ORDINARY COLLISION STRATA AND THEIR HILBERT FIBRES

We compute simultaneous collisions of arbitrary multiplicities, and allow higher-order perturbations of the total family. Besides describing its reduced components, we determine every resolved power system and exhibit positive-dimensional families in a fibre of the pencil incidence modification. The calculation uses the ordinary power ideals on the two-dimensional total base.

G.1. A local ideal calculation.

Lemma G.1 (Distinct tangent factors). *Let $\ell_1, \dots, \ell_c$ be pairwise nonproportional linear forms in two variables. For $0 \leq j < c$, the products of j distinct ℓ_i span all binary forms of degree j . Consequently, if R is a regular local complex surface ring with maximal ideal $\mathfrak{n} = (z, \tau)$, and a symmetric $c \times c$ matrix S satisfies*

$$S = zC + \tau D + O(\mathfrak{n}^2), \quad C \text{ nonsingular}, \quad \det(\alpha C + \beta D) \text{ squarefree},$$

then $I_j(S) = \mathfrak{n}^j$ for $j < c$, while $I_c(S) = (\det S)$.

Proof. Choose $j + 1$ of the linear forms. Their products obtained by omitting one factor are linearly independent: evaluation at the zero of the omitted factor kills all other products and not that one. Their number is $j + 1$, the dimension of the required binary-form space. For $j = 0$ the assertion is immediate.

The endomorphism $C^{-1}D$ is self-adjoint for C and has distinct eigenvalues. Its eigenspaces are orthogonal and none is isotropic, since otherwise C would be degenerate. A constant congruence therefore makes $\alpha C + \beta D$ diagonal with distinct linear entries. Its j -minors span $\text{Sym}^j(\mathfrak{n}/\mathfrak{n}^2)$. The entries of S lie in $\mathfrak{n}$, so $I_j(S) \subset \mathfrak{n}^j$ and $I_j(S) + \mathfrak{n}^{j+1} = \mathfrak{n}^j$. Nakayama's lemma applied to $\mathfrak{n}^j/I_j(S)$ proves equality. The argument may be made after completion and descended by faithful flatness. The determinant assertion is its definition. $\square$

Lemma G.2 (Block products and the first normal term). *For $V = U \oplus W$, multiplication gives an injection*

$$B_h(U) \otimes B_h(W) \longrightarrow B_h(V).$$

If $\dim U = r$, $Q \in \text{Sym}^2 U$ is nonsingular, and $S \in \mathfrak{n} \otimes \text{Sym}^2 W$ has a generically nonsingular linear term S_1 , the first nonzero normal term of $(Q + S)^p$, for $hr < p \leq h \dim V$, is

$$\binom{p}{hr} Q^{hr} S_1^{p-hr}.$$

For $p \leq hr$ its order-zero term Q^p is nonzero. These statements remain valid with a nonsingular formal unit block and after a formal congruence, with the corresponding invertible changes of coefficients.

Proof. The Cartan relations on each summand vanish in $B_h(V)$, so the algebra map is defined. Its source is Artin Gorenstein. Its socle is the product of the two socles, and that product is nonzero in $B_h(V)$: in determinant apolarity this is the nonzero mixed top derivative of $\det(X_U)^h \det(X_W)^h$, obtained by restricting to block diagonal matrices. A nonzero ideal in an Artin local Gorenstein algebra meets its socle: successive multiplication of a nonzero ideal by the maximal ideal ends in a nonzero subspace annihilated by that maximal ideal. The kernel is therefore zero.

Expand $(Q + S)^p$ in the injected tensor product. Powers of Q above hr vanish, whereas Q^{hr} is a nonzero socle generator. For $p > hr$ the term with the fewest factors of S consequently has precisely $p - hr$ such factors and is the displayed term. The unit block and a formal change of frame specialize to invertible changes on the associated graded coefficient space. This proves the last assertion as well. $\square$

G.2. The simultaneous collision theorem. Let $D_0 = \text{Spec } \mathbb{C}[[\tau]]$, let $\mathcal{R}$ be a rank-two pencil subbundle of $\text{Sym}^2 V \otimes_{D_0}$, and put $X = \mathbb{P}(\mathcal{R})$. Assume its generic pencil has n distinct spectral points. Its special pencil is assumed regular. Apart from simple spectral points, suppose it has distinct points $x_1, \dots, x_k$ of coranks c_a , where $2 \leq c_a \leq n - 1$. Write $r_a = n - c_a$.

At x_a , choose a parameter z_a on the special pencil and a unit $r_a \times r_a$ block. Its symmetric Schur complement is required to satisfy

$$S_a = z_a C_a + \tau D_a + O((z_a, \tau)^2), \quad C_a \text{ nonsingular}, \quad \det(\alpha C_a + \beta D_a) \text{ has } c_a \text{ simple zeros.} \quad (\text{G.1})$$

We call this an ordinary transverse cluster. Intrinsically, $C_a, D_a \in \text{Sym}^2 W_a$, where $W_a = (\ker(A(x_a) : V^* \rightarrow V))^*$. Their pencil is the first derivative of the form restricted to its radical. Taking a Schur complement does not alter that first derivative. Replacing the parameter changes the basis of this residual pencil and preserves its marked member $[C_a]$. Thus the hypothesis does not depend on the chosen complement or frame.

Theorem G.3 (Simultaneous ordinary collision fibres). *Under (G.1), the schematic closure of the generic lifted pencil in $\text{CQ}(V) \times D_0$ is*

$$Z = \text{Bl}_{\{x_1, \dots, x_k\}} X. \quad (\text{G.2})$$

It is regular and flat over D_0 . Its special fibre is the reduced nodal tree

$$C \cup F_1 \cup \dots \cup F_k,$$

where C is the lift of the special pencil and each $F_a \simeq \mathbb{P}^1$ meets C in one point. There are no embedded components and the tails are pairwise disjoint. The tail F_a is the complete-quadric lift of the pointed residual pencil $([\alpha C_a + \beta D_a], [C_a])$ on W_a . It lies in E_{r_a} , meets E_{n-1} in c_a distinct reduced points, meets no other boundary divisor, and attaches at the nonsingular member $[C_a]$.

Let H_q be the hyperplane bundle of the resolved q th exterior map, $1 \leq q \leq n-1$. Then

$$\deg_{F_a} H_q = (q - r_a)_+, \quad \deg_C H_q = q - \sum_a (q - r_a)_+. \quad (\text{G.3})$$

For the product polarization $\bigotimes_{q=1}^{n-1} H_q$ the Hilbert polynomial is $\binom{n}{2}m + 1$. In particular $\deg F_a = \binom{c_a}{2}$ and $\deg C = \binom{n}{2} - \sum_a \binom{c_a}{2}$.

For every $h \geq 1$ and $1 \leq p \leq hn$, set

$$b_a = (p - hr_a)_+, \quad s_p = (p - h(n-1))_+, \quad \delta_{a,p} = b_a - c_a s_p. \quad (\text{G.4})$$

These integers are nonnegative. The resolved p th power restricts to the complete linear system of degree $\delta_{a,p}$ on F_a . It is constant if that degree is zero, and otherwise the complete Veronese map into its linear span. Near x_a , with $\mathbf{n}_a = (z_a, \tau)$ and $f_a = \det S_a$, the exact ideals are

$$J_p(A) = \mathbf{n}_a^{\delta_{a,p}} (f_a)^{s_p}, \quad J_{hr_a+1}(A) = \mathbf{n}_a. \quad (\text{G.5})$$

Equalities are up to multiplication by a unit after a local frame is fixed, not up to radical or integral closure. Consequently the multiplication powers select the graph modification and every tail linear system. This description depends only on the first normal jet in (G.1); with the special pencil fixed, higher-order terms do not change its embedded special fibre or its power maps.

Proof. Near x_a the unit block and its Schur complement give $I_q(A) = \mathcal{O}$ for $q \leq r_a$ and $I_{r_a+j}(A) = I_j(S_a)$. Lemma G.1 therefore gives

$$I_{r_a+j}(A) = \mathbf{n}_a^j \quad (1 \leq j < c_a), \quad I_n(A) = (u_a f_a), \quad u_a \in \mathcal{O}^\times.$$

At every other point the rank is at least $n-1$, so all minor ideals of order at most $n-1$ are units. The determinant ideal is invertible. The simultaneous exterior graph is the blow-up of the product of its base ideals by Lemma 2.2. Near x_a this product is $\mathbf{n}_a^{\binom{c_a}{2}}$. A positive power has the same blow-up as the ideal itself, and the points are disjoint. These local identifications glue uniquely over X to (G.2). This computes the graph on the total surface; it is not an assertion that graph closure commutes with arbitrary base change.

The tautological family of genuine pencil lines embeds X into $\mathbb{P}(\mathrm{Sym}^2 V) \times D_0$. Hence the simultaneous graph embedding places Z in $\mathrm{CQ}(V) \times D_0$, with exactly the prescribed generic curve. The two charts $\tau = z_a v$ and $z_a = \tau w$ of each point blow-up show that Z is regular. In the first chart its special fibre is $z_a v = 0$. Both branches have multiplicity one and intersect transversally. In the second chart only the exceptional branch occurs. The parameter τ is a nonzerodivisor, so Z is flat over the DVR. The charts also rule out embedded components.

On F_a , divide the Schur block by the exceptional parameter. Its value is $\alpha C_a + \beta D_a$. All earlier unit-block data specialize to their values at x_a , independently of the direction. The complete-quadric chart thus identifies F_a with the lifted residual pencil in the fibre over $[A(x_a)]$. Its only residual rank drops have corank one and occur at the c_a simple determinant zeros. This proves the boundary assertions, including the attachment at $[\alpha : \beta] = [1 : 0]$. For $q = r_a + j < n$, the leading q -minors are the j -minors of this residual pencil. They have no common zero and have degree j . The lower maps are constant. On C , the common factor in the q -minors has order $(q - r_a)_+$ at x_a and no other zeros. This proves (G.3). The sum of the component degrees is $\binom{n}{2}$. The normalization sequence for a tree of $k + 1$ projective lines subtracts its k nodes, so its Euler characteristic is one. This proves the Hilbert polynomial and identifies the curve as the fibre in the Hilbert incidence space.

For the power ideals apply Theorem 4.1. If $p \leq hr_a$, the ideal is the unit ideal. Otherwise write $p = hr_a + hj + s$ with $0 \leq s < h$. For $j < c_a - 1$, the two minor ideals are $\mathfrak{n}_a^j$ and $\mathfrak{n}_a^{j+1}$, so the power ideal is $\mathfrak{n}_a^{hj+s}$. For $j = c_a - 1$ it is

$$\mathfrak{n}_a^{(c_a-1)(h-s)}(f_a)^s.$$

At $p = hn$ it is $(f_a)^h$. These cases are exactly (G.5). The value $p = hr_a + 1$ gives the point ideal for every h .

It remains to identify the complete coefficient space on F_a . By Lemma G.2, its first normal coefficients, for $p > hr_a$, are exactly those of the b_a th power of the residual pencil, multiplied by a nonzero unit-block socle vector. In the polynomial ring $\mathbb{C}[\alpha, \beta]$, Lemma G.1 gives $I_j = (\alpha, \beta)^j$ for $j < c_a$. The universal power identity gives the homogeneous ideal

$$(\alpha, \beta)^{\delta_{a,p}} \det(\alpha C_a + \beta D_a)^{s_p}.$$

All its original power generators have degree b_a ; therefore their span is its entire degree- b_a part. Remove the common determinant factor. The remaining space is exactly $H^0(\mathbb{P}^1, \mathcal{O}(\delta_{a,p}))$, proving the assertion about the linear system. This argument determines the map as well as its degree. All surviving leading coefficients use only the first normal jet; the higher-order terms vanish on the exceptional divisor. $\square$

G.3. A modular family over a fixed pencil. Let R_0 be a regular pencil whose spectral elementary divisors all have exponent one, with repeated spectral multiplicities $c_1, \dots, c_k \geq 2$. Thus these are semisimple collisions, not nontrivial Jordan blocks. At x_a keep the residual space W_a and the nonsingular tangent member $[C_a]$. Put

$$U_a = \{\ell \text{ a line through } [C_a] \text{ in } \mathbb{P}(\mathrm{Sym}^2 W_a) : \ell \text{ meets the determinant in } c_a \text{ simple points}\}. \quad (\text{G.6})$$

This is a nonempty open in $\mathbb{P}(\mathrm{Sym}^2 W_a / \langle C_a \rangle)$, of dimension $\binom{c_a+1}{2} - 2$.

Theorem G.4 (Residual-pencil moduli in an incidence fibre). *The product $U = \prod_a U_a$ parametrizes an algebraic flat family of distinct embedded Hilbert limits over the fixed pencil R_0 . Each is the fixed main curve with the tails prescribed by its residual lines, and each is realized by an ordinary transverse one-parameter smoothing of R_0 . Hence, for the normal incidence*

modification $\rho : \widehat{G} \rightarrow G$, one has

$$\dim \rho^{-1}(R_0) \geq \sum_a \left(\binom{c_a + 1}{2} - 2 \right). \quad (\text{G.7})$$

Distinctness is detected by the resolved multiplication power $J_{h(n-c_a)+1}$ on the a th tail. The statement describes a specified family in the fibre, not the entirety of the fibre or its normalization.

Proof. Since C_a is nonsingular, lines through $[C_a]$ are parametrized by the displayed projective quotient. The condition on determinant zeros is open and nonempty, as is seen by choosing D_a diagonal with distinct eigenvalues relative to C_a .

The complete-quadric morphism is an isomorphism over forms of rank at least $c_a - 1$. Each line in U_a stays in this open, and thus its universal line has an algebraic lift. The fibre of $\text{CQ}(V) \rightarrow \mathbb{P}(\text{Sym}^2 V)$ over $[A(x_a)]$ contains the residual $\text{CQ}(W_a)$: the nondegenerate first quotient form is fixed, and its radical carries the remaining complete form. This follows also by taking the unit block in the Schur charts used in the preceding proof. Thus the universal lifted lines give an algebraic family of embedded tails, each passing through the fixed attachment corresponding to $[C_a]$.

Adjoin the fixed main curve. The scheme-theoretic intersection of each tail family with that main curve is just this reduced section. Indeed the main curve has contact order one with E_{r_a} , since its local nonzero Smith exponents are 0 and 1, whereas the tail lies inside that divisor. The tails lie over different points, hence do not intersect each other. The union exact sequence

$$0 \longrightarrow \mathcal{O}_{C \cup \bigcup F_a} \longrightarrow \mathcal{O}_{C \times U} \oplus \bigoplus_a \mathcal{O}_{F_a} \longrightarrow \bigoplus_a \mathcal{O}_{\{\text{attachment}_a\} \times U} \longrightarrow 0$$

proves flatness over U : the two terms on the right are flat, so its kernel is flat. The degrees from (G.3) give the required fixed Hilbert polynomial. We therefore obtain a morphism U to the Hilbert scheme. The embedded fibre over $[A(x_a)]$ recovers its residual tail. Its first nonconstant exterior map, equivalently the power of index $h(n - c_a) + 1$, has degree one and recovers the line ℓ itself. Distinct parameters give distinct embedded curves.

To see that these Hilbert points belong to the graph closure defining $\widehat{G}$, choose a nonsingular member of R_0 . Semisimplicity makes its spectral eigenspaces orthogonal, so in a congruence frame the pencil is a direct sum of blocks $(z - x_a)C_a$ and simple blocks. Choose any representative D_a of each residual line and replace these blocks by $(z - x_a)C_a + \tau D_a$. Each cluster splits into c_a simple spectral points and distinct clusters stay separated. Theorem G.3 identifies the Hilbert limit with the constructed curve. This realizes every parameter by a smoothing, independently on all the blocks.

Let $\overline{G}$ denote the reduced Hilbert graph before normalization. The morphism from the smooth, reduced scheme U factors through its fibre over R_0 : the defining equations vanish at every closed point by the smoothing construction, hence vanish on U . Its geometric fibres are zero-dimensional by the distinctness just proved; its image consequently has dimension $\dim U$. Normalization $\widehat{G} \rightarrow \overline{G}$ is finite and surjective, and the preimage of this image has the same dimension. This gives (G.7). We use this finite preimage, rather than asserting that a nondominant map from a normal scheme must lift to a normalization. Such an assertion would not be a valid normalization universal property. $\square$

Example G.5 (Triple collision). For one semisimple triple collision, $c = 3$, the tail has exterior multidegree $(0, \dots, 0, 1, 2)$ and total degree three. At $h = 2$ its successive residual powers $b = 0, \dots, 6$ have degrees

$$0, 1, 2, 3, 4, 2, 0.$$

The residual lines through a fixed nonsingular ternary quadratic form form an open in $\mathbb{P}^4$, and give a four-dimensional family of distinct Hilbert limits over the same special pencil. Every

member has three distinct spectral points on its tail and one nonspectral attachment. This is a family of embedded limits, not just a count of branches of a normalization fibre.

The special finite failure algebra remains A_{R_0} throughout this family of limits. What varies is the total-family first-relation geometry: its intrinsic envelope produces the maximal-ideal centre, its successive products give the tail systems, and their degree-one member recovers the residual line. Thus the family supplies information not contained in its closed algebra alone. The construction is equivariant under source congruence and under reparametrization of the pencil, but the dimension bound concerns the embedded incidence space, not a quotient by its stabilizer.

APPENDIX H. SINGULARITIES OF THE UNIVERSAL POWER-ZERO SCHEMES

There is an explicit further consequence for all the universal ideals, which we record separately from the geometry of the full reciprocal normalization fibre. Multiplier ideals are taken on the affine smooth space M , and $I_{r+1}^{(a)}$ denotes symbolic power, with exponent zero interpreted as the unit ideal.

Corollary H.1 (Multiplier ideals and threshold). *For $1 \leq p \leq nh$, put $w_r = (p - hr)_+$ and $c_r = \binom{n-r+1}{2}$ for $0 \leq r < n$. For every real $b \geq 0$,*

$$\mathcal{J}(M, b \cdot J_p) = \bigcap_{r=0}^{n-1} I_{r+1}^{(\max\{0, \lfloor bw_r \rfloor + 1 - c_r\})}, \quad (\text{H.1})$$

$$\text{lct}_M(J_p) = \min_{\substack{0 \leq r < n \\ p > hr}} \frac{\binom{n-r+1}{2}}{p - hr}. \quad (\text{H.2})$$

Proof. Blow up the affine origin and then the strict rank loci. This is the affine complete-quadric resolution, whose charts are (5.4) with one additional scalar coordinate. Let E_0 be the divisor over the origin. The same minor calculation now includes $r = 0$, and the total transform of J_p is $\sum_{r=0}^{n-1} w_r E_r$. The discrepancy along E_r is $c_r - 1$: over the generic rank- r point the earlier blow-ups are isomorphisms, and the smooth rank- r locus has codimension c_r ; for $r = n - 1$ this is the strict determinant divisor and the discrepancy is zero. All boundary divisors have normal crossings, so this is a log resolution.

The defining pushforward formula for the multiplier ideal requires $\text{ord}_{E_r}(f) \geq \lfloor bw_r \rfloor + 1 - c_r$. At the generic rank- r point this valuation is the order in the regular local ring with maximal ideal I_{r+1} , so the condition on polynomials is exactly membership in the indicated symbolic power. Intersecting the conditions proves the first formula; the first value at which one becomes nontrivial proves the second. Equivalently this is the specialization, via Theorem 4.1, of the classical symmetric-determinantal multiplier-ideal calculation [32, §2.4 and Theorem 4.8]. $\square$

The ideal formula, graph identification and contact reconstruction are statements about the entire universal coefficient space. They do not supply the conductor of every multivariate common-divisor image or the Hilbert function of every full $F_{g,k}(W)$. Those objects remain in the preserved incidence theory; the present results connect their exact quadratic sections to a native global boundary and its spectral schemes.

APPENDIX I. SPECTRAL DIVISOR FIBRES AND THE RECOVERY OF DEGENERATION TYPE

The preceding completion concerns the native congruence action. To retain degeneration type before passage to a coarse quotient, we associate a tower of binary divisor incidences to a regular pencil. This construction is intrinsic after the unmarked inverse and is unchanged by a simultaneous change of the pencil and source frames.

Let $\Lambda = \mathbb{P}(R)$ be the pencil line and let $\mathcal{T}_R$ be the cokernel of the universal symmetric map on Λ . A regular pencil means that its determinant is not identically zero. Thus $\mathcal{T}_R$ is a torsion sheaf. Define $D_j(R)$ by

$$\mathcal{I}_{D_j(R)} = \text{Fitt}_j(\mathcal{T}_R), \quad 0 \leq j \leq n, \quad (\text{I.1})$$

where $D_n = \emptyset$. These are effective divisors on the smooth curve Λ . Write $s_j = \deg D_j$. If $s_j > 0$, let G_j be its binary equation and put

$$J_j = G_j H^0(\Lambda, \mathcal{O}_\Lambda(1)) \subset H^0(\Lambda, \mathcal{O}_\Lambda(s_j + 1)).$$

Changing the scalar representative of G_j does not change J_j . Let $\mathcal{F}_j(R)$ be the whole scheme fibre at J_j of the degree-one divisor normalization ν_1 , with $r = 2$ and $D = s_j + 1$. For $s_j = 0$ put $\mathcal{F}_j = \emptyset$.

Theorem I.1 (A fibre-theoretic Segre invariant). *There is a natural identification of finite schemes over Λ*

$$\mathcal{F}_j(R) \simeq D_j(R). \quad (\text{I.2})$$

At a spectral point x , write the positive elementary divisors of $\mathcal{T}_R$ as $\lambda_{x,1} \geq \dots \geq \lambda_{x,r_x} > 0$, and extend this sequence by zeroes. Then

$$(\mathcal{F}_j(R))_x = \text{Spec } \mathbb{C}[\epsilon]/(\epsilon^{b_{x,j}}), \quad b_{x,j} = \sum_{i>j} \lambda_{x,i}, \quad (\text{I.3})$$

with the empty-scheme convention when $b_{x,j} = 0$. Consequently

$$\lambda_{x,j+1} = b_{x,j} - b_{x,j+1}. \quad (\text{I.4})$$

The tower, with its maps to the same intrinsic spectral line, therefore determines the full Segre symbol. Conversely, the elementary-divisor partitions together with their spectral support on that line determine the tower. The tower retains the projective configuration of the support as well as the Segre symbol. It is an invariant of the unmarked finite failure algebra.

The construction is compatible with base change on every family stratum where the Fitting ideals in (I.1) define relative effective Cartier divisors of fixed degrees. On such a stratum $\mathcal{F}_j$ is finite locally free of degree s_j . These hypotheses are part of the relative statement; no assertion of flatness across a jump of s_j is intended.

Proof. Over the discrete valuation ring $\mathcal{O}_{\Lambda,x}$, Smith normal form gives diagonal entries with exponents $\lambda_{x,r_x}, \dots, \lambda_{x,1}$ and $n - r_x$ zero exponents. The minors of size $n - j$ have greatest common divisor of order $\sum_{i>j} \lambda_{x,i}$. This proves the formula for D_j , including its nonreduced structure. It is the usual Fitting-ideal description of elementary divisors, used here on the intrinsic spectral line.

The residual two-space $H^0(\mathcal{O}_\Lambda(1))$ is gcd-free. Hence Theorem M.1 identifies the degree-one normalization fibre at J_j with the scheme of linear factors of G_j . On an affine chart write a monic linear factor as $z - a$. Monic division gives $G_j = (z - a)H + G_j(a)$, so the entire factor scheme is $G_j(a) = 0$, not its reduction. These descriptions agree on overlaps of the projective line. They give (I.2) and the local algebra (I.3). Taking successive differences proves (I.4). The common map to Λ is essential: forgetting it separately in each member of the tower would discard the alignment of the spectral supports.

Pencil frame changes act on Λ by projective transformations; source congruence changes give isomorphic cokernels and preserve all Fitting ideals. The intrinsic inverse of Theorem I.8.3 therefore defines this invariant from the unmarked algebra. The equivalence with Segre data uses the classical elementary-divisor convention of [4, §5].

For a family with the stated Cartier and degree conditions, the same monic division argument works over any base algebra, including one with nilpotents. It represents the divisor's functor of points, so it commutes with base change. A relative effective divisor of fixed degree on a

projective-line bundle is finite locally free of that degree. Local choices of G_j differ by units and glue. Formation of Fitting ideals itself commutes with base change; requiring them to remain flat Cartier divisors is precisely what permits the additional finite-flat conclusion. $\square$

Example I.2 (Equal discriminant, different fibre towers). At a root of multiplicity four, the partitions $(3, 1)$ and $(2, 2)$ have the same zeroth fibre $\mathbb{C}[\epsilon]/(\epsilon^4)$. Their first fibres are respectively $\mathbb{C}$ and $\mathbb{C}[\epsilon]/(\epsilon^2)$; all higher fibres are empty there. Thus the discriminant alone does not detect the distinction, whereas the divisor-fibre tower does. Both are realized by regular symmetric pencils: for a block of size l , take the reversal matrix H_l and the symmetric pair $(H_l, H_l J_l(\alpha))$, where $J_l(\alpha)$ is a Jordan block. Direct sums realize every indicated partition. Additional simple blocks may be appended. The example is a statement on the regular-pencil stack, not a claim that distinct Segre strata always remain distinct after strictly semistable GIT identification.

Remark I.3 (The two divisor constructions). The primitive first relation of an actual pencil has exact common factor $(\det T)^{n-1}$, by Lemma I.J.1. Its common-factor degree does not jump when the pencil changes Segre type. The divisors D_j above instead lie on the recovered spectral line and define auxiliary binary first-relation subspaces J_j . Thus (I.4) relates pencil data to intrinsically induced divisor ramification, not to a nonexistent change in the conductor of the original exact-gcd locus. The identification supplies a moduli consequence of the all-binary-fibre theorem while respecting this distinction.

APPENDIX J. COMMON-DIVISOR STRATA AND INTRINSIC FAMILIES

The coefficient condition in Theorem I.G.1 is not imposed in this section. We instead retain the common divisor of a space of first relations. This geometric condition gives a larger smooth parameter space, includes relations without either factor symmetry, and has a scheme-theoretic inverse. The distinction between a stratum and the condition on its geometric points is essential when the parameter base is nonreduced.

Write $S_j(E) = \text{Sym}^j E$, where E is a complex vector space. For $r \geq 2$, denote by

$$X_{h,r}(E) \subset \text{Gr}(r, S_h(E))$$

the locus of subspaces with greatest common divisor equal to 1, up to a scalar. Empty Grassmannians are omitted throughout. When $h = 0$ this notation can also be used for $r = 1$.

Lemma J.1 (Universal common divisor). *Let $D = g + h$. The spaces of r degree- D forms whose exact greatest common divisor has degree g form a locally closed reduced subscheme Z_g of $\text{Gr}(r, S_D(E))$. Multiplication induces an isomorphism*

$$\mathbb{P}(S_g(E)) \times X_{h,r}(E) \xrightarrow{\sim} Z_g, \quad ([f], J) \mapsto fJ. \quad (\text{J.1})$$

In particular, the common-divisor line and the residual subbundle of a family with classifying map in Z_g are recovered uniquely and commute with arbitrary complex base change.

Proof. For each a , multiplication from $\mathbb{P}(S_a(E)) \times \text{Gr}(r, S_{D-a}(E))$ has closed image: its domain is projective. The union for $a > g$ is closed. Remove this union from the image for $a = g$, and give the resulting locally closed set its reduced structure. Its points are exactly those in the statement. The inverse image of this set under multiplication is $\mathbb{P}(S_g(E)) \times X_{h,r}(E)$. The same closed-image argument with $D = h$ shows that $X_{h,r}(E)$ is open.

The restricted multiplication map is proper, by base change. It is injective on geometric points over every algebraically closed extension of $\mathbb{C}$, since a greatest common divisor is unique up to scalar in a polynomial ring. Its differential is injective as well. Indeed, a tangent vector in its kernel satisfies

$$\dot{f}j + f\dot{j} \in fJ \quad (j \in J).$$

Consequently $f \mid \dot{f}j$ for every $j \in J$. For each irreducible factor of f , some element of J is not divisible by that factor. Comparison of its valuation gives $f \mid \dot{f}$. Equality of degrees makes $\dot{f}$ a scalar multiple of f ; then $\dot{J} = 0$ in $\text{Hom}(J, S_h(E)/J)$.

In characteristic zero the geometric-point injectivity and this differential calculation imply that the map is universally injective and unramified. A proper, universally injective, unramified morphism is a closed immersion [12, Lemma 41.7.2]. Here its image is all of the reduced target. The defining ideal of a closed immersion with this support is zero in a reduced scheme. This proves the isomorphism, including its assertion for nonreduced test schemes. Equivalently, one can check properness on the base change of the original projective multiplication map; that base change has the same underlying open preimage. Its source is reduced because it is this open subscheme of the original smooth product. $\square$

Remark J.2. A map from a nonreduced scheme is required to factor through the locally closed scheme Z_g , not merely to have geometric points in its underlying set. These are different requirements. For example, over $\mathbb{C}[\epsilon]/(\epsilon^2)$ the span of $x^2 + \epsilon y^2, xy$ specializes to a pair with common divisor x , but has no factor $x + \epsilon(ax + by)$. Comparison of the y^2 coefficient in the first form proves this. The stratum in Lemma J.1 excludes this transverse infinitesimal family. No use of a fibrewise radical can replace its scheme structure.

Now let $\dim V = \dim U = n \geq 3$, $E = V^* \otimes U$, and $\delta = \det T$. Denote by $G \subset \text{GL}(E)$ the stabilizer of the line $\mathbb{C}\delta$. By Lemma I.4.1,

$$G^\circ = (\text{GL}(V) \times \text{GL}(U))/\mathbb{G}_m, \quad G/G^\circ = \mathbf{Z}/2\mathbf{Z}. \quad (\text{J.2})$$

The second component exchanges the two tensor factors; the diagonal scalars in the displayed product act trivially on the matrix space. Fix $s \geq 1$ and $D = sn + h$. Inside Z_{sn} , impose the condition that the common-divisor line is the s -th power of a determinant form after a linear change of the n^2 variables. Call the resulting locally closed scheme $Z_{s,h,r}^{\det}$.

Theorem J.3 (Intrinsic determinant-divisor stratum). *There are canonical isomorphisms of schemes and algebraic stacks*

$$Z_{s,h,r}^{\det} \simeq \text{GL}(E) \times^G X_{h,r}(E), \quad [Z_{s,h,r}^{\det} / \text{GL}(E)] \simeq [X_{h,r}(E)/G]. \quad (\text{J.3})$$

They hold over arbitrary complex parameter schemes. The stack is smooth. At a residual relation J , its tangent complex, in degrees $-1, 0$, is

$$[\text{Lie}(G) \longrightarrow \text{Hom}(J, S_h(E)/J)], \quad X \longmapsto (j \mapsto Xj \bmod J). \quad (\text{J.4})$$

No invariance of J under either full factor group is assumed. The equivalence identifies stabilizers, infinitesimal families, and specializations whose classifying maps remain in this stratum.

Proof. The power map $\mathbb{P}(S_n(E)) \rightarrow \mathbb{P}(S_{sn}(E))$, $[f] \mapsto [f^s]$, is a closed immersion in characteristic zero. It is proper and geometrically injective, and its differential has kernel given by $sf^{s-1}\dot{f} \in \mathbb{C}f^s$, hence by $\dot{f} \in \mathbb{C}f$. The same closed-immersion criterion as above applies. The orbit of $[\delta]$ is the smooth locally closed homogeneous space $\text{GL}(E)/G$. The preimage of its power orbit under the first projection in (J.1) is therefore a locally closed subscheme. It is exactly $Z_{s,h,r}^{\det}$.

Pulling back along the G -torsor $\text{GL}(E) \rightarrow \text{GL}(E)/G$ trivializes the common-divisor form, leaving the unrestricted residual relation $J \in X_{h,r}(E)$. Descent gives the first isomorphism in (J.3); descent of equivariant maps and morphisms gives the second. This proof uses morphisms of schemes, not a comparison only on complex points. It also supplies local lifts of all isomorphisms over an fppf cover.

The space $X_{h,r}(E)$ is open in a Grassmannian and G is smooth. The quotient is algebraic and smooth [13, Theorem 97.17.2]; see also [14, Lemma 101.33.7]. For completeness, write a first-order residual subspace as the graph of $\epsilon\phi : J \rightarrow S_h(E)/J$. An infinitesimal change

$1 + \epsilon X$ replaces ϕ by $\phi + X|_J \bmod J$. This proves the tangent complex. Its kernel records infinitesimal automorphisms and its cokernel records first-order isomorphism classes. The same torsor description proves compatibility with specialization. No assertion is made about limits outside the specified common-divisor stratum. $\square$

The next step connects these parameter spaces with ungraded finite algebras. It is useful to state the family condition intrinsically, since an absolute nilradical is unsuitable over a nonreduced base. Put $e = \dim E$ and $\ell = \binom{e+D}{D} - r$. Consider finite locally free commutative unital algebras $\mathcal{A}$ of rank ℓ on a complex scheme T_0 , subject to the following conditions. The map $\epsilon(a) = \ell^{-1} \operatorname{tr}(M_a)$ is multiplicative. For $\mathcal{N} = \ker \epsilon$, one has $\mathcal{N}^{D+1} = 0$; all quotients of its power filtration are locally free; $\mathcal{E} = \mathcal{N}/\mathcal{N}^2$ has rank e ; and multiplication gives isomorphisms $\operatorname{Sym}^j \mathcal{E} \simeq \mathcal{N}^j/\mathcal{N}^{j+1}$ for $j < D$. In degree D its kernel has rank r . These conditions specify the maximal-lower-Hilbert-function stratum; they do not assert that arbitrary smoothings of a local algebra remain in it.

Proposition J.4 (Relative first-relation algebra). *Every such algebra is locally isomorphic, over its base, to*

$$\mathcal{A}(\mathcal{K}) = \operatorname{Sym} \mathcal{E} / ((\mathcal{K}) + \mathfrak{m}^{D+1}), \quad \mathcal{K} = \ker(\operatorname{Sym}^D \mathcal{E} \longrightarrow \mathcal{N}^D).$$

Its trace augmentation, $\mathcal{E}$, and $\mathcal{K}$ are intrinsic and compatible with arbitrary base change in this stratum. For two such algebras, taking the linear part gives a locally surjective map of isomorphism sheaves to the sheaf of isomorphisms of $(\mathcal{E}, \mathcal{K})$. Its kernel is the smooth substitution group whose underlying scheme in a local presentation is $\operatorname{Hom}(\mathcal{E}, \mathcal{N}^2)$.

After quotienting morphisms by this tangent-identity kernel and fppf stackifying, the algebras with first-relation map in $Z_{s,h,r}^{\det}$ have stack $[X_{h,r}(E)/G]$.

Proof. Locally lift a basis of $\mathcal{E}$ to $\mathcal{N}$. Nilpotence and multiplication on the successive quotients make the induced map from the truncated symmetric algebra surjective. A relation with a nonzero term of least degree $j < D$ would give a nonzero kernel in degree j of the associated graded algebra, which is impossible. Its entire kernel therefore lies in degree D and is precisely $\mathcal{K}$. Changing a lift by an element of $\mathcal{N}^2$ does not alter a product of D generators, since the difference has degree at least $D + 1$. This also proves the asserted uniqueness of the first relation.

In this presentation multiplication by a positive-degree element is strictly filtration-increasing. Its trace is zero over any base ring. Multiplication by a scalar has trace ℓ times that scalar, proving that the presentation's augmentation is the specified intrinsic trace augmentation. All filtration sequences are locally split sequences of vector bundles, so their kernels and quotients commute with base change.

An isomorphism must preserve trace and induces a linear map carrying $\mathcal{K}$ to $\mathcal{K}'$. Conversely such a map lifts locally to an isomorphism by the homogeneous presentations. All other lifts are obtained by substituting generators plus arbitrary elements of $\mathcal{N}^2$; the inverse is constructed degree by degree in a finite filtration. This gives the stated kernel and its smoothness, without replacing its substitution law by addition. After identifying morphisms with the same linear part, frames of $\mathcal{E}$ give the quotient $[Z_{s,h,r}^{\det}/\operatorname{GL}(E)]$. Theorem J.3 completes the proof. $\square$

Remark J.5. The original algebra stack is not this quotient stack: it has the additional nonlinear inertia just computed. The phrase *effective first-relation stack* will refer to the specified quotient on morphisms followed by stackification. The construction retains the full automorphism group before taking that quotient. It does not identify algebras with the same geometric fibres, nor does it discard nilpotent parameters. In particular, its relation to a stratum of first relations is stronger than a bijection of isomorphism classes over $\mathbb{C}$.

APPENDIX K. THE COMMON-DIVISOR BOUNDARY AND ITS NORMALIZATION

The exact-common-divisor stratum has a canonical completion which need not be normal. Its normalization records a divisor, and its failure to be an isomorphism is controlled by two different phenomena: choices of that divisor and overlap with the residual common factor. We establish these assertions for arbitrary spaces of forms. No determinant, Cauchy, or Pluecker equation is imposed in this section.

Let E have dimension $e \geq 2$, put $S_j = \text{Sym}^j E$ and $s_j = \dim S_j = \binom{e+j-1}{j}$, and fix

$$g, h \geq 1, \quad D = g + h, \quad 2 \leq r \leq s_h.$$

Write $\mathcal{G} = \text{Gr}(r, S_D)$. A *degree- g divisor incidence* over a complex scheme T_0 consists of a line subbundle $\mathcal{L} \subset S_g \otimes \mathcal{O}_{T_0}$ and a rank- r subbundle $\mathcal{K} \subset S_D \otimes \mathcal{O}_{T_0}$ such that

$$\mathcal{K} \longrightarrow (S_D \otimes \mathcal{O}_{T_0})/(\mathcal{L} \cdot S_h) = 0. \quad (\text{K.1})$$

Subbundle always means that the quotient is locally free. In particular, the form generating $\mathcal{L}$ is nonzero on every geometric fibre. Condition (K.1) is the vanishing of a bundle map, not a condition imposed only on closed points.

Proposition K.1 (Canonical divisor incidence). *The incidence functor is represented, with its full scheme structure, by*

$$\mathcal{X}_g = \mathbb{P}(S_g) \times \text{Gr}(r, S_h).$$

Its forgetful morphism is multiplication

$$\nu_g : \mathcal{X}_g \longrightarrow \mathcal{G}, \quad ([f], J) \longmapsto fJ. \quad (\text{K.2})$$

This morphism is finite. Its scheme-theoretic image Y_g is an integral projective scheme. If K is a geometric point of Y_g and $F_K = \gcd(K)$, the underlying set of $\nu_g^{-1}(K)$ is

$$\{[f] : f \mid F_K, \deg f = g\}, \quad (\text{K.3})$$

with residual space $J = K/f$. The incidence construction and its scheme-theoretic fibres commute with arbitrary complex base change.

Proof. On $\mathbb{P}(S_g)$, multiplication from the tautological line tensored with S_h to S_D has constant rank s_h . Indeed, multiplication by a nonzero polynomial over a field is injective. A nonzero maximal minor locally splits this map, so its image and cokernel are vector bundles, also after a nonreduced base change. The closed incidence (K.1) is therefore the relative Grassmannian of rank- r subbundles of $\mathcal{L} \otimes S_h$. Tensoring by $\mathcal{L}^{-1}$ identifies it canonically with the stated product. In more elementary terms, $\mathcal{K}$ has a unique inverse image in $\mathcal{L} \otimes S_h$; its quotient there is locally free because the quotient of $S_D/\mathcal{K}$ onto $S_D/(\mathcal{L} \cdot S_h)$ is a surjection of vector bundles. This proves the functorial assertion, including its scheme structure.

The source of ν_g is projective. Unique factorization in the polynomial ring over an algebraically closed field gives (K.3). A fixed polynomial has only finitely many divisors of a fixed degree, up to scalar. The morphism is thus proper with finite fibres and is finite [15, Tag 02LS]. Define Y_g by the kernel of $\mathcal{O}_{\mathcal{G}} \rightarrow (\nu_g)_* \mathcal{O}_{\mathcal{X}_g}$, as in [16, Tag 01R5]. Its integrality follows from the integrality of $\mathcal{X}_g$: on an affine open meeting the image, the coordinate ring of the image injects into the domain which is the coordinate ring of its finite inverse image. Reducedness is therefore a consequence of this incidence construction, not an extra choice. Representability of incidence gives the base-change assertion. $\square$

Remark K.2. The condition defining Y_g is the existence of a divisor of *degree* g . In several variables it is not the condition $\deg \gcd(K) \geq g$: an irreducible polynomial of larger degree need not have a degree- g factor. Nor do we assert that formation of the scheme-theoretic image, or normalization itself, commutes with arbitrary nonflat base change. The assertion for arbitrary

bases in Proposition K.1 concerns the incidence functor and the base change of its representing morphism.

We next retain the nilpotents of a fibre. This will distinguish a ramified divisor choice from several distinct divisor choices.

Lemma K.3 (Fibre equations and the overlap space). *Let $K = fJ$, set $H = \gcd(J)$, and put $c = \gcd(f, H)$ and $k = \deg c$. Then the tangent space of the scheme-theoretic fibre at $([f], J)$ is naturally, after the displayed representatives have been chosen,*

$$\ker(d\nu_g)_{([f], J)} \simeq S_k/\mathbb{C}c, \quad \dim \ker(d\nu_g) = s_k - 1. \quad (\text{K.4})$$

More explicitly, choose a projective coefficient chart $f(u) = f + \sum u_\alpha m_\alpha$, with one coefficient normalized to its value in f . Let $M(f(u))$ be the matrix of multiplication $S_h \rightarrow S_D$. Choose a set A_0 of s_h rows for which $M_{A_0}(f)$ is invertible, and let B_0 denote the remaining rows. For a matrix C whose columns form a basis of K , the full fibre on this chart is defined by

$$C_{B_0} - M_{B_0}(f(u))M_{A_0}(f(u))^{-1}C_{A_0} = 0, \quad \det M_{A_0}(f(u)) \neq 0. \quad (\text{K.5})$$

In particular its completed local ring is the power-series ring in the $s_g - 1$ variables u_α , modulo the entries in (K.5); no radical is taken.

Proof. For each column of C , its possible quotient by $f(u)$ is forced by the A_0 rows to be $M_{A_0}^{-1}C_{A_0}$. The remaining rows are exactly the condition that this quotient works. These quotients span a subbundle wherever the equations hold, by the incidence argument above. Thus elimination has introduced neither additional solutions nor discarded nilpotents, proving the fibre presentation over arbitrary coefficient rings in the chart.

A tangent vector is a pair $(\dot{f} \bmod \mathbb{C}f, \dot{J})$. It lies in the fibre precisely when

$$\dot{f}j + f\dot{j} \in fJ \quad \text{for all } j \in J.$$

It is necessary and sufficient that $f \mid \dot{f}j$ for every $j \in J$; the class of $\dot{j}$ modulo J is then forced. Taking the minimum valuation over a basis of J , for each irreducible polynomial, makes this equivalent to $f \mid \dot{f}H$. Writing $f = cf_1$, $H = cH_1$, with $\gcd(f_1, H_1) = 1$, gives $\dot{f} = f_1w$ with $w \in S_k$. The scalar tangent direction $\dot{f} \in \mathbb{C}f$ is exactly $w \in \mathbb{C}c$. This proves (K.4) and its dimension assertion. $\square$

Theorem K.4 (Normalization and the exact normal locus). *The finite map*

$$\nu_g : \mathbb{P}(S_g) \times \text{Gr}(r, S_h) \longrightarrow Y_g$$

is the normalization of the natural degree- g divisor locus. Its dimension is

$$\dim Y_g = s_g - 1 + r(s_h - r). \quad (\text{K.6})$$

For a geometric point K , the scheme Y_g is normal at K if and only if F_K has exactly one degree- g divisor $[f]$ and

$$\gcd(f, F_K/f) = 1. \quad (\text{K.7})$$

Normality at a point is also equivalent to smoothness there. Thus the two precise obstructions are more than one divisor choice, or overlap between the selected divisor and the residual common factor. The latter gives the fibre tangent dimension in (K.4), and (K.5) determines its higher nilpotents.

Proof. The locus of gcd-free r -planes in S_h is nonempty: a space containing x_1^h, x_2^h is one example, and it can be extended to every allowed rank r . It is open, since the loci possessing a positive-degree divisor are the finitely many closed images of projective multiplication maps. It is therefore dense in the irreducible Grassmannian. Its product with $\mathbb{P}(S_g)$ maps to the locus where the exact gcd has degree g .

This last locus is open in Y_g : remove the finitely many closed images for divisor degrees $a > g$. Call it U_g . Every fibre over a geometric point of U_g has one point and, by Lemma K.3 with $H = 1$, zero tangent space. A finite local algebra over an algebraically closed field with zero cotangent space is the field itself, by Nakayama's lemma. The fibre consequently has length one, not merely one point.

Here is the local argument which also settles the scheme structure. At a closed point of U_g , write A for the local ring of Y_g , and B for its finite inverse-image algebra. The map $A \rightarrow B$ is injective, and $B/\mathfrak{m}_A B = \mathbb{C}$, generated by 1. Nakayama's lemma, applied to the finite module B/A , gives $B = A$. The non-isomorphism locus of a finite morphism is closed, so the same conclusion holds throughout U_g . Hence ν_g is birational. The source is smooth and integral, thus normal. On affine opens, a finite birational extension $A \subset B$ with B integrally closed is the integral closure of A in the common fraction field: every element integral over A is integral over B and hence belongs to B . This proves the normalization assertion; compare [17, Tag 035E]. The dimension follows from the product description.

If Y_g is normal at K , its normalization is an isomorphism there, so its fibre is one reduced point. Formula (K.3) gives uniqueness of $[f]$, and (K.4) gives $s_k - 1 = 0$, equivalent to $k = 0$ because $e \geq 2$. Conversely uniqueness and $k = 0$ again make the entire fibre a single reduced point. The same finite local Nakayama argument shows that the normalization is an isomorphism near K . Its source is normal, so Y_g is normal at K . The source is smooth, so every normal point of the target is smooth; the converse holds for every complex variety. This proves both directions without replacing a nonreduced fibre by its set of points. $\square$

Corollary K.5 (Universality of the exact stratum). *The open subscheme $U_g \subset Y_g$ represents the unmarked relation families in the divisor incidence whose residual space is gcd-free on every geometric fibre. Over it the divisor line and residual subbundle are unique and commute with arbitrary base change. Moreover U_g , as a locally closed subscheme of $\mathcal{G}$, is the reduced exact-gcd stratum of Lemma J.1.*

Proof. The inverse image of U_g is $\mathbb{P}(S_g) \times X_{h,r}(E)$, and the proof above shows that its finite map to U_g is an isomorphism. Its source is reduced, so U_g has the reduced induced structure on its locally closed support. The incidence functor supplies the asserted universal property. A family in the ambient Grassmannian which only has the right geometric gcd need not factor through Y_g . In particular, the transverse dual-number family of Remark J.2 is not admitted by this corollary. The corollary is a scheme-theoretic identification, not a claim that fibrewise gcd tests detect nilpotent equations. $\square$

Example K.6 (A nonreduced fibre at a divisor collision). Take $E = \langle x, y, z \rangle$, $g = 1, h = 2, r = 2$, and $K = \langle x^2 y, x^2 z \rangle$. Its gcd is x^2 , so the only point of the normalization fibre has divisor x . In the chart $f = x + ay + bz$, division by this monic linear form amounts to substitution $x = -ay - bz$. The two remainders are

$$(ay + bz)^2 y, \quad (ay + bz)^2 z.$$

Their coefficient ideal is (a^2, ab, b^2) . Thus the full fibre is

$$\text{Spec } \mathbb{C}[a, b]/(a, b)^2, \tag{K.8}$$

of length three and tangent dimension two. It is not a reduced point, despite uniqueness of the geometric divisor. The family

$$K_t = x \langle xy + ty^2, xz + tz^2 \rangle$$

has independent generators for every t , exact gcd x when $t \neq 0$, and the displayed collision at $t = 0$. Consequently this nonreduced fibre occurs on the boundary of the exact stratum, not in an unrelated example.

Example K.7 (Several branches and a normal gcd jump). For the same g, h, r, E , let $K = xy\langle x, z \rangle$. The divisors x and y give two points of the normalization fibre. At both points the overlap is one, so the fibre consists of two reduced points. The point of Y_1 is nonnormal. By contrast, put $q = x^2 + yz$ and $K = xq\langle y, z \rangle$, now with $h = 3$. The polynomial q is irreducible and is not divisible by x . There is exactly one linear divisor of $\gcd K = xq$, and its overlap with the residual $\gcd q$ is one. This point of Y_1 is normal although its gcd degree is three. These two examples show why neither gcd degree nor the number of geometric lifts alone is a sufficient normality criterion.

Corollary K.8 (An intrinsic boundary for first-relation algebras). *On the maximal-lower-Hilbert-function stratum of Proposition J.4, the algebras whose intrinsic degree- D kernel has classifying map in Y_g have, after removing only tangent-identity substitutions, stack $[Y_g/\mathrm{GL}(E)]$. Its normalization is the finite representable morphism*

$$[\mathcal{X}_g/\mathrm{GL}(E)] \longrightarrow [Y_g/\mathrm{GL}(E)]. \quad (\text{K.9})$$

All members of the universal relation family give finite locally free algebras of rank $\binom{e+D}{D} - r$. This construction imposes no pencil image equations and permits the divisor and residual common factor to collide.

Proof. For every subbundle $\mathcal{K}$ classified by the Grassmannian, $\mathrm{Sym} E/((\mathcal{K}) + \mathfrak{m}^{D+1})$ is, as a module, the sum of S_j for $j < D$ and $S_D/\mathcal{K}$. This proves finite local freeness and the rank. The first-relation proposition identifies the quotient after the specified nonlinear inertia is removed. Incidence and multiplication are $\mathrm{GL}(E)$ -equivariant. Pulling (K.9) back along the smooth frame atlas $Y_g \rightarrow [Y_g/\mathrm{GL}(E)]$ gives precisely ν_g . Finiteness, representability, and the normalization identification follow by descent; equivalently normality and integral closure may be checked on these smooth atlases. This quotient is not claimed to classify smoothings outside the specified Hilbert stratum. $\square$

For the pencil first relations one can take $e = n^2$, $g = n(n-1)$, $h = 3n-4$, and $r = \binom{n+1}{2}$. Every pencil then lies in the exact-gcd open of this canonical boundary. The boundary theorem concerns the much larger space of all degree- g divisorial first relations, rather than a closure defined by recovering a pencil and reimposing its Pluecker equations. It determines its normalization and normal locus, but does not assert a classification of all its linear orbits or of all finite-flat algebra deformations.

APPENDIX L. FORMAL IMAGES AND MARKED DIVISOR GERMS

All rings and schemes in the following sections are over $\mathbb{C}$. Write $S_j = \mathrm{Sym}^j E$, and let $Y_g \subset \mathrm{Gr}(r, S_D)$ be the scheme-theoretic image of

$$\nu_g : \mathbb{P}(S_g) \times \mathrm{Gr}(r, S_{D-g}) \longrightarrow \mathrm{Gr}(r, S_D), \quad (f, J) \longmapsto fJ,$$

where $\dim E \geq 2$, $g \geq 1$, and $2 \leq r \leq \dim S_{D-g}$. The source is smooth, and ν_g is its finite normalization: a fibre records the degree- g divisors of the common divisor, and on the exact-degree- g locus there is one reduced lift. We shall retain the scheme structures of these fibres and of the image.

Lemma L.1 (Completed finite images). *Let $X \rightarrow M$ be a finite morphism of complex finite-type schemes, with X reduced and M smooth, and let Y be its scheme-theoretic image. Fix a closed point $y \in Y$, put $R = \hat{\mathcal{O}}_{M,y}$, and write $B_i = \hat{\mathcal{O}}_{X,x_i}$ for the points over y . Then*

$$\hat{\mathcal{O}}_{Y,y} = R / \bigcap_i \ker(R \rightarrow B_i).$$

This completed image is reduced. If the cotangent map from R to B_i is surjective, then $R \rightarrow B_i$ is surjective. If there are two such points, writing $I_i = \ker(R \rightarrow B_i)$ gives

$$\widehat{\mathcal{O}}_{Y,y} = (R/I_1) \times_{R/(I_1+I_2)} (R/I_2).$$

The formal schemes at the different x_i are disjoint. A local Artin point of the finite inverse image belongs to exactly one of them.

Proof. On an affine neighbourhood the image ring is the image of the ambient ring in a finite algebra. Exactness of completion on finite modules preserves this kernel. The completed finite algebra is the product of the B_i : the orthogonal idempotents of its Artin closed fibre lift uniquely through the powers of the maximal ideal. Reduced finite-type complex rings are excellent and analytically unramified, so their completions are reduced; alternatively this applies directly to the reduced image ring. For cotangent surjectivity, lift a basis of the maximal ideal of B_i modulo its square to R . The resulting lifts generate every successive associated-graded piece. Subtracting successive homogeneous lifts and taking their convergent sum expresses any element of B_i as the image of an element of R . The finite map factors through the completed image, so this is also a closed immersion into that image, not only into M . Finally the elementary exact sequence

$$0 \longrightarrow R/(I_1 \cap I_2) \longrightarrow R/I_1 \oplus R/I_2 \longrightarrow R/(I_1 + I_2) \longrightarrow 0$$

proves the fibre-product assertion. The idempotent decomposition gives the last assertion, including over nonreduced Artin bases. $\square$

Lemma L.2 (Coprime lifts and the split codimension). *If homogeneous forms a, b have coprime reductions over a local Artin base A , then $(a) \cap (b) = (ab)$ in $A[E]$. The relevant graded quotients and Koszul kernels are A -flat. In the split-divisor model, with $s_j = \binom{e+j-1}{j}$, $e \geq 2$, $h > g \geq 1$, and $r \geq 2$,*

$$c = r(s_h - s_{h-g}) - (s_g - 1) \geq (r-1)(s_g - 1) \geq 1.$$

Proof. Over the residue field the two forms form a regular sequence. In each degree their Koszul complex consists of finite free modules and has its prescribed ranks. The acyclic-complex criterion over an Artin local ring, proved by lifting splittings successively along its maximal ideal, gives exactness and flatness of the lifted cokernels. Thus b is a nonzerodivisor modulo a , proving the intersection identity. For the inequality, express $s_h - s_{h-g}$ as the sum of g consecutive first differences of the sequence s_j . These differences are nondecreasing for $e \geq 2$. Their sum is at least $s_g - s_0 = s_g - 1$. $\square$

In particular the two formal marked lifts in a coprime split model are separated before their images are compared. Their intersection functor is precisely $K_A = a_A b_A L_A$. Division gives a subbundle: multiplication by a fibrewise nonzero form is locally split, and the quotient of the ambient bundle by K_A is locally free. Combining this Artin-functor identification with Lemma L.1 justifies the completed-image fibre product without any assumption that a tangent-space comparison determines an intersection scheme.

APPENDIX M. A GLOBAL FACTORIZATION THEOREM FOR BINARY NORMALIZATION FIBRES

For this section E has dimension two. Set $N_0 = r(D+1-r)$ and $m = r-1$. Let

$$\mathcal{T}_s = Y_s \setminus Y_{s+1}, \quad 1 \leq s \leq D+1-r,$$

with the reduced locally closed structure; an inadmissible Y_j is empty. Binary forms split into linear factors, so the Y_j are nested. The multiplication map identifies

$$\mathcal{T}_s \simeq \mathbb{P}(S_s) \times X_{D-s,r},$$

where $X_{D-s,r}$ is the open Grassmannian of gcd-free residual spaces. The first projection is the algebraic common-divisor morphism, not a choice made independently on individual fibres.

For nonnegative a, b define $H_{a,b} = \mathbb{C}$ when $ab = 0$. Otherwise, put $k = \min(a, b)$ and $l = \max(a, b)$, and set

$$H_{a,b} = \mathbb{C}[q_1, \dots, q_k]/(c_{l+1}, \dots, c_{l+k}), \quad \sum_{j \geq 0} c_j z^j = (1 + q_1 z + \dots + q_k z^k)^{-1}. \quad (\text{M.1})$$

The weight of q_i is i . This grading is not the maximal-ideal adic grading when $k > 1$.

Theorem M.1 (Stratified finite flatness and all binary fibres). *For $g \leq s$, the restriction of ν_g over $\mathcal{T}_s$ is the base change, along the common-divisor morphism, of*

$$\mathbb{P}(S_g) \times \mathbb{P}(S_{s-g}) \longrightarrow \mathbb{P}(S_s), \quad (f, H) \mapsto fH.$$

It is finite locally free of degree $\binom{s}{g}$. If $\gcd(K) = \prod_{i=1}^b \ell_i^{s_i}$, with distinct ℓ_i and $\sum_i s_i = s$, its entire normalization fibre is

$$\prod_{\substack{0 \leq a_i \leq s_i \\ \sum_i a_i = g}} \bigotimes_{i=1}^b H_{a_i, s_i - a_i}. \quad (\text{M.2})$$

Every local factor is an Artin complete intersection and is Gorenstein. Its length, weighted Hilbert series, and socle weight are, respectively,

$$\prod_i \binom{s_i}{a_i}, \quad \prod_i \prod_{j=1}^{\min(a_i, s_i - a_i)} \frac{1 - z^{\max(a_i, s_i - a_i) + j}}{1 - z^j}, \quad \sum_i a_i (s_i - a_i).$$

The total length is $\binom{s}{g}$, including at arbitrary simultaneous root collisions. Over fixed s , the closure order of common-root partition strata is the order generated by merging parts.

Proof. Over $\mathcal{T}_s$ the common divisor G and residual subbundle L are defined scheme-theoretically by the inverse to the exact-divisor incidence. Locally on any base change, choose a member of L coprime to G on each relevant geometric fibre. There is such a member over $\mathbb{C}$, since only finitely many roots must be avoided. Regular-sequence cancellation, or monic division and its flat quotient, gives $f \mid GL$ if and only if $f \mid G$. This equivalence is an equality of incidence functors over nonreduced bases. The residual subspace is then forced. This proves the base-change description.

The displayed multiplication map is finite by properness and the finiteness of the divisors of a fixed binary form. Its source and target are smooth of dimension s . A finite map between these smooth spaces is flat: a regular system of parameters in the target is a system of parameters in each source local ring, hence a regular sequence; the local flatness criterion applies. Its degree can be computed at a squarefree form, where the $\binom{s}{g}$ divisors are reduced and distinct.

Coprime root clusters split uniquely over Artin bases by Hensel factorization. On the cluster with polynomial x^{s_i} , choosing a degree- a_i factor is the equation $f_i H_i = x^{s_i}$ with both factors monic. Eliminating one factor by reciprocal expansion gives (M.1). This proves the full product decomposition, not only its closed points. Its equations have their sole common zero at the origin: a monic factor of x^{a+b} over $\mathbb{C}$ is a power of x . Thus its k equations in k variables are a homogeneous system of parameters and a regular sequence. The Hilbert series is the quotient of the corresponding weighted polynomial Hilbert series. Taking its value at one gives $\binom{a+b}{a}$; its top weight is ab . A zero-dimensional complete intersection has a one-dimensional socle. Equivalently q_k^l , up to a nonzero scalar, represents the rectangular Schur class. The identification with the usual Grassmannian cohomology presentation is classical; a purely algebraic Schur-basis statement is [24, Theorem 2.7]. Tensoring the local factors proves all three formulas. The total length identity

is the coefficient of z^g in $\prod_i (1+z)^{s_i}$. Finally the exact-gcd stratum is the product with $X_{D-s,r}$ above. In the symmetric power of the projective line, root clusters specialize by coalescence, and every merging is realized by moving the corresponding roots together. This gives precisely the stated closure order within fixed gcd degree. $\square$

Remark M.2. The finite flat statement is on the exact-gcd stratum with its indicated scheme structure. It does not assert that normalization or formation of scheme-theoretic images commutes with an arbitrary nonflat base change of the ambient Grassmannian. It identifies a specific base change of the already constructed incidence morphism.

APPENDIX N. CONDUCTORS AND ALL BRANCH INTERSECTIONS ON THE SQUAREFREE LOCUS

Let $\mathcal{U} \subset \text{Gr}(r, S_D)$ be the open set on which the common divisor is squarefree. Its complement is the proper image of the incidence $K = \ell^2 L$. This condition concerns the common divisor, not every member of the subspace.

Theorem N.1 (Squarefree conductor stratification). *Let $K \in Y_g \cap \mathcal{U}$, let $s = \deg \gcd(K)$, and set*

$$t = s - g, \quad m = r - 1, \quad q = N_0 - ms.$$

Introduce s disjoint blocks $u_i = (u_{i1}, \dots, u_{im})$, and let J_j be the ideal generated by products of one variable from each of j distinct blocks. Put $J_0 = (1)$ and $J_{s+1} = 0$. Then

$$\widehat{\mathcal{O}}_{Y_g, K} \simeq \mathbb{C}[[z_1, \dots, z_q, u_{11}, \dots, u_{sm}]]/J_{t+1}. \quad (\text{N.1})$$

The branches are indexed by the subsets $A \subset \{1, \dots, s\}$ of cardinality g . The branch A has ideal $I_A = (u_{ij} : i \in A)$ in the ambient power-series ring. For any collection of branches, their full scheme-theoretic intersection has ideal $I_{\cup A}$; it is smooth of dimension $q + m(s - |\cup A|)$.

The conductor in (N.1) is J_t/J_{t+1} . Its restriction to branch A is $\prod_{i \notin A} (u_{i1}, \dots, u_{im})$. Consequently the global conductor sheaf on $Y_g \cap \mathcal{U}$ is exactly

$$\mathfrak{c}_{Y_g|U} = \mathcal{I}_{Y_{g+1}/Y_g}|U. \quad (\text{N.2})$$

In particular it is radical there. These local rings are seminormal. Their dimension, depth and multiplicity are

$$q + mt, \quad q + t, \quad \binom{s}{g}.$$

They are Cohen–Macaulay exactly when $s = g$ or $r = 2$. The Hilbert series of the nonsmooth factor in its standard grading is

$$\sum_{j=0}^t \binom{s}{j} ((1-z)^{-m} - 1)^j. \quad (\text{N.3})$$

The whole collection of branch intersections and conductor strata is invariant under permutation of the roots and descends from any ordered-root etale chart.

Proof. Choose $F_0 \in K$ with a simple zero at each of the s common roots, and complete it to a frame $F_0, \dots, F_m$. Hensel factorization gives moving roots $x_1, \dots, x_s$ in the completion. Use as coordinates the s root parameters and the ms evaluations $u_{ij} = F_j(x_i)$, with harmless invertible normalizing factors. These are part of a regular parameter system: the evaluation map $S_D/K \rightarrow \mathbb{C}^s$ is surjective because K vanishes on the common-root scheme, and $D \geq s - 1$. Thus $\text{Hom}(K, S_D/K)$ allows independent variation of all rs jets. The passage to the displayed coordinates is triangular with invertible diagonal, since F_0 has simple zeros. The other $N_0 - rs$ parameters and the s root parameters give exactly $q = N_0 - ms$ smooth parameters.

A marked degree- g divisor near this fibre chooses a subset A of these roots. Its incidence equations are exactly $u_{ij} = 0$ for $i \in A$, with no nilpotents suppressed. There are no other divisor lifts. Lemma L.1 therefore identifies the completed image ideal with $\bigcap_{|A|=g} I_A$. A monomial belongs to this intersection exactly when its block support meets every g -element subset, or equivalently has at least $s - g + 1 = t + 1$ blocks. This proves (N.1). Sums of the coordinate ideals give all intersections, including intersections of three or more branches.

For completeness compute the conductor inside the product of the branch rings. Its component on branch A consists of functions whose extension by zero on every other branch belongs to the image ring. The image of $\bigcap_{B \neq A} I_B$ modulo I_A is precisely $\prod_{i \notin A} (u_i)$. Indeed a surviving monomial has support outside A , and must meet every g -subset other than A . Replacing one element of A by each $i \notin A$ forces all those blocks to occur. Conversely their product meets every such subset. Hence a conductor monomial has exactly t surviving blocks; monomials on more blocks are zero. This is J_t/J_{t+1} . The same coordinates identify Y_{g+1} with J_t , proving the sheaf equality by faithful flatness of completion. The case $t = 0$ means that the conductor is the unit ideal and the next stratum is absent locally.

The ring is the ring of tuples on its coordinate branches that agree on every pairwise intersection. To verify this directly, compare the coefficient of each monomial on all branches where it survives; the pairwise agreements give a unique common coefficient. If an element b of the total fraction ring has b^2, b^3 in the image, it first belongs to the normalization, by integrality. Its two restrictions on any intersection have equal squares and cubes, hence are equal in that reduced ring. Thus b lies in the image. This proves seminormality by the square-and-cube criterion.

Formula (N.3) follows by sorting nonzero monomials by their block support. The top-dimensional contributions come from the $\binom{s}{t}$ supports of size t , giving dimension mt and multiplicity $\binom{s}{t}$. We supply the homological step for the depth, since branch counting alone does not give it. The simplicial complex of the squarefree monomial ideal J_{t+1} consists of sets using at most t blocks. Any induced subcomplex meeting b blocks is homotopy equivalent to the $(t-1)$ -skeleton of the simplex on those b blocks: collapse each block to a representative vertex. The two composites are contiguous to the identity, because adjoining the representatives never increases the number of blocks. If $b \leq t$ the complex is a simplex; otherwise its only reduced homology is in degree $t-1$, with rank $\binom{b-1}{t}$. This last assertion follows also by the elementary induction that adds the last vertex to a simplex skeleton.

The squarefree multidegree- W part of the Koszul complex on all sm variables is the augmented simplicial cochain complex of this induced subcomplex, with degree shifted by $|W| - 1$. Non-squarefree multidegrees are contractible in the relevant summands, by pairing faces using a variable whose exponent exceeds one. Consequently a nonzero positive Betti number has homological degree $|W| - t$. The full vertex set gives the nonzero degree $sm - t$, since $s > t$ when $g > 0$. The projective dimension is therefore $sm - t$, and the Auslander–Buchsbaum formula gives depth t . When $t = 0$ the quotient is simply $\mathbb{C}$. Adjoining q power-series parameters gives depth $q + t$ in every case, and comparison with $q + mt$ proves the Cohen–Macaulay criterion. All the constructions commute with relabelling roots, proving descent. $\square$

Corollary N.2 (Binary divisor-image intersections). *For admissible binary divisor degrees $a_1, \dots, a_v$, their scheme-theoretic intersection is $Y_{\max a_i}$. On the squarefree locus all multiple intersections of normalization branches, rather than only the degree images, are given by the union-of-subsets rule in Theorem N.1.*

Proof. The reduced binary images are nested as closed subschemes: a binary form of degree b has a divisor of each degree $a \leq b$, and reduced closed subschemes are determined by their closed points. Their ideals are thus nested, so their sum is the largest ideal. The assertion about branches has already been proved as an identity of coordinate ideals, without taking radicals. $\square$

APPENDIX O. AN INCIDENCE ATLAS FOR ALL BINARY ROOT COLLISIONS

The squarefree theorem has a uniform replacement when any number of root clusters collide. This replacement specifies the full completed image, not merely a transverse family in it.

Theorem O.1 (Simultaneous collision atlas). *Let $K \in Y_g$ be binary with exact common divisor $\prod_{i=1}^b \ell_i^{s_i}$ and $s = \sum_i s_i$. There are formal ambient coordinates consisting of the coefficients of monic polynomials P_i of degree s_i , polynomials R_{ij} of degree less than s_i ($1 \leq j \leq m$), and $N_0 - rs$ smooth parameters z . They are centred at $P_i = x_i^{s_i}$ and $R_{ij} = 0$. For every $\alpha = (a_i)$ with $0 \leq a_i \leq s_i$ and $\sum a_i = g$, let B_α be the regular power-series ring in the coefficients of monic A_i, B'_i of degrees $a_i, s_i - a_i$, polynomials U_{ij} of degree less than $s_i - a_i$, and the same z . The completed normalization and image are exactly*

$$T \longrightarrow \prod_{\alpha} B_{\alpha}, \quad P_i \mapsto A_i B'_i, \quad R_{ij} \mapsto A_i U_{ij}, \quad \widehat{\mathcal{O}}_{Y_g, K} = T / \bigcap_{\alpha} \ker(T \rightarrow B_{\alpha}). \quad (\text{O.1})$$

Here $T = \mathbb{C}[[\text{coeff}(P_i), \text{coeff}(R_{ij}), z]]$. Degree-zero monic factors are one, and negative-degree remainder spaces are zero. The formula is independent, up to isomorphism of completed incidence diagrams, of the frame, the selected F_0 , and the local root coordinates. It covers every higher and simultaneous binary collision.

Proof. Choose $F_0 \in K$ having exact order s_i at every common root. The required nonvanishing of its residual value excludes finitely many hyperplanes in K , so a choice exists. Complete it to a frame. Weierstrass preparation at the distinct roots produces P_i ; division of the remaining frame members produces R_{ij} . The map

$$S_D/K \longrightarrow \bigoplus_i \mathbb{C}[x_i]/(x_i^{s_i})$$

is surjective: its kernel contains K , and the usual interpolation map from S_D is surjective. Thus the Grassmannian tangent space permits independent variation of all rs jets. Division and preparation change these jets by an invertible triangular matrix, whose diagonal contains the nonzero residual values of F_0 . Their differentials are independent. Formal inverse coordinates supply the remaining $N_0 - rs$ parameters; this integer is nonnegative since $r \leq D - s + 1$.

A marked divisor near a chosen normalization point splits uniquely among the coprime root clusters. In cluster i its monic representative has degree a_i and divides both P_i and all R_{ij} . Monic division is equivalent, over every local Artin base, to the unique equations $P_i = A_i B'_i$ and $R_{ij} = A_i U_{ij}$. Conversely these equations give the marked divisor and the residual subbundle. Thus the normalized incidence functor is represented by the displayed regular ring. It is finite over T : monic factor coefficients are integral over the coefficients of P_i , while the remaining factors and quotients are uniquely determined by division. Lemma L.1 now gives the exact completed image and the intersection of kernels.

Changing a frame or a coordinate changes only a presentation of the same functor of a subbundle with a marked divisor. The mutually inverse changes of incidence data therefore induce isomorphisms of its complete representing rings and of the resulting finite-image diagrams. This also proves that the calculation describes the entire completed image. A point of that image over an Artin ring need not lift to its normalization; no such lifting assertion is used or implied. $\square$

APPENDIX P. THE SINGULAR FITTING IDEAL OF EVERY DOUBLE-ROOT MODEL

The atlas specializes for $s = 2, g = 1$ to the following ring, after completion of the square and suppression of smooth parameters:

$$A_m = R \oplus It \subset B = \mathbb{C}[[u_1, \dots, u_m, t]], \quad R = \mathbb{C}[[u_1, \dots, u_m, \Delta]], \quad I = (u_1, \dots, u_m), \quad \Delta = t^2.$$

Put $w_i = u_i t$. The defining equations are $u_i w_j - u_j w_i$ and $w_i w_j - \Delta u_i u_j$.

Theorem P.1 (Closed singular-ideal formula). *For every $m \geq 1$, the intrinsic singular Fitting ideal is*

$$\text{Fitt}_{m+1} \Omega_{A_m/\mathbb{C}}^{\text{cont}} = (I^{m+1} + \Delta I^m) \oplus I^m t = I^{m-1} (I^2, (w_1, \dots, w_m), \Delta I). \quad (\text{P.1})$$

The first equality is a decomposition as an R -submodule of B ; the right side is an ideal of A_m . It is not merely an ideal of Jacobian minors left unevaluated. For $m = 1$ it is $(u^2, w, \Delta u)$, and its radical is the conductor (u, w) for all m .

Proof. There are $2m + 1$ ambient variables and dimension $m + 1$, so the Fitting ideal consists of the m -minors of the Jacobian of the displayed relations. Evaluate this matrix in B using $w_i = u_i t$ and $\Delta = t^2$. A minor not using the Δ column has u -degree m . It vanishes at $t = 0$: the only remaining rows are the Koszul rows in the w columns, of rank at most $m - 1$. Hence its terms have a positive power of t , and belong to $\Delta I^m \oplus I^m t$. A minor using the Δ column has u -degree $m + 1$, and hence lies in $I^{m+1} \oplus I^{m+1} t$. This proves the first inclusion.

Use the m rows

$$u_1 w_j - u_j w_1 \quad (2 \leq j \leq m), \quad w_1^2 - \Delta u_1^2.$$

The columns $(w_1, w_2, \dots, w_m)$, $(u_1, w_2, \dots, w_m)$, and $(\Delta, w_2, \dots, w_m)$ give, up to sign and nonzero scalar, $u_1^m t$, Δu_1^m , and u_1^{m+1} . The ideal is invariant under simultaneous linear change of u and w . In characteristic zero the powers of linear forms span each symmetric power, so these three minors generate the three indicated spaces of all u -monomials. They generate the claimed R -module; it is an A_m -ideal since multiplication by $w_i = u_i t$ preserves it. This proves equality. The formula also immediately gives its radical, without discarding the nonreduced Fitting structure in the assertion itself. $\square$

APPENDIX Q. A STRUCTURAL DICHOTOMY FOR PURE-POWER DIVISOR FIBRES

Let $E = \mathbb{C}a \oplus W$, $d = \dim W \geq 1$, and $g, k \geq 1$. The factorization fibre of a^{g+k} records monic factors of degrees g and k . Interchanging these factors identifies its two descriptions; we may and shall take $g \geq k$. Write

$$Q_i \in \text{Sym}^i W \quad (1 \leq i \leq k), \quad C_0 = 1, \quad C_j = - \sum_{i=1}^k Q_i C_{j-i}, \quad C_j = 0 \quad (j < 0).$$

Let $P = \text{Sym}(\bigoplus_{i=1}^k (\text{Sym}^i W)^*)$, with its generators of weight i , and let $\mathfrak{n}$ be its variable ideal. Put

$$F_{g,k}(W) = P/\mathcal{I}, \quad \mathcal{I} = (\text{coeff } C_{g+1}, \dots, \text{coeff } C_{g+k}). \quad (\text{Q.1})$$

It is an Artin local algebra. Indeed its complex points are exactly the factorizations of a^{g+k} , hence the single origin by unique factorization. The polynomial and completed presentations coincide. If $K = a^{g+k} L$ with $\gcd(L) = 1$, cancellation by a member of L not divisible by a , as in Lemma L.2, identifies this with the full degree- g normalization fibre at K .

Theorem Q.1 (Minimal equations and complete-intersection classification). *The embedding dimension and minimal number of equations of $F_{g,k}(W)$ in a minimal regular local presentation are*

$$\text{edim } F_{g,k}(W) = \binom{d+k}{k} - 1, \quad (\text{Q.2})$$

$$\mu(\mathcal{I}) = \sum_{j=g+1}^{g+k} \binom{d+j-1}{j} = \binom{d+g+k}{d} - \binom{d+g}{d}. \quad (\text{Q.3})$$

More precisely the minimal relation module has the equivariant weighted decomposition

$$\mathcal{I}/\mathfrak{n}\mathcal{I} \simeq \bigoplus_{j=g+1}^{g+k} (\mathrm{Sym}^j W)^*. \quad (\text{Q.4})$$

Consequently this fibre is a complete intersection if and only if $d = 1$. For $d = 1$, it has length $\binom{g+k}{k}$, weighted Hilbert series $\prod_{i=1}^k (1 - z^{g+i})/(1 - z^i)$, and one-dimensional socle generated by Q_k^g , of weight gk . For $k = 1$ and arbitrary d ,

$$F_{g,1}(W) = \mathrm{Sym}(W^*)/\mathfrak{n}^{g+1}, \quad \text{length } F_{g,1}(W) = \binom{d+g}{g}, \quad \mathrm{Soc } F_{g,1}(W) = \mathrm{Sym}^g(W^*). \quad (\text{Q.5})$$

In particular the last algebra is Gorenstein precisely when $d = 1$. No Gorenstein classification for $d > 1, k > 1$ is needed for the theorem.

Proof. Since $g \geq k$, each defining form has no linear term in the coefficient variables. Thus the presentation is minimal and (Q.2) follows by summing $\dim \mathrm{Sym}^i W$. For $d = 1$, the regular-sequence argument in the proof of Theorem M.1 applies to the k reciprocal relations. In particular none of them belongs to $\mathfrak{n}\mathcal{I}$. It gives the stated Hilbert series, length, and the classical rectangular socle class.

For general W , taking coefficients of C_j gives an equivariant map $(\mathrm{Sym}^j W)^* \rightarrow \mathcal{I}/\mathfrak{n}\mathcal{I}$. Its source is irreducible. This map is nonzero: choose one coordinate x of W , keep only the coefficients of x^i in every Q_i , and set every other coefficient to zero. The coefficient of x^j maps to the j th binary reciprocal relation. Were it in $\mathfrak{n}\mathcal{I}$ before specialization, its specialization would be in the binary $\mathfrak{n}\mathcal{I}$, contradicting the preceding minimality. Irreducibility now makes the entire coefficient map injective. The different j have different weights, so their images are independent. They generate the relation module by the chosen presentation, proving (Q.4) and (Q.3).

If $d \geq 2$, the sequence $\binom{d+j-1}{j}$ strictly increases in j . Comparing its terms at $j = g+i$ and $j = i$ shows that the minimal number of relations is strictly greater than the embedding dimension. A zero-dimensional complete intersection in a minimal regular local presentation has exactly that many minimal relations as variables; hence these fibres are not complete intersections. This proves both directions of the classification. Finally when $k = 1$ the sole form is $C_{g+1} = (-Q_1)^{g+1}$. Its coefficients are, with nonzero multinomial factors, every monomial of degree $g+1$ in the d variables. This gives (Q.5), including the socle and length. $\square$

Corollary Q.2 (The ternary-pencil ramification fibre). *At the common limiting relation algebra with $e = 9$, $g = 6$ and $k = 2$, the normalization fibre has embedding dimension 44 and exactly*

$$\binom{14}{7} + \binom{15}{8} = 9867$$

minimal scalar equations. Their two weighted pieces have dimensions 3432 and 6435, in weights seven and eight. In particular this actual pencil-orbit boundary fibre is not a complete intersection.

Proof. Here $d = e - 1 = 8$. Apply (Q.2) and (Q.4). These are minimal equation counts, not the number of coefficient variables or the length of the fibre. $\square$

The binary factorization rings in this section are classical. The additional point of the multivariate argument is that polarization preserves every full irreducible coefficient space in the minimal relation module. Neither the Grassmannian identification nor a finite calculation in a few variables is being used as a claim of priority or as a substitute for that argument.

APPENDIX R. HOMOLOGICAL INVARIANTS OF THE DIVISOR FIBRES

We first extract all graded Betti numbers from the squarefree local model and then identify a determinant-apolar quotient of the multivariate fibres. A separate exact calculation treats an entire multivariate fibre, rather than only a quotient or its minimal relations. The grading conventions will be specified in each case.

Theorem R.1 (Betti numbers of the squarefree block model). *Let $S = \mathbb{C}[u_{ij} : 1 \leq i \leq s, 1 \leq j \leq m]$ with its standard multigrading, let $0 \leq t < s$, and put $A = S/J_{t+1}$, with the block-support ideal of Theorem N.1. For a nonempty squarefree multidegree F let $b(F)$ be the number of blocks it meets. Besides $\beta_{0,0} = 1$, the only nonzero multigraded Betti numbers are*

$$\beta_{i,F}^S(A) = \binom{b(F)-1}{t} \quad \text{if } b(F) > t \text{ and } i = |F| - t. \quad (\text{R.1})$$

For $t = 0$ this is the Koszul resolution of $\mathbb{C}$. For all t , the singly graded numbers are

$$\beta_{i,i+t}^S(A) = \sum_{b=t+1}^{\min(s,i+t)} \binom{s}{b} \binom{b-1}{t} [z^{i+t}]((1+z)^m - 1)^b \quad (i \geq 1). \quad (\text{R.2})$$

The ideal J_{t+1} has a $(t+1)$ -linear resolution. The projective dimension of A is $sm - t$. In the Cohen–Macaulay case $m = 1$, its type is $\binom{s-1}{t}$; it is Gorenstein exactly when $t = 0$ or $t = s - 1$. The same Betti numbers over the completed ambient regular ring apply to the local model after adjoining its smooth parameters.

Proof. The Stanley–Reisner complex consists of sets meeting at most t blocks. If $t \geq 1$, its induced subcomplex on F has, in each nonempty block, a simplex all of whose vertices have identical external links. Contract that simplex to one of its vertices: the map and the inclusion are contiguous, because adjoining the chosen vertex to any face does not change its block support. Applying this one block at a time gives a homotopy equivalence with the $(t-1)$ -skeleton of the simplex on $b(F)$ vertices. If $b(F) \leq t$ this simplex is contractible. Otherwise its reduced homology is concentrated in degree $t-1$, with dimension $\binom{b(F)-1}{t}$. The latter assertion follows by attaching its top faces successively, or from the augmented simplicial chain complex and the binomial identity for its Euler characteristic. Hochster’s multigraded formula now gives (R.1); see [26, Chapter 5]. If $t = 0$, every variable is in the ideal and the Koszul formula gives the same result directly.

For a support of size b , the coefficient in (R.2) counts choices of $i+t$ vertices meeting each selected block. Summing proves the formula. All ideal syzygies therefore have the indicated linear degrees. Taking F to be all sm vertices gives the nonzero last Betti number $\binom{s-1}{t}$ in homological degree $sm - t$; no larger degree occurs. When $m = 1$, Auslander–Buchsbaum gives depth t , equal to dimension, and the last Betti number is the Cohen–Macaulay type. Completion and adjoining independent regular parameters are flat operations preserving the minimal free resolution. These are classical squarefree-monomial methods. The assertion here is their full specialization to the normalization image model, not a claim of priority for Hochster’s formula or skeleton resolutions. $\square$

For the remaining results use the weighted reciprocal algebra $F_{g,k}(W)$ of (Q.1), where $d = \dim W$ and $g \geq k \geq 2$. Set all Q_i except Q_2 equal to zero and, in addition, impose every coefficient of Q_2^2 . Denote the resulting algebra by

$$B_d = \text{Sym}((\text{Sym}^2 W)^*) / (\text{coeff } Q_2^2). \quad (\text{R.3})$$

Its coefficient variables have reciprocal weight two. When referring to its ordinary polynomial grading, we instead give them degree one.

Theorem R.2 (A determinant-apolar quotient of every higher fibre). *There is a $\mathrm{GL}(W)$ -equivariant graded surjection $F_{g,k}(W) \twoheadrightarrow B_d$ for every $g \geq k \geq 2$. After fixing the standard coefficient/differential pairing, B_d is the apolar algebra of the determinant of a generic symmetric $d \times d$ matrix. Its polynomial-degree Hilbert function is*

$$\dim(B_d)_p = \frac{1}{d+1} \binom{d+1}{p} \binom{d+1}{p+1} \quad (0 \leq p \leq d). \quad (\text{R.4})$$

It is Gorenstein, its reciprocal-weight socle is $2d$, and

$$\text{length } F_{g,k}(W) \geq \text{length } B_d = \frac{1}{d+2} \binom{2d+2}{d+1}. \quad (\text{R.5})$$

For $k = 2$ and $g = 2$ or 3 , the quotient by the coefficients of Q_1 alone is already B_d . In particular the actual ternary-pencil fibre $F_{6,2}(\mathbb{C}^8)$ has length at least 4862, independently of its embedding dimension 44 and its 9867 minimal equations.

Proof. Under the stated substitution, the reciprocal series becomes $(1 + Q_2 z^2)^{-1}$. Its odd coefficients vanish and its even coefficients are powers of Q_2 . Since $g + 1 \geq 3$, imposing $Q_2^2 = 0$ annihilates all defining coefficients. This proves the surjection, functorially for linear changes of W . When $k = 2$ and $g = 2$ or 3 , the first even defining coefficient is Q_2^2 and there are no further independent equations after $Q_1 = 0$.

Choose coordinates $x_1, \dots, x_d$ on W and write $Q_2 = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$. Let $X = (X_{ij})$ be symmetric, with independent entries for $i \leq j$, and interpret y_{ij} as $\partial/\partial X_{ij}$. Then $Q_2(x)$ acts as directional differentiation in the rank-one symmetric direction xx^t . The determinant of $X + zxx^t$ is linear in z , so every coefficient of $Q_2(x)^2$ annihilates $\det X$. These coefficients are, up to nonzero scalars, y_{ii}^2 , $y_{ii}y_{ij}$, $y_{ij}^2 + 2y_{ii}y_{jj}$, $y_{ii}y_{jl} + y_{ij}y_{il}$ for distinct i, j, l , and $y_{ij}y_{kl} + y_{ik}y_{jl} + y_{il}y_{jk}$ for four distinct indices. They are exactly the quadratic generators in [27, Proposition 3.1 and Theorem 3.10]. That theorem proves equality with the full apolar ideal, not merely containment. Its Corollary 2.7 gives (R.4) and the Catalan length. Alternatively, that Hilbert function counts the independent minors obtained by differentiating $\det X$. Apolarity to one homogeneous form gives a perfect pairing between complementary degrees and a one-dimensional socle in degree d . Replacing polynomial degree by twice that degree proves the weight statement. The length inequality follows from the surjection. For $d = 8$ the Catalan number is 4862.

The apolar Hilbert function and its Gorenstein property are classical results of the cited determinant theory. The new identification in this setting is the functorial quotient of the entire family of normalization fibres. A quotient's Gorenstein property is not being attributed to the original fibre. $\square$

Proposition R.3 (The complete first multivariate two-factor fibre). *Let $A = F_{2,2}(\mathbb{C}^2)$ and write $Q_1 = ax + by$, $Q_2 = cx^2 + dxy + ey^2$. Then*

$$H_A^{\text{wt}}(z) = 1 + 2z + 6z^2 + 6z^3 + 7z^4, \quad \text{length } A = 22, \quad \dim \text{Soc } A = 7, \quad (\text{R.6})$$

$$H_{\text{grm } A}(z) = 1 + 5z + 6z^2 + 5z^3 + 5z^4. \quad (\text{R.7})$$

The weight grading gives a, b weight one and c, d, e weight two. In the minimal weighted free resolution over $S = \mathbb{C}[a, b, c, d, e]$, the shift polynomials $B_i(z) = \sum_j \beta_{i,j}^S(A) z^j$ are

$$\begin{aligned} B_0 &= 1, & B_1 &= 4z^3 + 5z^4, \\ B_2 &= 4z^5 + 12z^6 + 12z^7, & B_3 &= 21z^8 + 20z^9 + z^{10}, \\ B_4 &= 21z^{10} + 8z^{11}, & B_5 &= 7z^{12}. \end{aligned}$$

In particular the total Betti numbers are $(1, 9, 28, 42, 29, 7)$. This fibre is not Gorenstein. The assertion concerns this whole five-variable fibre, not the determinant-apolar quotient.

Proof. The nine initial equations are the coefficients of $-Q_1^3 + 2Q_1Q_2$ and $Q_1^4 - 3Q_1^2Q_2 + Q_2^2$. For graded reverse lexicographic order $e > d > c > b > a$, a Groebner basis is the following list (a scalar multiple does not change a member):

$$\begin{aligned} & (c, d, e)^3, \quad b(c, d, e)^2, \quad a(c, d, e)^2, \\ & b^2e - e^2, \quad b^2d - de, \quad b^2c - a^2e, \quad b^3 - 2be, \\ & 2abe - de, \quad 2a^2e + 2abd - 2ce - d^2, \quad 2abc - cd, \\ & 3ab^2 - 2ae - 2bd, \quad a^2d - cd, \quad a^2c - c^2, \\ & 3a^2b - 2ad - 2bc, \quad a^3 - 2ac. \end{aligned}$$

Here a monomial ideal in the list means all its minimal monomial generators. There are $10 + 6 + 6 + 12 = 34$ polynomials. Polynomial division gives both equality with the ideal of the nine equations and zero remainders for all their basis S-pairs. This is a finite exact arithmetic verification; the explicit input and all reduction and rank tests are supplied in the accompanying calculation. The standard monomials, grouped by weight, are

$$\begin{array}{c|l} 0 & 1 \\ 1 & a, b \\ 2 & a^2, ab, b^2, c, d, e \\ 3 & ac, bc, ad, bd, ae, be \\ 4 & c^2, cd, d^2, a^2e, ce, de, e^2. \end{array} \tag{R.8}$$

Reduction of products of these monomials gives $\dim A/\mathfrak{m} = 1$ and $\dim(\mathfrak{m}^i) = (21, 16, 10, 5, 0)$ for $1 \leq i \leq 5$. The common kernel of multiplication by a, b, c, d, e is exactly the last row of (R.8). This proves both distinct Hilbert series and the type.

For precision, the resolution calculation uses no numerical rank thresholds. In weight w and homological degree i , take basis pairs (J, m) with $J \subset \{a, b, c, d, e\}$ of size i , m a standard monomial, and $\text{wt}(J) + \text{wt}(m) = w$. The differential is

$$d(J, m) = \sum_{v \in J} (-1)^{\text{pos}(v, J) - 1} (J \setminus \{v\}, \text{NF}(vm)). \tag{R.9}$$

All entries are rational, explicitly determined by the displayed 34 polynomials. Exact row reduction and $\beta_{i,w} = \dim C_{i,w} - \text{rank } d_{i,w} - \text{rank } d_{i+1,w}$ give the six shift polynomials above. The calculation is fully specified by (R.8)–(R.9); the archived rank table makes each finite elimination independently repeatable. As additional checks, their alternating sum is $H_A^{\text{wt}}(z)(1-z)^2(1-z^2)^3$ and the last Betti space is the socle shifted by the total variable weight eight. These checks do not replace the exact rank computations. $\square$

APPENDIX S. SPECTRAL FITTING SCHEMES AS POWER-ZERO LOCI IN ONE RECIPROCAL SECTION

The algebra just calculated supplies a single geometric object in which all spectral Fitting schemes can be read simultaneously. This avoids constructing a separate binary form after each Fitting ideal has already been recovered.

Let V have dimension n and $R \subset \text{Sym}^2 V$ be any pencil. Fix $h \geq 1$ and put $\tilde{E} = \mathbb{C}a \oplus V^*$. Consider the divisor-incidence point

$$K_h = a^{2h+2} \tilde{E} \subset \text{Sym}^{2h+3} \tilde{E}. \tag{S.1}$$

The entire degree- $2h$ normalization fibre over K_h is $F_{2h,2}(V^*)$, because $\tilde{E}$ is gcd-free. Its section by the linear-factor coefficients is $B_{n,h}$ and $(B_{n,h})_1 = \text{Sym}^2 V$. Thus the given pencil defines a line $\Lambda = \mathbb{P}(R)$ in the projectivized degree-one space of this actual reciprocal section. This construction uses a new formal line $\mathbb{C}a$ and is functorial for isomorphisms of V .

For $0 \leq p \leq hn$ let $\mathcal{J}_p$ be the ideal sheaf on Λ locally generated by the coordinates of

$$\ell^p \in (B_{n,h})_p \otimes \mathcal{O}_\Lambda(p), \quad (\text{S.2})$$

where ℓ is the universal pencil member. Multiplying a local representative of ℓ by a unit does not change this ideal. We set $\mathcal{J}_0 = \mathcal{O}_\Lambda$. Let $\mathcal{T}_R$ be the cokernel of the universal symmetric matrix $V^* \otimes \mathcal{O}_\Lambda(-1) \rightarrow V \otimes \mathcal{O}_\Lambda$. This definition includes singular pencils.

Lemma S.1 (The local power-ideal formula). *Let $A(t)$ be a symmetric matrix over $\mathbb{C}[[t]]$ of rank r over $\mathbb{C}((t))$. Write its finite elementary exponents as $0 \leq e_1 \leq \dots \leq e_r$ and put $e_i = +\infty$ for $i > r$. For $1 \leq p \leq hn$, write $p = hq + s$, $0 \leq s < h$. The ideal generated by the coordinates of $\ell_{A(t)}^p$ in $(B_{n,h})_p \otimes \mathbb{C}[[t]]$ is*

$$\left(t^{h \sum_{i=1}^q e_i + s e_{q+1}} \right). \quad (\text{S.3})$$

An empty sum is zero, the last term is omitted if $s = 0$, and $t^{+\infty}$ means the zero ideal.

Proof. A symmetric matrix over this discrete valuation ring is congruent to

$$\text{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0).$$

Here is the required elementary reduction. Divide by the least valuation of an entry. A diagonal unit is a pivot; if only an off-diagonal entry is a unit, the value of the form on a sum of two basis vectors supplies a unit because 2 is invertible. Split off the resulting rank-one summand by elimination and continue. Every remaining unit has a square root in $\mathbb{C}[[t]]$, so the units can be removed from the diagonal. The finite exponents are the Smith exponents, as congruence is in particular equivalence.

Congruence acts on every graded piece of $B_{n,h}$ by an invertible matrix over $\mathbb{C}[[t]]$. It therefore preserves the ideal in question. The apolar embedding

$$(B_{n,h})_p \longrightarrow \mathbb{C}[X_{ij}]_{hn-p}, \quad D \longmapsto D(\det X)^h$$

is injective and splits as a map of complex vector spaces. After tensoring with $\mathbb{C}[[t]]$, the coordinates of a vector and of its image generate the same ideal. We may therefore calculate the coefficient ideal of $[z^p] \det(X + zA(t))^h$.

For diagonal $A(t)$, expand each of the h determinant factors in its chosen diagonal entries. Every term is divisible by $t^{\sum c_i e_i}$ with $0 \leq c_i \leq h$ and $\sum c_i = p$. The least such exponent is obtained by filling the smallest e_i to capacity h , giving (S.3). There is no cancellation at this least exponent: specialize X to $\text{diag}(x_1, \dots, x_n)$. Then

$$\det(X + zA(t))^h = \prod_i (x_i + zt^{e_i})^h$$

(with x_i^h for a zero diagonal entry of A). The monomial $\prod x_i^{h-c_i}$ has coefficient $\prod \binom{h}{c_i} t^{\sum c_i e_i}$, which is nonzero and has exactly the required valuation. If $p > hr$, every term is zero. This proves all the conventions, including the zero-ideal case. $\square$

Theorem S.2 (A reciprocal multiplication model for every spectral Fitting ideal). *For every complex quadratic pencil, regular or singular, and for $0 \leq j \leq n$, there is an equality of coherent ideal sheaves on its intrinsic line*

$$\mathcal{J}_{h(n-j)} = \text{Fitt}_j(\mathcal{T}_R)^h. \quad (\text{S.4})$$

In particular, when $h = 1$, the complete spectral Fitting schemes, including their multiplicities and any whole-line components, are exactly the power-zero schemes inside the one quadratic section of the normalization fibre at K_1 .

For a regular pencil and a spectral point x , set $b_{x,j} = \text{length } \mathcal{O}_{\Lambda,x} / \text{Fitt}_j(\mathcal{T}_R)_x$. The local length of the power-zero scheme is $hb_{x,j}$, and, for $0 \leq j < n$,

$$\lambda_{x,j+1} = \frac{1}{h} \left(\text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j),x}) - \text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j-1),x}) \right). \quad (\text{S.5})$$

Thus its multiplication, together with the embedded pencil line, recovers the full Segre data and their common projective support. For $n \geq 3$, the unmarked inverse makes this an invariant of the finite failure algebra. For a singular pencil, the construction recovers its normal rank and all its spectral Fitting ideals; it does not assert that these ideals alone recover the singular minimal indices.

Proof. At a point of the smooth curve Λ , pass to the completed discrete valuation ring. The ideal of $(n-j)$ -minors of a matrix with the stated Smith exponents is $(t^{e_1+\dots+e_{n-j}})$, with the zero-ideal convention. Apply Lemma S.1 with $p = h(n-j)$. The equality of completed ideals descends by faithful flatness. This proves (S.4) at every stalk and hence globally. This proof does not replace an ideal by its radical.

For a regular pencil the positive Smith exponents at x , in reverse order, are $\lambda_{x,1} \geq \lambda_{x,2} \geq \dots$. Consequently $b_{x,j} = \sum_{i>j} \lambda_{x,i}$. Equation (S.4) multiplies these lengths by h , so taking successive differences proves (S.5). The powers that vanish at the generic point similarly determine the normal rank in the singular case. Frame changes act on Λ by projectivities and congruences act equivariantly on the reciprocal construction. Theorem I.8.3 recovers precisely the needed pencil up to these changes, proving the intrinsic assertion. $\square$

Remark S.3 (Families and the scope of the ideal identity). For a vector bundle and pencil subbundle over any complex base, the quadratic-section algebra is obtained by applying the fixed $\text{GL}(V)$ -equivariant construction to local frames. Its graded pieces are vector bundles, and the coefficient ideal of each power map commutes with arbitrary base change: it is the image of the dual of a map between vector bundles. No flatness of its zero scheme is needed for that assertion. Formula (S.4) has been proved on each complex pencil line (and on a regular one-dimensional base by the same local argument). We do not infer an equality of those relative ideals on an arbitrary nonreduced parameter space from equality on its geometric fibres. Nor are the power-zero schemes asserted to be flat across a change in their length.

Corollary S.4 (The native specialization order in reciprocal coordinates). *On the fixed-discriminant, fixed-corank locus $X_{m,c}(A)$ of (I.Q.1), a specialization from the labelled partition tuple λ to μ exists if and only if, at every labelled root and every j , the local length of the corresponding power-zero scheme does not decrease. Equivalently the special power-zero divisor contains the generic one, using the common labelled spectral line. The reduced closure of a full $O(A)$ -orbit within this locus is therefore specified by these length inequalities. Each inequality can be written as the vanishing of the requisite jets of the coordinates of the power map (S.2). Every strict transition is realized by a flat family of the original finite failure neighbourhoods and is first distinguished at order $n^2 + 2n - 4$.*

Proof. For fixed multiplicity m_x , dominance $\mu_x \preceq \lambda_x$ is equivalent to $\sum_{i>j} \mu_{x,i} \geq \sum_{i>j} \lambda_{x,i}$ for every j . Use (S.5) and Theorem I.Q.2. In a local coordinate t at a labelled root, a coefficient ideal has colength at least b exactly when each of its generators is divisible by t^b ; this is the asserted finite jet condition. The closure order and the flat realization are the inherited full-orthogonal results, ultimately using Ohta's theorem. The new assertion is their expression inside

one reciprocal multiplication algebra. No reducedness assertion is made for the scheme cut out by the jet equations before taking its reduction. $\square$

Example S.5 (A rank profile with a genuinely different power-zero scheme). At a root of multiplicity four the partitions $(3, 1)$ and $(2, 2)$ have equal determinant and corank. In $B_{4,h}$ their sixth powers when $h = 2$ (in general their $3h$ th powers) cut out local schemes of lengths h and $2h$, respectively. Their $4h$ th powers have length $4h$ in both cases. Thus the nonreduced power-zero scheme, not just the set of non-Lefschetz elements, distinguishes them. The specialization $(3, 1) \rightsquigarrow (2, 2)$ increases the first of these lengths. Additional simple spectral blocks give the same distinction in all larger dimensions, with power indices $h(n - 1)$ and hn .

The representation theory and strong Lefschetz property used above are classical inputs, explicitly credited to Nagaoka–Wachi and, for $h = 1$, Shafiei. The geometric statements here are the exact section of the whole reciprocal family, its size and Loewy-depth consequences for actual first-relation boundary fibres, and the power-ideal identity on pencil lines. The last construction is an intrinsically induced reciprocal model; it is not a claim that the common extreme $\mathrm{GL}(E)$ boundary point remembers the starting pencil. Its embedded pencil line is essential.

The preceding relative caution concerns the valuation-ring proof. The following polynomial identity establishes the equality over arbitrary complex base algebras.

APPENDIX T. DESCENT AND COMPLETE-LOCAL DETAILS

The following statements supply the local-to-global steps used in Sections L–Q. They also specify which assertions concern a completed image, a normalization fibre, or a family on an exact-divisor stratum.

Lemma T.1 (Complete-local surjectivity and reduced completion). *In the cotangent-surjectivity assertion of Lemma L.1, the rings are Noetherian complete local $\mathbb{C}$ -algebras with residue field $\mathbb{C}$, and the homomorphism is local, continuous, and the identity on that residue field. Under these hypotheses cotangent surjectivity implies ring surjectivity. A reduced local ring essentially of finite type over $\mathbb{C}$ has reduced completion.*

Proof. Choose preimages in the source maximal ideal of generators of the target cotangent space. Their degree- q products span every quotient $\mathfrak{m}_B^q / \mathfrak{m}_B^{q+1}$. Given an element of B , first lift its residue, then its error successively in degrees $q = 1, 2, \dots$. The correction in degree q lies in $\mathfrak{m}_R^q$. Their sum converges in R , and continuity and separatedness imply that its image is the desired element of B . This is not a statement about arbitrary noncomplete local homomorphisms.

Finite-type complex rings are excellent. Quasi-excellent local rings are Nagata and have geometrically reduced formal fibres; see [29, Tags 07QV and 0BJ0]. For a reduced local ring A , the injection into the product of its finitely many minimal-prime quotients remains injective after flat completion. Each quotient is a Nagata domain and its completion is reduced. Thus $\hat{A}$ is a subring of a product of reduced rings. This justifies the reduced completed-image step of the earlier lemma. $\square$

Lemma T.2 (Coprime cluster products over Artin bases). *Let A be a local Artin $\mathbb{C}$ -algebra. If the reduction of a monic polynomial has a factorization into pairwise coprime monic factors, there is a unique factorization into monic lifts of the same degrees. In a factorization fibre, the functor of divisors with prescribed degrees in the distinct residual clusters is the product of the cluster divisor functors. Its coordinate algebra is therefore the tensor product of their Artin coordinate algebras.*

Proof. For two coprime factors p, q , the linear correction map

$$(\dot{p}, \dot{q}) \longmapsto q\dot{p} + p\dot{q}, \quad \deg \dot{p} < \deg p, \quad \deg \dot{q} < \deg q,$$

is an isomorphism on coefficient spaces over the residue field. Its kernel is zero by coprimeness, and source and target have equal dimension. It is therefore an isomorphism over A as well. Across a square-zero extension of Artin rings it uniquely corrects a lifted product to the given polynomial. Induction on the nilpotence order and then on the number of clusters proves existence and uniqueness. Apply the same argument to both the marked divisor and its complement. The cluster product and decomposition constructions are mutually inverse on every Artin base; Yoneda gives the asserted product of finite schemes. Passing to inverse limits gives the corresponding assertion on complete local bases. $\square$

Lemma T.3 (The local choice and descent in the binary stratum). *The base-change identification of Theorem M.1 holds as an identification of incidence functors, not only on closed fibres. The required choice of a member of the residual space coprime to the common divisor can be made on a Zariski open cover of the exact-gcd stratum; the choices are auxiliary and the resulting identifications glue uniquely.*

Proof. Write a point of the stratum as (G, L) in $\mathbb{P}(S_s) \times X_{D-s,r}$. The vanishing of a member of L at one of the finitely many roots of G is a proper linear condition on that member. Over $\mathbb{C}$ a finite union of these proper hyperplanes does not cover L . Choose a linear combination avoiding them, lift its coefficients on a bundle-trivializing neighbourhood, and invert its resultant with G . This gives the required Zariski neighbourhood. These neighbourhoods cover the stratum and may be pulled back by any base change.

On the marked-divisor incidence, pass locally to a monic coordinate for the proposed divisor f . The quotient by f is finite free. On every geometric fibre f divides G , so the chosen residual member is invertible in that quotient, by coprimeness. Its multiplication matrix has invertible determinant locally, hence it is invertible also over a nonreduced base. Thus $f \mid GL$ implies $f \mid G$ as an equation on that base, and the converse is immediate. Monic division forces the remaining subspace. The identification is independent of which coprime residual member established cancellation, so it glues on overlaps. No lifting of arbitrary image points to the normalization is asserted.

The finite factorization map used there is flat by miracle flatness: its target local rings are regular, its source local rings are Cohen–Macaulay, it is finite, and dimensions agree. These are exactly the hypotheses of [29, Tag 00R4]. This names the flatness input separately from the incidence-functor argument. $\square$

Lemma T.4 (The coordinate equalizer and coherent conductor). *For the block ideals I_A of Theorem N.1, the completed image ring is the equalizer of the branch rings with their restrictions to all pairwise intersections. This assertion uses the coordinate monomial structure. The completed conductor calculation there implies equality of the corresponding coherent ideal sheaves on the squarefree open.*

Proof. A monomial has a coefficient on each branch where its support survives. If it survives on two branches, it also survives on their coordinate intersection. Pairwise agreement therefore assigns it one well-defined coefficient. Collecting these coefficients gives one formal series modulo the intersection of the I_A ; conversely such a series has compatible restrictions. Formal convergence causes no issue, since there are finitely many monomials in each degree. The claim would be false for general configurations of ideals without this coordinate argument.

Write $A \subset B$ for the local finite normalization algebra. The conductor is $\text{Ann}_A(B/A)$, a coherent ideal. Flat completion preserves the exact finite-module sequence and its annihilator: compute the latter as the intersection of the kernels of multiplication on a finite generating set. The completed finite algebra is the product of the normal branch rings. Thus the conductor and the ideal of Y_{g+1} have equal completed stalks at every squarefree closed point. Faithful flatness

gives equal stalks, and coherence on a finite-type complex scheme gives equality of the ideal sheaves. $\square$

Proposition T.5 (Finiteness and uniqueness of the collision diagram). *Every branch map in Theorem O.1 is finite, and the completed incidence diagram is determined, up to the unique isomorphism respecting its incidence data, by its marked-divisor functor. It is independent of the chosen frame and root coordinates.*

Proof. For each cluster $P_i = A_i B'_i$, the coefficients of the monic A_i are integral over the coefficients of P_i . This follows from the finite factorization morphism, or by expressing them as elementary symmetric functions of a selected subset of the roots in a finite splitting algebra. Division by a monic polynomial expresses every coefficient of B'_i polynomially in those of A_i and P_i . Likewise $R_{ij} = A_i U_{ij}$ expresses all quotient coefficients in terms of A_i and R_{ij} . Consequently the branch algebra is generated over the ambient coefficient algebra by finitely many integral coefficients of the A_i , so it is finite. Completing preserves this finite-module assertion.

The equations represent the functor of a subbundle with a marked divisor and its uniquely determined quotient. A change of frame or local coordinate gives inverse natural transformations of this same functor and of its forgetful map to the ambient Grassmannian. The complete representing rings and their maps are uniquely isomorphic by Yoneda. Their scheme-theoretic images consequently agree as completed diagrams. This proves coordinate independence without identifying arbitrary Artin points of an image with lifted Artin points of its normalization. $\square$

Partition closures. For a fixed partition $(s_1, \dots, s_b)$, the proper map $(\mathbb{P}^1)^b \rightarrow \mathbb{P}(S_s)$ sending (ℓ_i) to $\prod \ell_i^{s_i}$ has closed image. Its open set of distinct ℓ_i is dense. At the boundary precisely some of these points coincide, and the associated parts add. Every such coincidence can be approached from distinct points. This proves exactly the merging closure order in the fixed-degree symmetric product; compare the coincident-root constructions of [19, 20]. It is not a closure classification for all analytic singularities of the completed collision atlas.

Equivariance of the singular Fitting calculation. The group used in Theorem P.1 is $\mathrm{GL}_m(\mathbb{C})$, acting simultaneously on the columns u and w and fixing Δ . It preserves the presentation and therefore its intrinsic Fitting ideal. In the normalization the three displayed minors span, under this group, respectively $\mathrm{Sym}^m(\mathbb{C}^m)t$, $\Delta \mathrm{Sym}^m(\mathbb{C}^m)$, and $\mathrm{Sym}^{m+1}(\mathbb{C}^m)$, with the coordinate/dual convention determined by the action on u . To see spanning directly, their translates are nonzero scalar multiples of the corresponding powers of all linear forms; polarization in characteristic zero spans each symmetric power. Multiplication by R gives exactly the modules asserted in (P.1). This is stronger than invoking invariance without identifying the representations.

Specialization of minimal relations. For the surjective coefficient specialization $\pi : P \rightarrow P'$ to one coordinate in Theorem Q.1, one has $\pi(\mathfrak{n}) = \mathfrak{n}'$ and $\pi(\mathcal{I}) = \mathcal{I}'$. Consequently $\pi(\mathfrak{n}\mathcal{I}) = \mathfrak{n}'\mathcal{I}'$. A coefficient whose image is a minimal binary relation cannot lie in $\mathfrak{n}\mathcal{I}$. This explicitly justifies the nonzero irreducible coefficient map used in that proof. All reciprocal series and socle weights in these arguments use the weight of a Q_i coefficient equal to i . They do not claim ordinary maximal-ideal Hilbert functions; (R.7) gives one such function separately.

APPENDIX U. CONDUCTORS AND LOCAL MODELS OF DIVISOR COLLISIONS

We retain the notation of Section K. All completed local rings below are completed at closed complex points. For a reduced local ring A with finite normalization B , its *conductor* is $\mathfrak{c} = \mathrm{Ann}_A(B/A)$, regarded also as an ideal of B . We use the intrinsic singular subscheme defined by $\mathrm{Fitt}_{\dim A} \Omega_{A/\mathbb{C}}$; on completions, differentials mean continuous differentials. The conductor and this Fitting ideal need not agree. Our first result determines both on a multivariate stratum. The second identifies the way two branches merge when a common binary root becomes double.

The formal-image and separated-germ steps below are isolated in Lemma L.1. Lemma L.2 gives the Artin cancellation and a direct positive lower bound for the split codimension. Theorem O.1 supplies the full Weierstrass incidence functor in every collision order; Theorem P.1 evaluates the singular ideal for all ranks.

U.1. Two coprime divisor choices.

Theorem U.1 (The split-divisor stratum). *Suppose $h > g$, $2 \leq r \leq s_{h-g}$, and*

$$K = abL, \quad [a], [b] \in \mathbb{P}(S_g), \quad L \in \text{Gr}(r, S_{h-g}), \quad \gcd(L) = 1.$$

Assume that a, b are coprime and that the only degree- g divisors of ab , up to scalar, are a and b . Put

$$\begin{aligned} t &= 2(s_g - 1) + r(s_{h-g} - r), \\ c &= r(s_h - s_{h-g}) - (s_g - 1), \quad d_g = t + c. \end{aligned}$$

Then $c \geq 1$, and there are formal coordinates in which

$$\hat{\mathcal{O}}_{Y_g, K} = \mathbb{C}[[z_1, \dots, z_t, x_1, \dots, x_c, y_1, \dots, y_c]] / (x_i y_j : 1 \leq i, j \leq c). \quad (\text{U.1})$$

Its normalization is $\mathbb{C}[[z, x]] \times \mathbb{C}[[z, y]]$. The conductor is

$$\mathfrak{c} = (x_1, \dots, x_c, y_1, \dots, y_c), \quad \hat{\mathcal{O}}_{Y_g, K} / \mathfrak{c} = \mathbb{C}[[z]], \quad (\text{U.2})$$

and the intrinsic singular subscheme has ideal $\mathfrak{c}^c$. The two smooth branches meet scheme-theoretically in a smooth t -dimensional subscheme. The local ring has multiplicity two and depth $t + 1$; it is Cohen–Macaulay exactly when $c = 1$.

Proof. We first determine the intersection as a scheme, not by comparing closed points or tangent dimensions. The normalization has precisely two points above K , namely (a, bL) and (b, aL) . At either point, the overlap in Lemma K.3 is zero. The map from the completed local ring of the ambient Grassmannian to each completed source ring is consequently surjective: its cotangent map is surjective, and completeness and Nakayama’s lemma lift successive homogeneous terms. Thus each source completion is a closed smooth formal branch of the image.

Here is the required assertion over infinitesimal bases. Let A_0 be a local Artin $\mathbb{C}$ -algebra. A point on both formal branches carries unique marked lifts a_{A_0}, b_{A_0} near a, b . If homogeneous polynomials have coprime reductions, those reductions form a regular sequence in the polynomial ring over $\mathbb{C}$. Its graded Koszul complex is exact. In each degree the polynomial modules are finite free; lifting a splitting of the complex, or applying the local criterion for flatness successively, shows that its lift over A_0 is exact and that the quotient is A_0 -flat. In particular

$$(a_{A_0}) \cap (b_{A_0}) = (a_{A_0} b_{A_0}). \quad (\text{U.3})$$

This also follows by noting that b_{A_0} is a nonzerodivisor modulo a_{A_0} and solving a relation between them. Applying (U.3) to the columns of the relation subbundle gives, uniquely,

$$K_{A_0} = a_{A_0} b_{A_0} L_{A_0}.$$

The residual module is a subbundle: multiplication by $a_{A_0} b_{A_0}$ is a split injection in degree D , and the quotient of the ambient free module by K_{A_0} is free. The kernel of the induced surjection onto the complementary quotient is therefore free. This proves that the formal intersection is represented by the product of the two divisor charts and the Grassmannian chart for L , with no nilpotents omitted.

The differential of this product parametrization is injective. Indeed, an infinitesimally constant product gives

$$(\dot{a}b + a\dot{b})l + ab\dot{l} \in abL \quad (l \in L).$$

Taking valuations at the factors of a , and using $\gcd(a, b) = \gcd(L) = 1$, forces $a \mid \dot{a}$; similarly $b \mid \dot{b}$. These are the two projective scalar directions, and then $\dot{L} = 0$. The intersection is thus a smooth formal closed subscheme of dimension t .

Let R_0 be the ambient completed local ring and I_a, I_b the ideals of these branches. Finiteness of the normalization, exactness of completion on finite modules, and the definition of the scheme-theoretic image give

$$\widehat{\mathcal{O}}_{Y_g, K} = R_0 / (I_a \cap I_b) = (R_0 / I_a) \times_{R_0 / (I_a + I_b)} (R_0 / I_b).$$

This is the elementary conductor-square construction; see also [21, Lemma 1.3 and §1.3.2]. All three quotient rings are regular. Choosing compatible regular parameters for the two surjections gives (U.1). The branch dimension is $d_g = s_g - 1 + r(s_h - r)$, so its codimension along the intersection is the asserted c . It is positive: if $c = 0$, both branch ideals would coincide locally with the intersection ideal, contradicting the two distinct normalization branches of a reduced local ring.

In the fibre-product description, an element belongs to the conductor exactly when both its restrictions to the intersection vanish. This gives (U.2). The Jacobian of the relations $x_i y_j$ has nonzero entries only among the x_i and y_j . The Fitting ideal in question is its ideal of c -minors. Every such minor belongs to $\mathfrak{c}^c$. Conversely, choosing the c columns indexed by the y_j and one row (i_j, j) for each j gives every monomial $x_{i_1} \cdots x_{i_c}$. The symmetric choice gives every degree- c monomial in the y 's. Mixed monomials vanish in the ring. Thus the ideal is exactly $\mathfrak{c}^c$, including its nonreduced structure.

Finally use

$$0 \longrightarrow \widehat{\mathcal{O}}_{Y_g, K} \longrightarrow \mathbb{C}[[z, x]] \oplus \mathbb{C}[[z, y]] \longrightarrow \mathbb{C}[[z]] \longrightarrow 0,$$

where the last arrow is the difference of restrictions. The depth lemma gives depth $t + 1$; for $c = 1$ this equals the dimension. For $c > 1$ it is strictly smaller. Additivity of leading Hilbert coefficients gives multiplicity two, since the last module has smaller dimension than the two regular branches. $\square$

Remark U.2. The hypotheses are nonempty in every number of variables: for $e = 2$ one may take $g = 1$, and for $e \geq 3$ one may take two distinct irreducible forms of degree g . The theorem treats an explicitly specified stratum, not all pairs of overlapping degree- g divisors. In particular, it does not substitute a binary factorization rule for multivariate divisibility. Its high-codimension branch intersections need not be normal-crossing divisors.

U.2. A double common root. For $m \geq 1$, put $R_m = \mathbb{C}[[u_1, \dots, u_m, \Delta]]$ and $I = (u_1, \dots, u_m) \subset R_m$. Define

$$A_m = R_m \oplus It \subset B_m = \mathbb{C}[[u_1, \dots, u_m, t]], \quad \Delta = t^2. \quad (\text{U.4})$$

The right side of (U.4) is a subring, not a square-zero extension: $(It)^2 = \Delta I^2$. For $m = 1$ it is the ordinary pinch-point ring.

Theorem U.3 (Collision of two binary divisor choices). *Let $e = 2$, $g = 1$, $D = h + 1$, and $2 \leq r \leq D - 1$. At a point $K_0 = \ell^2 L$ with $L \in \text{Gr}(r, S_{D-2})$ and $\gcd(L) = 1$, set*

$$m = r - 1, \quad q = r(D - 1 - r) + 1.$$

Then

$$\widehat{\mathcal{O}}_{Y_1, K_0} \simeq A_m[[z_1, \dots, z_q]]. \quad (\text{U.5})$$

Writing $w_i = u_i t$, a complete presentation of A_m is

$$\mathbb{C}[[u_1, \dots, u_m, \Delta, w_1, \dots, w_m]] / (u_i w_j - u_j w_i, w_i w_j - \Delta u_i u_j), \quad (\text{U.6})$$

where the first relations have $i < j$ and the second have $i \leq j$. Its normalization is B_m , and its conductor is

$$\mathfrak{c} = (u_1, \dots, u_m, w_1, \dots, w_m), \quad \mathfrak{c}B_m = IB_m.$$

The induced map on conductor quotients is

$$\mathbb{C}[[\Delta]] \longrightarrow \mathbb{C}[[t]], \quad \Delta \longmapsto t^2. \quad (\text{U.7})$$

Thus the two reduced divisor lifts merge through a ramified double cover of the conductor. The completed local ring has dimension $q + m + 1$ and depth $q + 2$. It is Cohen–Macaulay if and only if $r = 2$. For $r = 2$, its intrinsic singular ideal, with the smooth variables suppressed, is $(w, \Delta u, u^2)$, whereas its conductor is (u, w) .

Proof. Choose an affine coordinate x with the root of ℓ at zero. Some member $F_0 \in K_0$ has multiplicity exactly two there, since $\gcd(L) = 1$. Extend it to a frame $F_0, \dots, F_{r-1}$. In the completed local ring of the ambient Grassmannian the formal Weierstrass theorem writes the universal F_0 as a unit times

$$P(x) = x^2 + ax + b.$$

Divide each F_i , $i > 0$, by this monic polynomial. Its remainder is $u_i x + v_i$. The $2r$ functions $a, b, u_1, v_1, \dots, u_m, v_m$ are part of a regular parameter system. To see this rather than assume versality, note that K_0 has zero two-jet at the root, while $S_D/K_0 \rightarrow \mathbb{C}[x]/(x^2)$ is surjective. The Grassmannian tangent space $\text{Hom}(K_0, S_D/K_0)$ therefore allows independent variations of both jets in every frame column. Passing from these jets to the displayed remainders is an invertible triangular operation, with leading coefficient $F_0/x^2(0) \neq 0$. Their differentials are consequently independent.

Set

$$t = x + a/2, \quad \Delta = a^2/4 - b, \quad w_i = au_i/2 - v_i.$$

The incidence of a common root near zero is now exactly

$$\Delta = t^2, \quad w_i = u_i t.$$

There are no other normalization points above K_0 , because its exact common divisor is ℓ^2 . Consequently the image of this completed incidence is the completed image defining Y_1 . The ambient dimension is $N_0 = r(D + 1 - r)$. Of the $2r$ parameters above, a is free, as are the other $N_0 - 2r$ parameters. This accounts for $q = N_0 - 2r + 1$ smooth variables in (U.5).

It remains to justify elimination scheme-theoretically. The relations in (U.6) reduce any series to $p + \sum_i w_i p_i$ with $p, p_i \in R_m$. If its image in B_m is zero, the decomposition $B_m = R_m \oplus R_m t$ gives $p = 0$ and $\sum_i u_i p_i = 0$. The first syzygies of the regular sequence $u_1, \dots, u_m$ are generated by the Koszul syzygies. They are precisely the relations $u_i w_j - u_j w_i$. This proves the presentation without taking a radical. The same argument works first over polynomial rings and then by exact completion.

The extension $A_m \subset B_m$ is finite, since $t^2 = \Delta$, and birational, since $t = w_1/u_1$ in the fraction field. The regular ring B_m is thus the normalization. For $p + vt \in R_m \oplus It$, multiplication by t stays in A_m if and only if $p \in I$. As $1, t$ generate B_m , the conductor is $I \oplus It$, proving the quotient map (U.7). Equivalently,

$$A_m = B_m \times_{\mathbb{C}[[t]]} \mathbb{C}[[\Delta]], \quad (\text{U.8})$$

where the first map sets $u = 0$ and the second sends Δ to t^2 . This is a specific conductor square, not an assumption that the normalization descends through its reduced fibres.

As an R_m -module, $A_m = R_m \oplus I$. For $m = 1$, the ideal I is free. For $m > 1$, the exact sequence $0 \rightarrow I \rightarrow R_m \rightarrow \mathbb{C}[[\Delta]] \rightarrow 0$ gives $\text{depth}_{R_m} I = 2$. Hence A_m has depth two for every $m \geq 1$. The maximal ideal of R_m generates an ideal with maximal radical in A_m , so this is also its local

ring depth. Adding the q smooth parameters gives the assertion. When $m = 1$, the equation is $w^2 - \Delta u^2 = 0$; its partial derivatives generate $(w, \Delta u, u^2)$ in characteristic zero. $\square$

Corollary U.4 (The split and ramified parts of the same model). *On the binary locus where the common divisor has degree exactly two, there are two strata. Over two distinct roots the formal local ring has the split form (U.1), with $t = q + 1$ and $c = r - 1$. Over a double root it has (U.5). The support of the singular locus in the latter model is the conductor, and the normalization fibre at a collision is $\text{Spec } \mathbb{C}[t]/(t^2)$. For arbitrary m , its singular subscheme is given without radicalization by the m -minors of the Jacobian of the displayed m^2 relations in (U.6).*

Proof. The assertion at distinct roots is Theorem U.1. In the collision model the normalization is an isomorphism wherever some u_i is invertible. On $u = w = 0$, a nonzero value of Δ gives two distinct points after an étale square-root extension, while $\Delta = 0$ gives the stated length-two fibre. The exact normality criterion proves the assertion about singular support. There are $2m + 1$ nonsmooth coordinates and dimension $m + 1$; the cotangent presentation therefore gives the m -minors as claimed. This formula specifies the whole singular scheme, even where it is thicker than its conductor support. $\square$

Remark U.5 (Descent to intrinsic first relations). On the maximal-lower-Hilbert-function stratum of Corollary K.8, these statements compute the completed smooth-atlas singularities of the intrinsic first-relation algebra stack after tangent-identity rigidification. The conductor is equivariant, and formation of its finite-module annihilator and of the Fitting ideals commutes with flat base change. Thus the calculations descend through the frame atlas. This does not identify these rings with completed local rings of a coarse orbit space, where stabilizers can introduce further singularities.

APPENDIX V. INTERSECTIONS OF DIVISOR-DEGREE IMAGES

Here the total relation degree D is fixed. Write $Y_a \subset \text{Gr}(r, S_D)$ for the image admitting a divisor of degree a ; empty Grassmannians are omitted. In several variables these images are not generally nested. Their reduced intersections nevertheless have a finite description by common parts of two marked divisors.

Proposition V.1 (The common-part formula). *For $1 \leq a, b < D$ and $0 \leq j \leq \min(a, b)$, let M_j be the scheme-theoretic image of*

$$\begin{aligned} \mathbb{P}(S_j) \times \mathbb{P}(S_{a-j}) \times \mathbb{P}(S_{b-j}) \times \text{Gr}(r, S_{D-a-b+j}) &\longrightarrow \text{Gr}(r, S_D), \\ (c, u, v, L) &\longmapsto cuvL. \end{aligned} \quad (\text{V.1})$$

A negative last degree or insufficient dimension gives the empty image, and $\mathbb{P}(S_0)$ is a point. Then

$$(Y_a \cap Y_b)_{\text{red}} = \left(\bigcup_j M_j \right)_{\text{red}}. \quad (\text{V.2})$$

On the locus of two marked divisors f_a, f_b with exact common part of degree j , the corresponding factorization is $c = \gcd(f_a, f_b)$, $f_a = cu$, $f_b = cv$, with $\gcd(u, v) = 1$. In particular, at a relation $K = a_0^m L$ with a_0 a nonzero linear form and $\gcd(L) = 1$, the set of positive divisor degrees is exactly $\{1, \dots, m\}$. For the boundary point in Theorem W.1, $m = D - q_n$.

Proof. Each map in (V.1) is defined on a projective scheme: multiplication by a nonzero form has constant rank, so the subspace construction is a morphism. Its image is closed and admits the two divisors cu and cv , proving one inclusion. Conversely, if K has divisors f_a, f_b , unique factorization gives their common part c of some degree j . Their least common multiple is cuv and divides every member of K . Division yields L of degree $D - a - b + j$, of the same rank.

This proves equality on geometric points. Over $\mathbb{C}$, a reduced closed subscheme of a finite-type scheme is determined by its closed points, giving (V.2). The last assertion follows because every homogeneous divisor of a power of one linear form is a power of that form. $\square$

The reduction in (V.2) is essential to the statement of this global formula. It is not a prescription to reduce the individual Y_a , their normalization fibres, or their singular subschemes. The full intersection of the two formal branches on the coprime stratum was computed over Artin bases in Theorem U.1; Theorem U.3 computes the thickening when those branches collide. Thus the degree-poset description and the local scheme calculations have distinct, explicit roles.

APPENDIX W. A COMMON RAMIFIED BOUNDARY OF PENCIL FAILURE ORBITS

The fixed tensor presentation of a pencil has exact common divisor δ^{n-1} . Varying the pencil alone therefore stays in the exact-divisor locus. The intrinsic finite algebra permits a larger, linear coordinate-change action: changing the basis of its entire cotangent space. We now describe a boundary point produced by such changes. The generic members are genuine pencil failure algebras. The central member has a larger common divisor, and its normalization fibre is nonreduced. Thus the boundary of Section K occurs inside the orbit closures of the motivating algebras themselves.

Let $n = \dim V = \dim U \geq 3$, $E = V^* \otimes U$, and put

$$e = n^2, \quad N = \binom{n+1}{2}, \quad r = \binom{N}{2}, \quad g = n(n-1), \quad h = 3n-4, \quad D = g+h.$$

For a pencil R , write $K_R = \delta^{n-1} C_R \subset \text{Sym}^D E$ as in Lemma I.J.1. Define the projective orbit closure

$$Z_R = \overline{\text{GL}(E) \cdot K_R} \subset \text{Gr}(r, \text{Sym}^D E).$$

It is independent of a chosen basis of the cotangent space. Since Y_g is closed and invariant, $Z_R \subset Y_g$.

The group in Z_R is the full $\text{GL}(E)$, not the $\text{PGL}(V)$ congruence group of the pencil. Fixing the standard flag, scalar direction and complement specifies the displayed common point; no choice-independent distinguished moduli point is asserted.

Theorem W.1 (A common point in full first-relation orbit closures). *Set*

$$q_n = \begin{cases} 3, & n = 3, \\ 4, & n \geq 4, \end{cases} \quad k_n = h - q_n.$$

There is an explicitly constructed point K_∞ belonging to Z_R for every quadratic pencil R , with

$$K_\infty = a^{g+k_n} L_n, \quad L_n \subset \text{Sym}^{q_n} E, \quad \text{rank } L_n = r, \quad \gcd(L_n) = 1, \quad (\text{W.1})$$

where $a \in E$ is a nonzero linear form. The flag pencil $R_ = \langle x_1^2, x_1 x_2 \rangle$ admits a finite-flat family of first-relation algebras over $\mathbb{A}^1$, with special relation space K_∞ and every nonzero fibre isomorphic to its failure algebra. Together with Theorem I.L.1, this gives a chain of two explicit finite-flat specializations from the failure algebra of every R to the same boundary algebra.*

The fibre of ν_g over K_∞ has a single geometric point, marked by a^g , and is nonreduced. Its tangent dimension is

$$\binom{e + k_n - 1}{k_n} - 1. \quad (\text{W.2})$$

In particular K_∞ lies in the singular locus of Y_g and outside the exact-degree- g locus. For $n = 3$ this tangent dimension is 44.

The construction requires the exact osculating order of one elementary flag embedding. We include the calculation to distinguish the theorem from a computationally suggested degeneration.

Lemma W.2 (The flag coefficient filtration). *Write a matrix of independent linear coordinates as*

$$T = aI_n + B, \quad B_{11} = 0.$$

For $C_ = C_{R_*}$, the filtration by vanishing order at $T = I_n$ on the chart $T_{11} = 1$ has last nonzero graded piece in degree q_n . It has a nonzero degree-zero piece. Hence an adapted basis $c_1, \dots, c_r$ has orders $0 \leq d_j \leq q_n$, attaining both endpoints, and*

$$\lim_{\tau \rightarrow 0} \tau^{-d_j} c_j(aI_n + \tau B) = a^{h-d_j} p_j(B), \quad (\text{W.3})$$

where the displayed initial forms are linearly independent.

Proof. The complementary-minor identity for $\text{Sym}^2 T$ identifies the vector of primitive coefficients, up to a fixed invertible coordinate change, with

$$\delta(T)^3 (v^2 \wedge vw), \quad v = T^{-1}e_1, \quad w = T^{-1}e_2. \quad (\text{W.4})$$

Indeed $\det(\text{Sym}^2 T) = \delta^{n+1}$, the complementary minors have order two, and division of δI_{N-2} by δ^{n-1} leaves δ^3 . This identity is initially on $\delta \neq 0$; both sides are the same polynomial vector of degree h .

A standard affine flag chart has representatives

$$v = e_1 + \sum_{i=2}^n z_i e_i, \quad w = e_2 + \sum_{i=3}^n y_i e_i.$$

Every coordinate of $v^2 \wedge vw$ is a polynomial of degree at most four. The map from GL_n to this flag chart is smooth, and restricting to $T_{11} = 1$ remains smooth: the scalar subgroup is contained in each flag stabilizer and is transverse to this chart. Normalizing v, w only multiplies the whole wedge by a common unit. Smooth pullback preserves the order of a nonzero function, as does multiplication by a unit. The coordinates in (W.4) are linearly independent by Lemma I.6.2 and the nonzero matrix-coefficient inclusion in (I.6.6). Thus no nonzero linear combination can have order exceeding four. For $n \geq 4$, the coordinate indexed by $e_3^2 \wedge e_3 e_4$ is

$$z_3^2(z_3 y_4 - z_4 y_3),$$

which is nonzero and has order exactly four.

For $n = 3$, put $z = z_2$, $w = z_3$, $t = y_3$. In the ordered basis $e_1^2, e_1 e_2, e_1 e_3, e_2^2, e_2 e_3, e_3^2$, the two factors are

$$(1, 2z, 2w, z^2, 2zw, w^2), \quad (0, 1, t, z, zt + w, wt).$$

Their fifteen wedge coordinates have the same linear span as

$$\begin{aligned} &1, z, w, t, z^2, zw, w^2, zt, wt, z^2 t, zwt, w^2 t, \\ &z^2 w - z^3 t, \quad zw^2 - z^2 wt, \quad w^3 - zw^2 t. \end{aligned} \quad (\text{W.5})$$

This is obtained by subtracting the two coordinates with a common quadratic term and rescaling by 2. Their lowest homogeneous terms are distinct monomials, all of degrees at most three, and some have degree three. Thus the third jet is injective on this coefficient space and the second is not. This proves $q_3 = 3$. The coordinate $e_1^2 \wedge e_1 e_2$ is 1 in every dimension, proving the order-zero assertion. Finally, homogeneity of degree h gives (W.3); an adapted basis is by definition a basis on each associated-graded piece, so its initial forms are independent. $\square$

Proof of Theorem W.1. The linear substitution $T \mapsto aI_n + \tau B$ is invertible for $\tau \neq 0$. Divide the adapted coefficient c_j by its exact power τ^{d_j} after substitution. There are no negative powers: all its terms have B -degree at least d_j . The resulting columns span a rank- r subbundle $J_\tau \subset \text{Sym}^h E \otimes \mathbb{C}[\tau]$. Independence at zero follows from the lemma, and away from zero from the invertible substitution. The ideal of its maximal minors in $\mathbb{C}[\tau]$ has no common closed zero, and is therefore the unit ideal. On each maximal-minor open a corresponding square submatrix is invertible and splits the image. These opens cover $\text{Spec } \mathbb{C}[\tau]$, proving the subbundle assertion over the whole line, including all scheme points. Set

$$K_\tau = \det(aI_n + \tau B)^{n-1} J_\tau.$$

Multiplication by the nonzero homogeneous polynomial in every fibre again has constant rank. Consequently K_τ is a subbundle, and

$$\mathcal{A}_\tau = \text{Sym } E \otimes \mathbb{C}[\tau] / (K_\tau, \mathfrak{m}^{D+1})$$

is finite locally free of rank $\binom{e+D}{D} - r$. For $\tau \neq 0$, its relation space is a linear transform of K_{R^*} .

At zero the marked factor becomes a^g , while $J_0 = \langle a^{h-d_j} p_j(B) \rangle$. It contains a nonzero multiple of a^h , so its gcd is a power of a . It also contains a polynomial with exact a -adic order $h - q_n$, and every column is divisible by that power. Therefore $\gcd(J_0) = a^{k_n}$, proving (W.1). Notice that $0 < k_n < g$ for every $n \geq 3$.

The closed-flag specialization of Theorem I.L.1 and its failure-algebra lift imply $K_{R^*} \in Z_R$. A closed invariant subset containing a point contains that point's orbit closure. Hence the just-constructed $K_\infty \in Z_{R^*}$ belongs to every Z_R . This proves the assertion about the two-stage specialization without replacing it by an unproved assertion about one fixed one-parameter subgroup acting on every initial pencil.

The sole degree- g divisor of a^{g+k_n} is a^g . Its overlap with the residual gcd is a^{k_n} . Lemma K.3 gives (W.2); the fibre is supported at one point and has positive tangent space. The normalization criterion then proves singularity. With $n = 3$, $e = 9$ and $k_n = 2$, the dimension is $\binom{10}{2} - 1 = 44$. $\square$

W.1. The complete normalization fibre at the limiting algebra. The last theorem is not restricted to the tangent space of its fibre. The full local algebra can be given uniformly.

Proposition W.3 (Pure-power divisor fibre). *Let $E = \mathbb{C}a \oplus W$, $g \geq k \geq 1$, and $K = a^{g+k}L$ with $\gcd(L) = 1$. Introduce coefficient variables for $q_i \in \text{Sym}^i W$, $1 \leq i \leq k$, and define forms c_j recursively by*

$$c_0 = 1, \quad c_j = - \sum_{i=1}^k q_i c_{j-i}, \quad c_j = 0 \ (j < 0). \quad (\text{W.6})$$

The entire fibre of ν_g over K , supported at its unique point, is the Artin algebra

$$\mathbb{C}[[\text{coefficients of } q_1, \dots, q_k]] / (\text{coefficients of } c_{g+1}, \dots, c_{g+k}). \quad (\text{W.7})$$

Its tangent dimension is $s_k(E) - 1$.

Proof. Choose $p \in L$ not divisible by a . Over any local Artin base, a divisor f near a^g is monic in a , and the quotient by f is flat over the base. The reductions of f, p are coprime. The regular-sequence argument used in (U.3) makes p a nonzerodivisor modulo f . Thus $f \mid a^{g+k}p$ is equivalent to $f \mid a^{g+k}$, functorially on these bases. Once this condition holds, it holds for every member of K .

Polynomial division gives a unique monic residual factor $H = a^k + q_1 a^{k-1} + \dots + q_k$. The equation $fH = a^{g+k}$ determines the coefficients of f successively as $c_1, \dots, c_g$ in (W.6). The remaining k product coefficients vanish if and only if $c_{g+1}, \dots, c_{g+k}$ vanish. This follows successively by comparing the product with the reciprocal series $(1 + \sum q_i a^{-i})^{-1}$. Hence (W.7) represents the full

infinitesimal fibre, without reduction. Finiteness of ν_g makes it Artin; there is no other geometric divisor, so completion loses no part of the fibre. Since $g \geq k$, these equations have no linear terms in the coefficient variables. The number of coefficient variables is $\sum_{i=1}^k \dim \operatorname{Sym}^i W = \dim \operatorname{Sym}^k E - 1$, as asserted. $\square$

Example W.4 (The nine-generator limit). For $n = 3$, the construction can be written with four of the nine linear coordinates, denoted a, u, v, w . The other five coordinates remain generators of the algebra. Up to nonzero scalar changes of the relation basis, $K_\infty = a^8 L_3$, where

$$L_3 = \langle a^3, a^2u, a^2v, a^2w, au^2, auv, av^2, auw, avw, \\ u^2v, u^2w, uv^2, uvw, v^3, v^2w \rangle. \quad (\text{W.8})$$

Here one may take $u = B_{21}$, $v = B_{31}$, $w = B_{32}$; the monomials are the lowest terms in (W.5). The relation space has dimension 15 and exact gcd a^8 . The selected divisor has degree $g = 6$, and its fibre has 44 coefficient variables: eight coefficients of a linear form Q_1 and thirty-six of a quadratic form Q_2 in W . Its defining ideal consists of all coefficients of the two forms

$$c_7 = -Q_1^7 + 6Q_1^5Q_2 - 10Q_1^3Q_2^2 + 4Q_1Q_2^3, \\ c_8 = Q_1^8 - 7Q_1^6Q_2 + 15Q_1^4Q_2^2 - 10Q_1^2Q_2^3 + Q_2^4.$$

This is a full presentation of the normalization fibre. Its embedding dimension 44 is not being identified with its length, nor is this fibre algebra being identified with the completed local ring of Y_g .

Remark W.5 (Relation to pencil invariants). The action defining Z_R is the intrinsic $\operatorname{GL}(E)$ action on first relations, not merely the congruence action on V . The flag pencil is singular: every member has rank at most two. It is therefore outside the regular-pencil open set treated by [4, Theorem 1.1 and §5], where Segre symbols organize classification and degenerations. The theorem above does not purport to classify that poset or all orbits in Z_R . Its conclusion is instead a uniform common boundary and an exact ramification algebra for genuine finite failure algebras. The increase from g to $g + k_n$ in the intrinsic gcd is precisely what is invisible in a fixed-determinant pencil specialization.

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AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

For the present revision, GPT-6 Astra Pro assisted specifically with the incidence normalization and overlap criterion in Section K, the universal-property proof in Section I.K, the explicit specialization in Section I.L, and the associated exact tests and exposition. These arguments require independent human verification. No statement here represents a completed journal review or a formal proof-assistant certification.

Additional disclosure for the present revision. GPT-6 Astra Pro assisted specifically with the conductor and collision models, the scalar-matrix degeneration, the flag-jet calculation, the reciprocal-polynomial fibre presentation, literature comparisons, and exact computational checks. The proofs above state the complete mathematical arguments; finite computations are consistency checks, not universal proof certificates. No journal decision or historical priority determination is inferred from the build and source audits.

ASSISTANCE IN THE PRESENT REVISION

An AI assistant helped formulate and write the binary conductor stratification, the simultaneous incidence atlas, the evaluated singular Fitting ideal, and the minimal-relation-module argument for pure-power fibres. It also prepared the source-preserving reorganization and finite exact consistency checks. The written proofs, rather than those finite checks, are the mathematical basis of the statements and are submitted to independent scrutiny.

Revision 154. AI assistance was used for the polarized moduli compatibility, spectral divisor-fibre construction, homological calculations and proof expansions. The fixed five-variable Betti table uses exact computer-assisted rational elimination with its complete finite inputs recorded. The general theorems rely on the written arguments and the explicitly attributed classical results, not on regression counts or source-preservation receipts.

Revision 155. AI assistance was used to formulate and draft the exact quadratic coefficient-section theorem and the reciprocal power-ideal description of spectral Fitting schemes, to check their finite examples, and to prepare the companion-paper organization. Classical apolar and orthogonal-orbit results are explicitly attributed. The general claims rely on the written proofs; finite exact tests and source preservation are not formal proof certificates.

Revision 157. AI assistance was used in deriving and drafting the universal power-ideal formula, its complete-quadric graph interpretation, the contact-divisor and resolved-pencil statements, and their exact finite checks. Classical determinantal, apolar and complete-quadric results are attributed at their points of use. The proofs, rather than the computations, support the general statements. No formal proof certification or journal endorsement is claimed.

Scope of the new intrinsic construction. The source-normalized envelope is an algebra bundle with an intrinsic projective power diagram. It is not a claimed preferred subquotient of the sharp failure algebra. Singular-pencil classification is read through the classical complex symmetric Kronecker form. The explicit Hilbert tail is a boundary calculation for the stated transverse family, not a classification of every fibre of the incidence compactification. Exact computations accompanying the source check finite instances; they do not certify the general proofs.

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